FLAGGED_149

Proofs, counterexamples, and remaining obstructions
in cote 149 and the opening of cote 142

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12 November 2026 No priority is claimed

Status: INCOMPLETE. We distinguish proved statements, exact literature deductions, refutations, and explicitly unresolved assertions. In particular, we do not prove the remaining positive-characteristic point-section assertion or the section and birational lifting assertions encountered in the sweep. The report and ledgers specify the failed steps and next mathematical obligations. The presence of an *Open* result never supplies a hypothesis for a preceding proof.

Notation concordance. Groups are profinite and homomorphisms continuous unless otherwise stated. A geometric base point and paths fix actual homomorphisms; changing those paths gives the stated inner-conjugacy ambiguity. Further local variables are defined at their first use. This table fixes the recurrent manuscript symbols on the first page.

Our notation	Meaning and manuscript notation	Introduction / relevant scans
G_K	$\mathrm{Gal}(K^{\mathrm{sep}}/K)$; his E_K, later $\mathcal{E}_K$	149: 3–4; 105–106
E_U	$\pi_1(U)$; his E_U, $\mathcal{E}_U$	149: 24, 53; 107–108
Π_U, $P_{L/K}$	Geometric kernels; his π, Π_U, $\pi_{L/K}$	149: 9–13, 24, 53, 113–114
Γ, Γ°	Arithmetic group and $\ker\chi$; his Γ, Γ^0, E°	149: 8–9, 45, 54–56
T, $T_{p'}$	$\widehat{\mathbb{Z}}(1)$, $\prod_{\ell\neq p}\mathbb{Z}_\ell(1)$; his T_∞, T	149: 53–54, 75–78, 100
I_c, I_v	Cusp or valuation inertia; his L_i, I_x, I_V	149: 53–55, 117–124
D_c, D_v	Decomposition; his $Z(L_i)$ or N_x, N_V	149: 54, 117–127
Γ_v, $\kappa(v)$	Additive value group and residue field; his V, $k(x)$ data	149: 121–126
$\widehat{B}$	Specified completion with its canonical map $B\to\widehat{B}$	149: 81–85; 142: 9–14
$A(S)$, $A_0(S)$	Oriented homeomorphisms and identity component; his $\mathcal{A}_S, \mathcal{A}_S^0$	142: 4–8
$M(S)$, G_u, H_u	Mapping classes and arrow stabilizer/projection; his $\mathcal{T}_S, G_u, H_u$	142: 8–14

A type (g,ν) is hyperbolic when $2g-2+\nu>0$. The symbol (p') on an arithmetic curve group replaces its geometric kernel by the maximal prime-to-p quotient and retains the full arithmetic quotient. "Page" in a passage citation means the handwritten manuscript page; a scan continuing that page is identified as such.

Scope and method

We read both supplied transcriptions of cote 149, scans 1–136, and the opening of cote 142, scans 1–22, with scans 4, 6, 5, 7 in their reading order. The first-pass transcriptions and selected images are evidence for the wording, not premises of the mathematics [Groa, Grob]. The readings ledger records every mathematical correction and every doubtful alternative on which our arguments bear. An inspected image supplies editorial evidence; it does not make a reading final.

Each result is labelled *Proved*, *Known*, *Refuted*, or *Open*. "Known" includes an exact statement of the opened input and the comparison maps needed to derive the manuscript assertion. "Refuted" identifies the assertion being refuted and proves the replacement stated there. A heading that combines a true part with a refuted overstatement is classified in the coverage table at the level of each manuscript row. The result identifiers are stable labels, not a chronological account of discovery.

The attached *CYCLOTOMIC_KERNEL* return is used only where the prompt expressly permits it or where we identify its precise proved statement and check the comparison being used. We distinguish its proof text from primary literature. The bibliography records opened versions; all numerical references are to those versions. We make no novelty claim. Computations verify finite examples and displayed formulas; none is used to extrapolate a profinite assertion from finitely many tests.

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1 A. The group-theoretic assertions of section 1

A1. The exact cohomological dimensions and the real exception *[Known]*

Let $K/\mathbb{Q}$ be finitely generated of transcendence degree n, let k be the algebraic closure of $\mathbb{Q}$ in K, and put $P = G_{K\overline{k}}$ and $E = G_K$. With continuous cohomology of discrete torsion modules,

$$\operatorname{cd}_\ell P = n \quad \text{for every prime } \ell, \qquad \operatorname{cd} P = n,$$

$$\operatorname{cd}_\ell E = n+2 \quad (\ell \neq 2), \qquad \operatorname{cd}_2 E = \begin{cases} n+2 & K \text{ not formally real,} \\ \infty & K \text{ formally real.} \end{cases}$$

In every case $\operatorname{vcd}_2 E = \operatorname{cd}_2 G_{K(i)} = n+2$. Thus the equalities proposed in F1 hold, with precisely this exception at two (149, section 1, manuscript p. 6, scan 13). For a general function field, the relevant condition is an ordering of K itself, not just a real embedding of its constant number field.

Proof. The opened upper-bound theorem is [Mil13, Theorem 15.2]: a field of transcendence degree d over F has cohomological dimension at most $\operatorname{cd} F + d$. For number fields the precise input is [Mil06, Theorem I.4.10(c)]: in degrees at least three, cohomology with finite coefficients is the product of the cohomology at real places. Thus a totally imaginary number field has dimension at most two. Apply these bounds to $K\overline{k}/\overline{k}$ and to $K(i)/k(i)$ to obtain n and $n+2$, respectively.

We give the lower bound and the comparison to K. On a smooth n-dimensional model over k, choose a closed point and a complete regular flag through it. The successive orders along the flag define a valuation of rank n, value group $\mathbb{Z}^n$ with lexicographic order, and residue field a finite extension k'/k. The same construction over $\overline{k}$ has residue field $\overline{k}$. Its henselization is algebraic over the original function field, hence its absolute group embeds as a closed subgroup of that field's absolute group.

At each rank-one stage, a henselian discrete valuation with residue characteristic zero has inertia $\widehat{\mathbb{Z}}(1)$: finite extensions of the strict henselization are obtained by roots of a uniformizer, since units have all roots by Hensel's lemma. The Hochschild–Serre sequence therefore has only rows zero and one. The Kummer class of a uniformizer restricts to the generator of the inertia row. Cup product with that class splits the residue map

$$H^{j+1}(L, \mu_\ell^{\otimes(r+1)}) \longrightarrow H^j(\kappa, \mu_\ell^{\otimes r}) \qquad (r \in \mathbb{Z}).$$

This proves in particular that a nonzero residue class remains nonzero after successive cups by the n uniformizers. Starting with 1 in $H^0(\overline{k}, \mathbb{Z}/\ell)$ gives degree n in the geometric case. Starting with a nonzero class in $H^2(k', \mu_\ell)$ gives a nonzero class of degree $n+2$ with coefficients $\mu_\ell^{\otimes(n+1)}$ in the arithmetic case. Such a class exists: localization onto $H^2(k'_v, \mu_\ell) \simeq \mathbb{Z}/\ell$ is surjective by [Mil06, Theorem I.2.1 and Corollary I.4.16], taking v a finite place and leaving another place available. Monotonicity of cohomological dimension for closed subgroups supplies both lower bounds. This construction also handles $n = 0$, with no uniformizers.

The exact finite-index theorem we opened is Serre's unnumbered Theorem in section 1 of[Ser65]: for a profinite group without an element of order ℓ, an open subgroup has the same ℓ-cohomological dimension. By Artin–Schreier (A2), G_K has no odd-prime torsion. It has no involution exactly when K is not formally real: an involution fixes a real closure, and any ordering has a real closure whose absolute group has order two. Serre's theorem applied to $G_{K(i)} \subset G_K$ proves the finite values above. When K is formally real, the closed subgroup of order two has nonzero mod-two cohomology in arbitrarily high degrees, so $\operatorname{cd}_2 G_K = \infty$. This also proves the virtual formula. □

1.1 Finite order, free groups, and surface groups

A2. Finite-order elements and complex conjugation *[Known]*

For a field F of characteristic zero, every nonidentity element of finite order in $G_F = \mathrm{Gal}(F^{\mathrm{sep}}/F)$ has order two. Its centralizer is the subgroup of order two that it generates. In $G_{\mathbb{Q}}$ all such elements are conjugate to the complex conjugation determined by $\overline{\mathbb{Q}} \subset \mathbb{C}$. Every open subgroup of $G_{\mathbb{Q}}$ has trivial centralizer in $G_{\mathbb{Q}}$. This answers F2 (149, section 1, manuscript p. 6, scan 14).

Proof. The exact Artin–Schreier input is the finite-absolute-group theorem: a nontrivial finite absolute Galois group has order two; its fixed field is real closed, of characteristic zero, and its algebraic closure is obtained by adjoining $\sqrt{-1}$. We opened this statement in [Gey12, section 3, Example 2, p. 22]. For $z \in G_F$ of finite order, the fixed field of $\langle z \rangle$ has absolute group $\langle z \rangle$, so the theorem applies.

Put $R = (F^{\mathrm{sep}})^{\langle z \rangle}$. An element commuting with z preserves R and restricts to an F-automorphism of R. The order on R is determined by its squares. This automorphism therefore preserves the order. For each polynomial in $F[T]$, its finitely many roots in R form an ordered finite set, and an order-preserving permutation of this set is the identity. Since R/F is algebraic, the automorphism fixes every element of R. Thus it belongs to $\mathrm{Gal}(R(\sqrt{-1})/R) = \langle z \rangle$. The reverse inclusion is immediate.

The ordered field $\mathbb{Q}$ has its unique usual order. Its algebraic real closures are uniquely isomorphic as ordered extensions: a root is mapped to the corresponding ordered root of its minimal polynomial. Extending such an embedding to algebraic closures shows that the subgroups of order two in $G_{\mathbb{Q}}$ are conjugate.

For the last assertion we use the opened [MT22, Theorem 2.7 and Proposition 1.2]: a nontrivial closed normal subgroup of the absolute Galois group of a Hilbertian field has trivial centralizer. We use the author version dated 15 February 2022, in which the theorem number is 2.7. We also checked Statement B2 of the supplied return[GPT26b]; its normal centralizer argument agrees with this exact primary theorem. The normal core N of an open $H \subset G_{\mathbb{Q}}$ is open and nontrivial. The field $\mathbb{Q}$ is Hilbertian [Gey12, section 4, Theorem 1]. Hence $C_{G_{\mathbb{Q}}}(H) \subset C_{G_{\mathbb{Q}}}(N) = 1$. In particular $Z(H) = 1$, including the open normal subgroups mentioned on scan 14. This establishes every input in the manuscript's finite-centralizer argument, without assuming its conclusion. □

A3. Open subgroups, torsion, and centers of profinite surface groups *[Proved]*

Let P be the full profinite completion of either a free group of finite rank $r \geq 2$, or the fundamental group of a closed oriented surface of genus $g \geq 2$. Every open subgroup has trivial centralizer in P, and P has no nonidentity element of finite order. An open subgroup of index d is the full profinite completion of a free group of rank $1 + d(r-1)$, respectively a surface group of genus $1 + d(g-1)$. The free rank-one exception is $\widehat{\mathbb{Z}}$, whose center and the centralizers of all its open subgroups equal the whole group. These statements answer F3 (149, section 1, manuscript pp. 7 and 13, scans 15 and 28).

Proof. Write $P = \widehat{\Lambda}$. For an open subgroup H, its inverse image $\Lambda_H \subset \Lambda$ has index d. Its full profinite topology is the induced one: for a finite-index subgroup $V \subset \Lambda_H$, the intersection of its finitely many conjugates in Λ is a finite-index subgroup of Λ contained in V. Consequently $\widehat{\Lambda_H} \to H$ is an isomorphism. The finite covering of the rose with r loops, respectively of the oriented surface, has Euler characteristic multiplied by d. A spanning tree in the graph gives the stated free basis; the classification of compact oriented surfaces gives the stated genus. Its abelianization is therefore a finite product of $\widehat{\mathbb{Z}}$ and is torsion-free.

This proves torsion-freeness of P itself by a finite-quotient argument. For a nontrivial finite-order $z \in P$, choose open normal N with $z \notin N$ and put $V = N\langle z \rangle$. The image of z in V^{ab} must

be zero, because this abelianization is torsion-free. But $V \to V/N$ is a nontrivial cyclic quotient generated by zN, so it factors through V^{ab} and gives a contradiction.

We also give the center argument explicitly. Let $N \triangleleft P$ be open and $D = P/N$. The corresponding regular covering has cellular chain modules $\mathbb{Q}[D]^r \to \mathbb{Q}[D]$ in the graph case, and $\mathbb{Q}[D] \to \mathbb{Q}[D]^{2g} \to \mathbb{Q}[D]$ in the closed surface case. The H_0 representation is trivial in both cases; H_2 is also trivial in the oriented surface case. Additivity of characters in these finite complexes gives

$$H_1 \simeq \mathbb{Q} \oplus \mathbb{Q}[D]^{r-1}, \qquad H_1 \simeq \mathbb{Q}^2 \oplus \mathbb{Q}[D]^{2g-2},$$

respectively, as rational D-representations. These identities follow from the character equalities and semisimplicity of finite-group representations in characteristic zero. The representations are faithful: a nonidentity $d \in D$ has trace 1, respectively 2, whereas the identity has strictly larger trace. Comparison with the maximal pro-ℓ quotient $N^{\mathrm{ab},(\ell)} \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ gives the same action for any prime ℓ. A central $z \in P$ acts trivially on N^{ab}, hence has trivial image in D. This holds for every N, so $z = 1$.

Apply the same argument to any open normal $N \subset H$. The kernel of $C_P(N) \to P/N$ is $Z(N) = 1$, so the centralizer is finite. Torsion-freeness makes it trivial, and $C_P(H) \subset C_P(N)$ finishes the proof. The literature statements with exactly these full profinite hypotheses are Nakamura's Corollary (1.3.4)[Nak94a], and Mochizuki–Tamagawa's Proposition 1.4 and Remarks 1.1.5, 1.2.2 [MT08]; we opened their statements. The above argument supplies the center, open-subgroup, and torsion reductions used here. For completeness, the nonhyperbolic geometric curve types in characteristic zero have groups 1 for $(0,0)$ and $(0,1)$, $\widehat{\mathbb{Z}}(1)$ for $(0,2)$, and $\widehat{\mathbb{Z}}^2$ for $(1,0)$. The latter two have their entire group as the centralizer of every open subgroup. The local meridian is trivial for $(0,1)$; in $(0,2)$ the two meridians generate the same group, with opposite orientations. These statements follow directly from the sphere, punctured-sphere, and torus presentations. □

A4. Inertia at distinct lifted cusps *[Proved]*

Let C be a smooth connected curve over an algebraically closed field of characteristic zero, of hyperbolic type (g, r) with $r > 0$, and let $P = \pi_1(C)$. In the normalization of its smooth compactification in the universal pro-étale cover, each cusp c has inertia $I_c \simeq \widehat{\mathbb{Z}}(1)$. Distinct lifted cusps have $I_c \cap I_d = 1$. Moreover

$$N_P(I_c) = I_c, \qquad C_P(H) = I_c \quad (1 \neq H \subset I_c).$$

Here subgroups are actual chosen subgroups; changing the universal lift conjugates all statements simultaneously. This supplies the loop inputs requested with F3.

Proof. For a complex curve the local punctured disk gives $I_c = \widehat{\mathbb{Z}}(1)$; finite covers and algebraically closed base change transport the statement to the stated fields. To see injectivity without discarding the one-cusp case, pass to a finite unramified cover of the compact curve when $g > 0$: the quotient of its surface group by a nonzero map to $\mathbb{Z}/3$ gives at least three preimages of every cusp. For $g = 0$ hyperbolicity already gives at least three cusps. A cusp in a curve with at least two punctures maps injectively into its abelianized geometric group: the puncture part is

$$\widehat{\mathbb{Z}}(1)^S / \widehat{\mathbb{Z}}(1)(1, \ldots, 1).$$

This formula follows by abelianizing the usual oriented surface presentation; the only relation among meridians is their sum.

For distinct pro-cusps c, d, a finite cover separates their images. Refine it, as just described, until the compactification has at least three punctures. If $1 \neq h \in I_c \cap I_d$, a positive integer power of h lies in the open subgroup of this covering and is still nontrivial, by torsion-freeness

of $\widehat{\mathbb{Z}}$. Its two images in the displayed abelianization belong to different coordinate lines. Those lines have zero intersection for $|S| \geq 3$: comparison of a third coordinate in a multiple of the diagonal forces that multiple, and then both coordinates, to be zero. Injectivity of the inertia line gives a contradiction.

An element normalizing I_c, or centralizing its nontrivial subgroup, therefore fixes the lifted cusp. Its stabilizer is I_c, since the residue field at a geometric cusp is algebraically closed. Conversely I_c is abelian. This proves both equalities. For comparison we opened [Moc04, Lemma 1.3.7], which states commensurable terminality for both the full profinite group and each pro-ℓ group; the argument here also establishes the stronger intersection assertion used in this paper. □

A5. Extensions and inverse limits *[Proved]*

The property $C_P(H) = 1$ for every open subgroup $H \subset P$ passes to extensions of profinite groups and to directed inverse limits with surjective transition maps. Consequently every successive extension of nonabelian free profinite groups and profinite hyperbolic surface groups has this property. The limit statement fills the omission between scans 12 and 15 (149, section 1, manuscript pp. 5–7).

Proof. For $1 \to A \to B \xrightarrow{q} D \to 1$, let $H \subset B$ be open. An element centralizing H has image centralizing the open subgroup $q(H)$ of D, so belongs to A. It then centralizes $H \cap A$, open in A, so is the identity.

For $P = \varprojlim P_i$, the projections q_i are continuous surjections of compact groups and are open. Thus $q_i(H)$ is open for every open H. Every coordinate of an element of $C_P(H)$ belongs to $C_{P_i}(q_i(H)) = 1$, so the element is the identity. For the manuscript's models the surjectivity is concrete: a connected finite étale cover of a normal integral scheme stays connected on a dense open, since the function field of the cover is unchanged. Hence $\pi_1(V) \to \pi_1(U)$ is surjective for dense opens $V \subset U$. Every finite separable extension of the function field is finite étale over a sufficiently small open of its normalization. Finite-cover correspondence therefore gives

$$G_{k(U)} \simeq \varprojlim_{V \subset U} \pi_1(V).$$

The identical construction after algebraic closure of the constant field gives the geometric comparison. These are isomorphisms of profinite groups, not merely bijections of abstract groups. □

A6. Theorem 1 and its strengthening *[Known]*

Let $K/\mathbb{Q}$ be finitely generated, let k be the algebraic closure of $\mathbb{Q}$ in K, and fix an algebraic closure $\overline{K}$ with $\overline{k} \subset \overline{K}$. Put $E = G_K$ and $P = G_{K\overline{k}}$. Every open subgroup of E, respectively P, has trivial centralizer in E, respectively P. For $\operatorname{trdeg}(K/k) > 0$ one has, more strongly,

$$C_E(H) = 1 \quad \text{for every open } H \subset P, \qquad G_k \longrightarrow \operatorname{Out}(P) \text{ is injective.}$$

This is Theorem 1 and the Remarque (149, section 1, manuscript pp. 6 and 10, scans 13 and 21; F4).

Proof. The Hilbertian normal-centralizer theorem used in A2, independently verified in[MT22, Theorem 2.7 and Proposition 1.2] and checked against the attached return[GPT26b], applies to K: it is an infinite finitely generated field and hence Hilbertian. It also applies to $K\overline{k}$ when the transcendence degree is positive, because this field is finite over a rational function field of positive transcendence degree over $\overline{k}$. The required Hilbertian facts are [Gey12, section 4,

Theorem 2(a),(d),(e)]. Applying that theorem to the normal core of each open subgroup proves the two first assertions. In degree zero $P = 1$ and its assertion is tautological.

Here is the reduction to a curve, including its comparison maps. Choose $t \in K$ transcendental over k, and let F be the relative algebraic closure of $k(t)$ in K. The extension $F/k(t)$ is finite, k is algebraically closed in F, and K/F is regular. Let C/k be the smooth projective geometrically connected curve with function field F, and choose a nonempty affine hyperbolic open $U \subset C$. Restriction on separable closures and then restriction to covers unramified on U give continuous surjections

$$P \twoheadrightarrow G_{F\overline{k}} \twoheadrightarrow \pi_1(U_{\overline{k}}).$$

The first is surjective because regularity makes K linearly disjoint from every finite separable extension of F; the second follows from finite-cover correspondence. Both maps are induced over k, so commute with the outer G_k-actions. An inner action on P descends to an inner action on this quotient. The opened Hoshi–Mochizuki [HM11a, Theorem C(ii)] says that, for a hyperbolic curve over a perfect characteristic-zero field, the kernel of the full outer representation is contained in the kernel of the outer map to $G_{\mathbb{Q}}$; in particular it is zero over a number field. Here k is a number field. This proves injectivity of $G_k \to \mathrm{Out}(P)$. Since $Z(P) = 1$, the kernel of this action identifies with $C_E(P)$: an inner action of a lift e can be corrected by its unique inducing element of P to centralize P.

Finally let $H \subset P$ be open. There is an open normal subgroup $E_0 \subset E$ with $N = E_0 \cap P \subset H$: this follows from the subspace topology on the closed subgroup P. The group N is a nontrivial closed normal subgroup of $E_0 = G_{K_0}$, where K_0/K is finite. It is nontrivial because P is infinite and N has finite index in P. The same theorem gives $C_{E_0}(N) = 1$, hence $C_E(H) \cap E_0 = 1$. Therefore $C_E(H)$ embeds in the finite group E/E_0. By A2 a nonidentity element of this centralizer would be an involution whose centralizer is its own order-two group. It cannot centralize the infinite group H. Thus $C_E(H) = 1$. This last argument supplies the step needed after shrinking the arithmetic group in the manuscript; membership in the shrunken group was not silently assumed. □

2 B. Rigidity, the translation error, and Kummer theory

B1. The rational-point group of a semiabelian variety *[Proved]*

Let K be finitely generated over its prime field and let $1 \to T \to G \to A \to 1$ be a semiabelian variety over K. As an abstract abelian group $G(K)$ is a direct sum of a finite group and a free abelian group, whose rank can be infinite. Consequently

$$\bigcap_{n \geq 1} nG(K) = 0.$$

In characteristic $p > 0$ the exact prime-to-p variant is

$$\bigcap_{(n,p)=1} nG(K) = G(K)[p^\infty].$$

The finite-generation assertion on scan 34 is false: already $\mathbb{G}_m(\mathbb{Q}) = \mathbb{Q}^*$ has one independent generator for each rational prime. This answers the erroneous input of F6 (149, section 1, manuscript p. 16, scan 34).

Proof. Let L/K be a finite splitting field of T. We first prove that L^* is the direct sum of its finite roots of unity and a free abelian group. Write k_0 for the algebraic closure of the prime field in L. A normal projective geometrically integral model Z/k_0 has

$$1 \longrightarrow k_0^* \longrightarrow L^* \xrightarrow{\mathrm{div}} \mathrm{Div}(Z).$$

The kernel consists of global invertible regular functions on the proper integral model, hence of constants. The divisor group is free abelian; its subgroup im(div) is free, so this sequence splits as a sequence of abstract abelian groups onto its image. For finite k_0, its multiplicative group is finite. For a number field k_0, the analogous divisor sequence over its ring of integers, together with the opened unit theorem[Mil20a, Theorem 5.1], expresses k_0^* as finite torsion plus a free abelian group. Thus the same holds for L^*.

The injection $T(K) \hookrightarrow T(L) \simeq (L^*)^{\dim T}$ now proves the same assertion for $T(K)$: its torsion is finite and its quotient by torsion is a subgroup of a free abelian group, hence free. The exact Mordell–Weil–Lang–Néron statement needed for the quotient is [Mil08, Theorem 16.7]: an abelian variety over a field finitely generated over the prime field has a finitely generated rational-point group. Hence the image B of $G(K) \to A(K)$ is finitely generated.

Choose finitely many lifts in $G(K)$ of generators of B, and let H be the subgroup they generate. The finitely generated subgroup $H \cap T(K)$ is contained in a finitely generated direct summand T_0 of $T(K)$ which also contains its torsion: in a free basis only finitely many coordinates occur. Write $T(K) = T_0 \oplus T_1$. Then

$$G(K) = (H + T_0) \oplus T_1.$$

Indeed every element is a sum from these groups, and an element of their intersection belongs to $H \cap T(K) + T_0 \subset T_0$ and to T_1, so vanishes. The first summand is finitely generated, and the second is free. The assertions about intersections of multiples follow coordinate by coordinate. For a finite abelian group, multiplication by integers prime to p is invertible on its p-primary part and eventually kills every other primary part. The rational-prime valuations on $\mathbb{Q}^*$ verify the advertised counterexample to finite generation. □

B2. Kummer comparison and Theorem 3 in characteristic zero *[Proved]*

Let $K/\mathbb{Q}$ be finitely generated, let X/K be geometrically connected of finite type, and let $i : X \hookrightarrow G$ be an immersion into a semiabelian variety. Both maps (*) and (**) on scans 32–33 are injective: K-points have distinct geometric conjugacy classes of sections, already after abelianizing the geometric kernel. This is Theorem 3 in its stated embedding generality (149, section 1, manuscript pp. 15–17, scans 32–35; F6).

Proof. For $n \geq 1$, multiplication by n on G is a finite étale torsor under $G[n]$. Pullback along $x \in G(K)$ gives its Kummer class

$$\delta_n(x) \in H^1(G_K, G[n]), \qquad \ker \delta_n = nG(K).$$

The geometric monodromy of this torsor is abelian. Therefore its pullback by a point section depends only on the section in $E_G/\overline{[\pi_1(G_{\overline{K}}), \pi_1(G_{\overline{K}})]}$. For two point sections this finite-level comparison is exactly $\delta_n(x - y)$: with division points $n\widetilde{x} = x$ and $n\widetilde{y} = y$, the two cocycles differ by $\sigma(\widetilde{x} - \widetilde{y}) - (\widetilde{x} - \widetilde{y})$. Consequently equality of the abelianized section classes puts $x - y$ in every $nG(K)$, and B1 gives $x = y$.

For completeness the converse comparison, over any characteristic-zero field, also holds. The natural map $\pi_1(G_{\overline{K}}) \to T(G) = \varprojlim_n G[n]$ is an isomorphism: after descent to a finitely generated subfield and embedding in $\mathbb{C}$, a complex semiabelian group has fundamental group $\mathbb{Z}^{2\dim A + \dim T}$, and its multiplication coverings form a cofinal system of finite connected coverings. Algebraically closed base change and the finite-cover comparison transport the statement back. If $x - y \in \bigcap_n nG(K)$, the finite nonempty sets of K-rational division points, indexed by divisibility, have a compatible point. To check this without a surjectivity assumption, every finite family of compatibility conditions is solved at one common multiple; compactness of the product of the finite sets then solves all conditions. Thus the full Tate-module Kummer cocycle is a coboundary. This rechecks the equivalence stated as D7 in the attached return.

Applying the morphism of arithmetic extensions induced by i sends equal section classes for X to equal classes for G. Equality of $i(x), i(y)$ implies $x = y$ because an immersion is a monomorphism. Passage from full sections to their geometric abelianizations is the displayed quotient homomorphism, so injectivity of (**) proves that of (*).

Every prime is needed for a theorem uniform in K, G. On omitting a prime q, take $K = \mathbb{Q}(\zeta_q)$ and $G = \mathbb{G}_m$. The points 1 and ζ_q differ by q-torsion, which is divisible by every integer prime to q. Their prime-to-q Kummer classes coincide. More exactly, B1 shows that the kernel on retaining a set Σ of primes is the torsion of $G(K)$ supported outside Σ. Thus the manuscript's Remarque is a uniform necessity statement; it does not say that every individual torsion-free example requires every prime. □

B3. Corollary 1(a), with the translation in the correct group *[Proved]*

Let k be algebraically closed, let X be a reduced connected finite-type k-scheme, and let $f, g : X \to Y$ be morphisms, where $i : Y \hookrightarrow G$ is an immersion into a semiabelian variety. For any prime $\ell \neq \operatorname{char} k$, equality $H_1(f, \mathbb{Z}_\ell) = H_1(g, \mathbb{Z}_\ell)$ implies a unique $a \in G(k)$ with $i \circ g = \tau_a \circ i \circ f$. This proves Corollary 1(a), including its one-prime refinement, on scans 24–25 (149, section 1, manuscript pp. 11–12; F5). The printed/transcribed $a \in Y$ must be corrected to $a \in G$: with $Y = \mathbb{G}_m \setminus \{1\}$ and $f = g$, the difference is $1 \notin Y$.

Proof. Put $h = i \circ g - i \circ f$. The addition law on a commutative algebraic group induces addition on its abelianized prime-to-characteristic fundamental groups: this follows by pulling back the $[\ell^n]$ Kummer torsors, whose cocycles add. Hence $H_1(h, \mathbb{Z}_\ell) = 0$. We show that such a map $h : X \to G$ is constant.

It suffices first to consider a smooth connected curve U mapping to X. The composite $U \to G \to A$ extends to the smooth proper compactification C by properness of A. After subtracting its value at a chosen point, it factors through a homomorphism $J_C \to A$: the exact opened universal property is [Mil08, III, Proposition 6.1]. The map $H_1(U, \mathbb{Z}_\ell) \to H_1(C, \mathbb{Z}_\ell)$ is surjective, since a connected finite étale cover of C stays connected on a dense open. The Abel map identifies the latter module with $T_\ell J_C$[Mil08, III, Proposition 9.1 and Corollary 9.6]. Thus the homomorphism $J_C \to A$ is zero on $T_\ell J_C$ and is zero. For this last implication, a positive-dimensional image would have a kernel of smaller dimension; a finite union of translates of that kernel contains at most a constant times ℓ^{2nd} points of order dividing ℓ^n, with $d < \dim J_C$, and cannot contain all $\ell^{2n \dim J_C}$ such points for every n.

The map $U \to G$ therefore lies in one torus fibre. Translate it into $T \simeq \mathbb{G}_m^r$. The inclusion of T induces an injection on ℓ-adic homology: the exact sequences on torsion give $0 \to T_\ell T \to T_\ell G \to T_\ell A \to 0$ (surjectivity uses divisibility of $T(k)$). Each coordinate is a unit on U, and its map on a meridian at a boundary point is multiplication by its integer order there, as is seen from the torsor $z^{\ell^n} = h$. Zero homology therefore forces every boundary order to vanish. The coordinate extends to an invertible function on proper C and is constant. This proves the curve assertion.

A nonconstant morphism on an irreducible finite-type component is nonconstant on some integral curve: take an affine open, choose a nonconstant coordinate from the image, and complete it to a transcendence basis; after localization, the finite map to affine space has a curve lying over a line on which that coordinate varies. Normalize a nonempty open of this curve to apply the preceding argument. Thus h is constant on every irreducible component of X. Their intersection graph is connected, and constants on intersecting components coincide. Finally, equality on all closed points of a reduced finite-type scheme implies equality of the morphisms, since the ideal of their closed equalizer has zero radical and is zero. This proves constancy on X. Any $x \in X(k)$ then gives $a = h(x)$ and proves uniqueness. □

B4. The open-subset translation lemma and its exact curve correction *[Refuted]*

The lemma of scan 37 is false for the hyperbolic type $(1,1)$. For a smooth hyperbolic curve $Y = C \setminus D$ over an algebraically closed field, with its generalized Abel map $\alpha : Y \hookrightarrow J_D^1$, the following is the exact correction on curve types: a translation sending a nonempty open of $\alpha(Y)$ into $\alpha(Y)$ is trivial for every hyperbolic type except $(1,1)$; for type $(1,1)$ every translation has that property. Full setwise stability $\alpha(Y) + u = \alpha(Y)$ forces $u = 0$ in every hyperbolic type. Here the field has characteristic zero; the same proof uses prime-to-characteristic cohomology in arbitrary characteristic. This answers the lemma and the homological proof in F5 (149, section 1, manuscript pp. 18–19, scans 37–40).

Proof. Let $E : y^2 = x^3 - x$, with origin O at infinity, and put $Y = E \setminus \{O\}$. Its generalized Jacobian for the reduced one-point modulus is E itself. Choose $u = (0,0)$, a nonzero point of order two, and $U = E \setminus \{O, u\}$. Translation by u interchanges O, u and therefore preserves U, so $U + u \subset Y$ although $u \neq 0$. On U it is explicitly

$$(x, y) \longmapsto (-x^{-1}, yx^{-2}).$$

Substitution verifies the equation of E and applying this map twice gives the identity. The script `code/AB_checks.py` checks these identities symbolically and checks the point permutation over $\mathbb{F}_5$. In general, for $Y = E \setminus \{c\}$ and any $u \in E$, the open $E \setminus \{c, c-u\}$ proves the asserted property of every translation.

We prove the positive statement and identify the failed step. Translation on the generalized Jacobian induces the identity on $H_1(-, \mathbb{Z}_\ell)$ for ℓ invertible in the field. The generalized Abel map identifies $H_1(Y, \mathbb{Z}_\ell)$ with $T_\ell(J_D)$; this follows from the degree-zero divisor construction, or from the compact Jacobian Tate module and the meridian exact sequence

$$0 \longrightarrow \mathbb{Z}_\ell(1)^D / \mathbb{Z}_\ell(1) \longrightarrow H_1(Y, \mathbb{Z}_\ell) \longrightarrow T_\ell(J_C) \longrightarrow 0.$$

The translation on an open part therefore extends to an automorphism $f : C \xrightarrow{\sim} C$ whose action on $T_\ell(J_C)$ is the identity. The extension is birational because the translated map has inverse given by the opposite translation on its image; a birational map between smooth proper curves extends uniquely in both directions.

For $g(C) \geq 2$, the group of automorphisms of C is finite: its faithful pluricanonical projective representation is an algebraic group with tangent space $H^0(C, T_C) = 0$, since $\deg T_C = 2 - 2g < 0$. Trivial action on a Tate module implies triviality of the induced Jacobian homomorphism by the torsion-count argument in B3. Its action on the Jacobian is faithful: if an automorphism acts trivially there, its Abel image is translated by a torsion point a. The cyclic group generated by a nontrivial such translation acts freely on C. Its quotient is étale of degree $m > 1$, while invariance of H_1 makes the quotient have the same genus, by the averaging projector on rational ℓ-adic cohomology. The unramified formula $2g(C) - 2 = m(2g(C/\langle f \rangle) - 2)$ contradicts $g(C) > 1$. Thus $f = 1$ and the immersion gives $u = 0$.

For the remaining hyperbolic types assume $r = |D| \geq 2$ (and $r \geq 3$ in genus zero). Shrink U so $J = C \setminus U$ contains both D and $f^{-1}D$. For the two maps $U \to Y$, the kernels on the meridian module $M_J = \mathbb{Z}_\ell(1)^J / \mathbb{Z}_\ell(1)$ are generated by the coordinates in $J \setminus D$ and $J \setminus f^{-1}D$. Equality of the homology maps makes these submodules equal. For subsets of cardinal at most $|J| - 2$, their coordinate spans in M_J are equal only when the subsets themselves are equal: choose a coordinate outside their union when available, or compare the coordinates in the two complements directly; a nonzero multiple of the diagonal has all coordinates nonzero and cannot express one missing coordinate using fewer than $|J| - 1$ others. Hence $f(D) = D$. It follows that f acts trivially on $H_1(Y)$ and therefore on M_D. The permutation representation on M_D is faithful for $|D| \geq 2$; for two points the transposition is multiplication by -1, which is different from 1 on $\mathbb{Z}_\ell$. Thus f fixes each point of D. In genus zero three fixed points force $f = 1$. In

genus one its trivial Jacobian action makes it a translation, and a translation with a fixed point is trivial. Again $u = 0$ follows.

When $|D| = 1$, both killed subsets have $|J| - 1$ elements, and each of their spans is all of M_J. This is precisely the exceptional case recognized algebraically on scan 39 but not eliminated on scan 40: shrinking U does not change the number $|D| = 1$. For the explicit elliptic example the two maps even induce the same map on H_1, whereas the full fundamental-group maps differ. A meridian about u is killed by the inclusion $U \to Y$ and maps to the nontrivial meridian about O under translation. A conjugation cannot turn the identity into a nonidentity. This locates the exact nonabelian information lost in the manuscript's homological argument.

For full stability in type $(1, 1)$, the translation must preserve the unique missing point, so it is trivial. This completes the classification. □

B5. Corollaries 1(b), 2(c)–(d), and 3 with all comparison maps *[Proved]*

A hyperbolic polycurve means a finite tower of smooth pairs whose geometric curve fibres have $2g - 2 + r > 0$, including proper fibres. Over a field finitely generated over $\mathbb{Q}$, rational points of a hyperbolic polycurve have distinct geometric conjugacy classes of full point sections, and a point section has trivial centralizer in the geometric group. For a reduced geometrically connected finite-type source, the arithmetic outer fundamental-group map therefore determines a morphism into that polycurve. The same morphism rigidity holds for a target immersed in a semiabelian variety, already after geometrically abelianizing both extensions.

Over an algebraically closed field of characteristic zero, two maps $f, g : X \to Y$ with equal geometric outer maps are equal in each of these cases: Y is a hyperbolic polycurve and f has open image on fundamental groups; or Y is immersed in a semiabelian variety and $C_{\pi_1(Y)}(f_*\pi_1(X)) = 1$. In particular a dominant map from an irreducible reduced source into a hyperbolic polycurve is rigid in this sense. These are Corollaries 1(b), 2(c)–(d), and 3 (149, section 1, manuscript pp. 11–15, scans 24–31; F5).

Proof. We first prove the centralizer statement for a curve V/K and any section $s : G_K \to E_V$. Write $\Delta = \pi_1(V_{\overline{K}})$. For $1 \neq z \in C_\Delta(s(G_K))$, choose a characteristic open normal subgroup $N \subset \Delta$ excluding z; such subgroups are cofinal because Δ is finitely generated. Put $H = N\overline{\langle z \rangle}$. Both N and H are stable under $s(G_K)$, so $Hs(G_K)$ defines a geometrically connected finite étale curve cover V_H/K. The image of z in H^{ab} is nonzero, since its image in the finite cyclic quotient H/N is nonzero. But it is fixed by G_K. The abelian group H^{ab} is the full Tate module of the generalized Jacobian of V_H, with its natural G_K-action. Its invariants are zero: a compatible system of rational torsion points would be infinitely divisible in the finite torsion group supplied by B1. This contradiction proves the assertion. The identification with the Tate module is the compact Jacobian comparison together with the meridian module; it is compatible with the arithmetic action because inner conjugation acts trivially on abelianization.

Point injectivity for a curve follows from B2 and its generalized Abel immersion. Prove point injectivity and the point-section centralizer assertion for a tower by induction. For its last map $Y \to B$, C4 supplies the exact sequence and, for $b \in B(K)$, the actual comparison isomorphism

$$E_{Y_b} \xrightarrow{\sim} E_Y \times_{E_B} s_b(G_K).$$

It is the identity on the geometric fibre kernel and on G_K. Equal point sections for $y, z \in Y(K)$ first imply equality of their images $b \in B(K)$. The projection of a geometric conjugator then centralizes $s_b(G_K)$, hence is one by induction. It lies in the geometric fibre kernel; the displayed map reduces the equality to point sections of the curve Y_b, where B2 applies. For a centralizer element of s_y, its projection is likewise one, and the curve centralizer assertion kills the remaining fibre element. This gives both induction steps. No immersion of the entire tower was used.

For arithmetic morphism rigidity, restrict two equal extension maps along a point section s_x after each finite field extension K'/K. The Cartesian field-base-change comparison $E_{X_{K'}} = E_X \times_{G_K} G_{K'}$ makes their values the sections of $f(x), g(x)$. Point injectivity gives equality at all such points. These include all geometric closed points. Reducedness and separatedness give equality of the maps, as in B3. For a semiabelian immersed target the same proof uses B2 and works on abelianized geometric kernels. If "conjugation by $\pi_1(Y)$" on scan 30 means the entire arithmetic group, its arithmetic image centralizes G_K, because both maps lie over G_K and the source surjects onto G_K. A6 gives $Z(G_K) = 1$, so the conjugator is geometric after all.

Now descend the geometric data to a field K finitely generated over $\mathbb{Q}$ inside the given algebraically closed field. Algebraically closed base change preserves these full geometric fundamental groups. Use the given geometric conjugator to make the restrictions of the two arithmetic maps $F, G : E_X \rightrightarrows E_Y$ equal on Δ_X. For $e \in E_X$, the element $G(e)F(e)^{-1}$ lies in Δ_Y and centralizes $F(\Delta_X)$: conjugate an element $F(\delta) = G(\delta)$ by both sides and use equality on $e\delta e^{-1} \in \Delta_X$. Thus triviality of this centralizer makes $F = G$. Arithmetic rigidity just proved gives $f = g$. For a polycurve, A3, A5, and C4 give trivial centralizers of open geometric subgroups, so this proves case (c) as well as (d).

Finally, a dominant morphism $f : X \to Y$ with X integral and Y normal has open fundamental-group image. The restriction map $G_{k(X)} \to G_{k(Y)}$ has image the open subgroup belonging to the finite relative algebraic closure of $k(Y)$ in $k(X)$. The map $G_{k(Y)} \to \pi_1(Y)$ is surjective, because a connected finite étale cover of a normal integral scheme has connected generic fibre. The resulting open subgroup of $\pi_1(Y)$ lies in $f_*\pi_1(X)$ by functoriality of the generic point. A subgroup containing an open subgroup is open. This proves Corollary 1(b). The argument proves these geometric statements for reduced connected sources in the stated cases; the normality assumption on X in Corollary 2 is unnecessary for this proof. The tentative margin on scan 24 that removes every dominance or image condition is false, as the author himself observes on scan 26. For $Y = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ and any nonempty reduced connected source, the constant maps with values 2 and 3 are distinct and both induce the trivial geometric fundamental-group homomorphism. Their image centralizer is all of $\pi_1(Y)$, so B5 states the precise additional hypotheses used by the surviving corollaries. This records the withdrawn assertion as sweep row ZAB7 (manuscript pp. 11–12). □

2.1 Characteristic p: full and geometrically prime-to-p groups

B6. The full point-section theorem in positive characteristic *[Proved]*

Let $K/\mathbb{F}_p$ be finitely generated. If a geometrically connected finite-type K-scheme X admits an immersion into a semiabelian variety, then its K-points give distinct geometric conjugacy classes of sections, already after abelianizing the full geometric étale fundamental group. Thus both (*) and (**) of Theorem 3 remain injective with the full groups. This proves the full-group assertion and the finite-field specialization margin on scan 35 (149, section 1, manuscript p. 17; F7).

Proof. First let A/K be an abelian variety and let $a \in A(K)$ have the same abelianized section class as 0. Spread A, a over a smooth integral finite-type $\mathbb{F}_p$-scheme S with function field K, shrinking so that A_S is an abelian scheme. If $a \neq 0$, the complement of the equalizer of $a, 0$ is a nonempty open and contains a closed point s. Its residue field $k_s = \mathbb{F}_q$ is finite and $a_s \neq 0$. Set $R = \mathcal{O}^{\mathrm{h}}_{S,s}$ and $L = \operatorname{Frac} R$. The inclusion $K \hookrightarrow L$ induces restriction of section classes.

The map $F_q - 1 : A_s \to A_s$ is a finite étale surjection with constant kernel $H = A_s(\mathbb{F}_q)$. Its derivative is -1, hence it is étale; properness makes it finite, and its image is nonempty, open and closed in the connected variety A_s. Its geometric kernel consists of the q-Frobenius-fixed

points. Its fibre above 0 is a trivial H-torsor, whereas its fibre above $a_s \neq 0$ has no $\mathbb{F}_q$-point, because $F_q(z) - z = 0$ for every such point z.

The precise opened proper henselian comparison is this: for a proper scheme over a henselian local ring, restriction to the closed fibre is an equivalence on finite étale covers [The26d, Lemma 58.9.1, Tag 0A48]. It lifts the Lang cover to a finite étale H_R-torsor $V \to A_R$; full faithfulness lifts the action and torsor identity. The pullback to the zero section is trivial, by the henselian equivalence on finite étale R-schemes[The26d, Lemma 58.8.2, Tag 09ZS]. This torsor has abelian geometric monodromy. Equality of the geometrically abelianized point-section classes therefore makes its pullbacks by $0, a \in A(L)$ isomorphic. Hence a^*V has an L-point. A finite étale scheme over the normal domain R with an L-point has an R-point: its coordinate values are integral over R and in L, and thus belong to R. Reduction gives an $\mathbb{F}_q$-point of the Lang fibre above a_s, a contradiction. Thus $a = 0$.

For $1 \to T \to G \to A \to 1$, equal abelianized point sections for $x, y \in G(K)$ give $d = x - y \in \bigcap_{(n,p)=1} nG(K) = G(K)[p^\infty]$, by the étale Kummer torsors and B1. Projection to A and the preceding argument put d in $T(K)$. A torus has no nonidentity geometric p-power torsion points, since $z^{p^a} - 1 = (z-1)^{p^a}$ after splitting it. Hence $d = 0$. Push the point sections along the immersion $X \hookrightarrow G$ to obtain the assertion for X, as in B2. These comparisons use finite étale covers throughout; a non-étale multiplication-by-p torsor has not been treated as an étale cover. □

B7. What geometrically prime-to-p abelianization loses *[Refuted]*

For $K/\mathbb{F}_p$ finitely generated and G/K semiabelian, the kernel of the prime-to-p Kummer map on $G(K)$ is exactly $G(K)[p^\infty]$. Consequently prime-to-p (**) is false even for hyperbolic once-punctured elliptic curves. Without hyperbolicity, prime-to-p (*) is false too. We retain the full arithmetic quotient G_K and replace only the geometric kernel by its maximal prime-to-p quotient. This decides the explicit assertion of scan 35 (149, section 1, manuscript p. 17; F7).

Proof. For $(n,p) = 1$, $[n] : G \to G$ is a finite étale torsor under $G[n]$. The comparison in B2 gives kernel $nG(K)$ at this level. The prime-to-p abelian geometric group of a semiabelian variety is $\prod_{\ell \neq p} T_\ell G$: the multiplication covers yield these quotients, as follows from the torus and abelian-variety Kummer sequences. The compatible-division-point compactness argument of B2 therefore identifies the whole kernel with the intersection of these finite-level kernels, namely $G(K)[p^\infty]$ by B1.

Take $E/\mathbb{F}_5 : y^2 = x^3 + 3x + 2$. Its discriminant is nonzero and

$$E(\mathbb{F}_5) = \{O, (1,1), (1,4), (2,1), (2,4)\}.$$

Thus this group has order five. The script `code/AB_checks.py` verifies the list and the orders by the group law. Put $P = (1,1), Q = (1,4)$. For $Y = E \setminus \{O\}$ the reduced one-point modulus has generalized Jacobian E, and the Abel map induces

$$\pi_1(Y_{\overline{\mathbb{F}}_5})^{\mathrm{ab},(5')} \simeq \prod_{\ell \neq 5} T_\ell E.$$

Indeed the meridian module for a single puncture is zero. Hence P, Q have equal prime-to-five abelianized section classes on the hyperbolic curve Y. For $X = E$ itself the geometric fundamental group is abelian, so the same points disprove prime-to-five (*) in the embedding generality of Theorem 3.

For a torus the displayed kernel is zero. Hence any variety immersed in a torus has injective prime-to-p (**) and (*). This includes smooth genus-zero curves other than the projective and affine lines: with a chosen rational point their generalized Jacobian is a torus and the Abel map

is an immersion. Passing to a finite extension to split the punctures is legitimate: section equality restricts along $G_{K'} \hookrightarrow G_K$, and inequality of points persists. □

B8. Nonabelian recovery for punctured genus-one curves *[Proved]*

For $K/\mathbb{F}_p$ finitely generated and $Y = C \setminus D$ a smooth geometrically connected genus-one curve with nonempty reduced finite boundary, prime-to-p (*) is injective. Thus every hyperbolic genus-one type is covered, including B7 where (**) fails. This proves part of the covering argument on scan 36 (149, section 1, manuscript p. 17; F7).

Proof. Suppose $P, Q \in Y(K)$ give equal section classes. Use P as origin of $C = E$. Abelianization and B1 imply that $Q - P \in E(K)$ has p-power order, say p^a. Suppose it is nonzero. Choose $(m, p) = 1$ with $m^2 > p^a$, and form

$$[m] : Y_m = E \setminus [m]^{-1}D \longrightarrow Y.$$

Its geometric Galois group $E[m]$ has order m^2 prime to p. Pass to one finite extension L/K over which $E[m]$ and every puncture are rational. The point $P' = 0$ lies above P. Equal point sections give a lift $Q' \in Y_m(L)$ with the same section class as P': rational lifts are fixed points of G_L in the geometric fibre; after conjugating the sections to be equal, the fixed fibre point for P is also fixed for Q. This proves the comparison rather than assuming points lift over K.

The geometric prime-to-p group of Y_m is the corresponding open subgroup of that of Y. Indeed, intersect a finite prime-to-p Galois cover of Y_m with its finitely many conjugates under $E[m]$. The resulting Galois group over Y is an extension of a subgroup of a product of prime-to-p groups by $E[m]$, and still has prime-to-p order. Thus the induced topology is the full required topology.

Put $D_m = [m]^{-1}D$. Abelianization on Y_m and B1 imply $[Q' - P'] \in J_{D_m}(L)[p^\infty]$. The kernel of $J_{D_m} \to E$ is a torus and has no geometric p-torsion. The order of this class therefore equals the order of $Q' - P'$ in $E(L)$. Multiplication by m sends the latter to $Q - P$ and is invertible on p-primary groups; the order is exactly p^a. The divisor description of the reduced modulus Jacobian consequently gives a rational function h with

$$\operatorname{div}(h) = p^a(Q' - P'), \qquad h|_{D_m} = 1.$$

The modulus condition requires the same nonzero value at all boundary points, and rescaling gives the value 1. The degree of h is p^a, yet $h - 1$ vanishes at all $|D|m^2 \geq m^2 > p^a$ distinct points of D_m, away from its poles. Equality of zero and pole degrees is a contradiction. Hence $P = Q$. □

B9. The remaining prime-to-p point question *[Open]*

We have not settled prime-to-p (*) for every hyperbolic curve over a field finitely generated over $\mathbb{F}_p$, nor the implication on scan 36 that compatible p-primary differences on all prime-to-p covers force equality. The remaining range in this work consists of proper and affine genus at least two with nonzero p-primary Jacobian differences. Genus zero and punctured genus one are settled by B7–B8. B14–B15 additionally separate the two specific genus-two pairs over $\mathbb{F}_3$ and $\mathbb{F}_5$ investigated below. The reduction proved here sends every collision over a finitely generated field to a collision over a finite field, with the same geometric genus and number of punctures. Thus the remaining universal question reduces exactly to smooth proper curves of genus at least two over finite fields; already its universal genus-two case remains undecided in this work. This retains the unresolved portion of F7 (149, section 1, manuscript p. 17, scans 35–36).

Verified partial result and obstruction. The actual compactification comparison. Write $\Sigma = \{\ell : \ell \neq p\}$. For an affine hyperbolic curve U of genus at least two and its smooth proper compactification $j : U \hookrightarrow C$, functoriality of the geometric prime-to-p quotient gives a continuous homomorphism

$$j_* : E_U^\Sigma \longrightarrow E_C^\Sigma \quad \text{over } \mathrm{id}_{G_K}.$$

It sends the geometric kernel into the geometric kernel and sends the point section at $R \in U(K)$ to its point section on C. Thus an equality $s_Q = \mathrm{int}(\delta)s_P$ with δ geometric gives the same equality on C, conjugated by $j_*(\delta)$. The open immersion keeps P, Q distinct. Consequently every affine collision of genus at least two gives a proper collision of the same genus. Proper curves are already among the curves in question; together with B7–B8 this reduces the universal problem to the proper genus-at-least-two case. The finite-field comparison below then gives the claimed exact range.

Equal section classes do imply the covering condition: on every finite extension where a chosen lift P' is rational, the fixed-fibre argument of B8 produces Q' on any finite geometrically prime-to-p cover, and Kummer theory gives $[Q' - P'] \in J_{Y'}(L)[p^\infty]$. The words "cover of X" on scan 36 must here refer to a cover of Y, the target containing P, Q. This is evidence for editorial correction, not a silently chosen reading. Here "p-torsion" means p-primary torsion; no uniform exponent one has been established.

Reduction to finite fields, with the arithmetic maps. Let U/K be a hyperbolic curve, where $K/\mathbb{F}_p$ is finitely generated, and let $P \neq Q$ have equal geometrically prime-to-p section classes. A smooth proper compactification with finite étale boundary is part of the smooth-pair definition of a hyperbolic curve. Even when only its geometric compactification is initially given, its equations, boundary, and open immersion descend from $\overline{K}$ to a finite extension K'/K. Replacing K by K' preserves the distinct points and the equality of their section classes. The actual field-base-change map is

$$E_{U_{K'}}^\Sigma \xrightarrow{\sim} E_U^\Sigma \times_{G_K} G_{K'}, \qquad \Sigma = \{\ell : \ell \neq p\};$$

for a separable extension this is the finite-étale base-change description, and for a purely inseparable extension it follows from the universal homeomorphism $U_{K'} \to U$. A finite extension is handled by these two steps. In every group here the arithmetic quotient is the full absolute Galois group.

Spread the compactification, its boundary, and the two sections over a smooth integral affine scheme S of finite type over a finite field, with function field K. Shrinking S gives a smooth proper geometrically connected curve C/S, a finite étale divisor $D \subset C$, and disjoint sections P, Q of $\mathcal{U} = C \setminus D$. These are finite-presentation conditions; the equalizer of the two sections is closed and misses the generic point, so removing it ensures disjointness. The genus and boundary degree are constant. Choose a closed point $s \in S$, put

$$R = \mathcal{O}_{S,s}^h, \qquad L = \mathrm{Frac}(R), \qquad k = \kappa(s).$$

The field k is finite. The ring R is a regular henselian local domain. The finite-étale equivalence for a henselian local ring identifies $\pi_1(R)$ with G_k[The26d, Tag 09ZS]. The homomorphism $r : G_L \to G_k$ induced by $\mathrm{Spec}(L) \to \mathrm{Spec}(R)$ is surjective: a connected finite étale cover of the normal integral scheme $\mathrm{Spec}(R)$ is normal and integral, and hence remains connected over L.

Here are the required relative groups, without replacing the arithmetic quotient by a prime-to-p quotient. Set

$$K_R = \ker\big(\pi_1(\mathcal{U}_R) \to \pi_1(R)\big), \qquad N_R = \ker(K_R \to K_R^\Sigma), \qquad E_R = \pi_1(\mathcal{U}_R)/N_R.$$

This is exactly the construction of SGA1 XIII, 4.0. Proposition 4.3 gives the homotopy sequence; Examples 4.4 verify its hypotheses for a smooth proper family with relative normal-crossings boundary and a section. Section 4.5.3 identifies the relative pro-Σ kernel after every connected

base change, in the source's words "canoniquement isomorphe à l'image inverse" [G+71, Exposé XIII, 4.0, Proposition 4.3, Examples 4.4, 4.5.3]. Our finite étale boundary is such a relative divisor; the section P supplies the required section. With compatible base points and paths supplied by P, the closed- and generic-fibre inclusions give

$$1 \longrightarrow \Delta_s^{\Sigma} \longrightarrow E_R \longrightarrow G_k \longrightarrow 1, \qquad j_s : E_{U_s}^{\Sigma} \xrightarrow{\sim} E_R,$$
$$\mathrm{sp} = j_s^{-1} j_L : E_{U_L}^{\Sigma} \longrightarrow E_{U_s}^{\Sigma},$$
$$\theta : E_{U_L}^{\Sigma} \xrightarrow{\sim} E_{U_s}^{\Sigma} \times_{G_k} G_L, \qquad e \longmapsto \big(\mathrm{sp}(e), \mathrm{pr}(e)\big).$$

Indeed, j_s is an isomorphism on both the geometric kernel and the arithmetic quotient. The map j_L is an isomorphism on the geometric kernel by 4.5.3; consequently θ is an isomorphism on its kernel and quotient, proving the asserted isomorphism of profinite groups. These descriptions also prove surjectivity of sp.

Naturality for the two actual sections of $\mathcal{U}_R/R$ gives

$$\mathrm{sp} \circ s_{P_L} = s_{P_s} \circ r, \qquad \mathrm{sp} \circ s_{Q_L} = s_{Q_s} \circ r.$$

The equality of the original section classes restricts to L. Thus, after compatible geometric choices, $s_{Q_L} = \mathrm{int}(\delta) s_{P_L}$ for $\delta \in \Delta_L^{\Sigma}$. Applying sp and using surjectivity of r gives

$$s_{Q_s} = \mathrm{int}\big(\mathrm{sp}(\delta)\big) s_{P_s}.$$

The conjugator belongs to Δ_s^{Σ}, and $P_s \neq Q_s$ by construction. We have produced a finite-field collision of the same type. Conversely a finite field is itself finitely generated over $\mathbb{F}_p$. This proves equivalence of the two universal injectivity assertions, separately for each geometric curve type.

The opened finite-field result of Saïdi–Tamagawa is precisely [ST09, Proposition 1.8(vi), Corollary 1.11(iii)]: for a hyperbolic curve and a cofinite retained set of primes, the closed-point decomposition map has a finite exceptional set. The introduction, p. 138, explicitly says "almost injective". The 2017 refinement retains the exceptional set[ST17, Definitions 1.1–1.4, Theorem 1.9]. Both texts were opened. Equality of point sections implies conjugacy of decomposition groups, so a possible coincidence lies in that finite set. Neither statement makes it empty. The Isom theorem reconstructing the curve does not supply the missing implication about this point map.

We can prove exactly why the finite exceptional set does not vanish merely by passing to covers. Use the notation of [ST09, Remark 1.2, Definition 1.5, Proposition 1.8]. For a hyperbolic U over a finite field k, write Π_U for its geometrically prime-to-p arithmetic group and $\widetilde{U}$ for the associated universal pro-cover. For a chosen pro-point x, let $D_x \subset \Pi_U$ be its decomposition subgroup. The exceptional pro-point set $\widetilde{E}_U$ consists exactly of the x for which there is $y \neq x$ with D_x, D_y commensurate. For an open subgroup $H \subset \Pi_U$, with corresponding curve V, the same pro-cover serves V, and its decomposition subgroup is $D_x \cap H$. Intersection with H preserves and reflects commensurability: each $D_x \cap H$ is open in D_x, and a common finite-index intersection in H is also one in Π_U. Hence

$$\widetilde{E}_V = \widetilde{E}_U, \qquad E_V = \mathrm{preimage}(E_U) \quad \text{in } V.$$

The second equality uses the Π_U-stability in Proposition1.8(v) and the orbit descriptions $E_U = \widetilde{E}_U/\Pi_U$, $E_V = \widetilde{E}_U/H$. For a geometrically connected degree-d cover, on geometric point sets it gives $|E_V(\overline{k})| = d|E_U(\overline{k})|$. Finiteness for each cover therefore supplies no shrinking argument or uniform bound.

Moreover E_U is empty exactly when the point-section map is injective over every finite constant extension of k. To prove the nontrivial direction, take distinct exceptional pro-points x, y and put $Z = D_x \cap D_y$. The cusps are excluded by Proposition1.8(ii), so the projections of D_x, D_y

to G_k are injective. The subgroup $G_0 = \mathrm{pr}(Z)$ is open in G_k; on restricting to $\mathrm{pr}^{-1}(G_0)$, both decomposition subgroups become exactly Z, by injectivity of these projections. Over the finite constant extension fixed by G_0, the two images are rational points with the same section. They are distinct, since Proposition1.8(iii) says commensurate decomposition groups cannot occur at distinct pro-points above the same geometric point. Conversely equal point-section classes can be represented by equal actual decomposition groups after choosing the corresponding pro-lifts, and give an exceptional pair. Thus an actually nonempty exceptional set would give a counterexample after a finite extension; the cited results do not assert that such a set is nonempty.

The first concrete proper test we attacked was the smooth model $y^3 - y = x + x^{-1}$ over $\mathbb{F}_3$. It has genus two and two rational points P, Q over $0, \infty$, with $\mathrm{div}(x) = 3P - 3Q$. At each pole, $1/y$ is a uniformizer and $v(1/(y+c) - 1/y) = 2$ for $c \in \mathbb{F}_3^*$; the two different exponents are four, giving $2g - 2 = 3(-2) + 8 = 2$. B14 now separates these sections by an explicit étale double cover over $\mathbb{F}_{27}$: no lift of Q has 3-primary Jacobian difference from the chosen lift of P. This test is resolved, and is not retained as an undecided example or as a counterexample.

The second explicit genus-two test is the smooth projective model over $\mathbb{F}_5$ of

$$C: \quad y^2 = x^5 + (x^2+1)^2, \qquad P = (0,1), \quad Q = \infty.$$

The polynomial on the right is squarefree: its derivative is $4x(x^2+1)$, whose zeros are not zeros of the polynomial. It has degree five, so the compactification is a smooth genus-two curve with its unique rational point Q at infinity. At P, x is a uniformizer and

$$(y - x^2 - 1)(y + x^2 + 1) = x^5$$

gives $\mathrm{div}(y - x^2 - 1) = 5P - 5Q$: the second factor is nonzero at P, and $y - x^2 - 1$ has pole order five at infinity. Thus $[P - Q]$ has exact order five, since a function with divisor $P - Q$ would have degree one. The squarefreeness and finite point calculation are included in `code/AB_checks.py`. B15 separates their prime-to-five section classes. Explicitly, write $x^5 + (x^2+1)^2 = (x-1)H(x)$, where $H(x) = x^4 + 2x^3 + 2x^2 - x - 1$. The étale double cover obtained by adjoining $v^2 = x - 1$ has the elliptic quotient $E : w^2 = H(x)$, with $w = y/v$ and $|E(\mathbb{F}_5)| = 8$. A chosen lift of P maps to $(0,3)$, while the two lifts of Q map to the two infinity points. Both resulting elliptic differences are nonzero of 2-power order, contradicting the necessary 5-primary condition. The complete cover and quotient verification is B15. Thus neither explicit pair is left undecided. The remaining statement is the universal higher-genus finite-field assertion, including genus two with nonzero p-primary Jacobian differences.

An exact finite test in odd-characteristic genus two. The two constructions also give a verified next test, with its remaining obligation explicit. The canonical linear system of a genus-two curve has dimension one and degree two and is base-point free: a base point R would contradict Riemann–Roch, which gives $h^0(K_C - R) = h^0(R) = 1$. It defines a separable degree-two map to $\mathbb{P}^1$ when p is odd. After a finite constant extension splitting its six branch points, a projective change of coordinate puts infinity outside that finite set. In these actual coordinates write the curve as $z^2 = A_2(x)B_4(x)$ with six distinct branch points, choosing any pair for A_2. There are fifteen choices of the pair. The curve $D : v^2 = A_2(x),\ w^2 = B_4(x)$ is a connected *étale* double cover, with elliptic quotient $E : w^2 = B_4(x)$; the proof is the same even-divisor and conic-quotient calculation as B14, with the displayed branch polynomials in place of its particular ones. The two involutions have hyperelliptic product, so the exact norm identity $f^*f_* + g^*g_* = [2]$ holds on J_D. Make a chosen lift P' and both lifts Q' rational over one finite field. For $d = [Q' - P']$, the two norms are $[Q - P]$ and $[g(Q') - g(P')]$. A non-p-primary elliptic norm already separates that lift, by the finite Lang construction of B14. When both norms have respective p-power orders p^a, p^b, put $N = p^{\max(a,b)}$. The norm identity gives $Nd \in J_D[2]$. Since p is odd, d is p-primary exactly when $Nd = 0$: in the forward direction Nd is both p-primary and killed by two, and in the reverse direction N is a power of p. The last equality is an exact divisor-class test; it is

equivalent to a nonzero section of the degree-zero line bundle $\mathcal{O}_D(NP' - NQ')$, and hence is decidable by a finite Riemann–Roch computation. This identifies the residual obstruction in each of the fifteen first double-cover tests. We have not proved that one of those tests always separates a distinct pair, or that successive deeper covers eliminate every pair for which they fail. The examples B14–B15 verify success for their two specified pairs; no universal success is inferred from them.

Compatibility of the fifteen lifts and their exact joint kernel. There is a further proved reduction of these fifteen tests. Work in odd characteristic, put $J = J_C$, and use the Abel map $j : C \to J$, $R \mapsto [R - P]$. Its map on geometric fundamental groups identifies the abelianization with that of J [Mil08, III, Proposition 9.1]. This opened proposition works over a separably closed field; only its prime-to-characteristic case $n = 2$, proved there by Lemmas 9.2–9.3, is used here. Consequently

$$Z = C \times_{j,J,[2]} J, \qquad \pi : Z \longrightarrow C, \qquad G = J[2](\overline{k}) \simeq \mathbb{F}_2^4$$

is a connected étale Galois cover of degree sixteen and genus seventeen. Its fifteen index-two quotients are precisely the fifteen double covers just considered. Here is the exact identification with the branch-pair list. Over $\overline{k}$, Kummer theory identifies connected double covers with the nonzero classes in $J_C[2]$[Mil08, III, Lemma 9.2]. The fifteen classes $[W_i - W_j]$ of distinct Weierstrass points are nonzero and distinct as unordered pairs. Nonzero follows because a function with divisor $W_i - W_j$ would have degree one. They are killed by two since $2W_i$ and $2W_j$ are fibres of the hyperelliptic map. For distinctness, equality of two such classes gives $W_i + W_l \sim W_k + W_j$. If these effective degree-two divisors differ, Riemann–Roch makes them canonical fibres. A canonical fibre supported on Weierstrass points is $2W$, forcing $i = l$ and $k = j$, hence the same unordered pair. Equality of the effective divisors themselves gives that pair as well. Since $|J_C[2]| = 16$, the list is exhaustive. In the degree-six model with infinity unramified, the divisor of $(x - a_i)(x - a_j)$ is $2W_i + 2W_j - 2\infty_+ - 2\infty_-$; its Kummer class is $[W_i + W_j - \infty_+ - \infty_-] = [W_i - W_j]$. Thus the displayed square-root construction gives exactly these quotients. All the groups, covers and points below are defined over one finite extension of the original finite field. This follows by descending their finite algebraic data from its algebraic closure.

For a hyperplane $H \subset G$, put

$$B_H = J/H, \qquad \rho_H : J \to B_H, \qquad [2] = \lambda_H \rho_H, \qquad D_H = C \times_{j,J,\lambda_H} B_H.$$

Here $\lambda_H : B_H \to J$ is an isogeny of degree two, $f_H : D_H \to C$ is the corresponding double cover, and

$$r_H : Z \longrightarrow D_H, \qquad (R, b) \longmapsto (R, \rho_H(b))$$

is the quotient by H, of degree eight. Choose $\widetilde{P} = (P, 0)$ and let $P_H = r_H(\widetilde{P})$. Changing the chosen lift of P in an individual double cover applies its deck automorphism to the two tested differences and does not change whether a p-primary lift exists.

Let $c = [Q - P]$ be p-primary. Multiplication by two is an automorphism of the p-primary torsion group, so there is a unique p-primary b with $2b = c$; explicitly $b = mc$ for an integer m inverse to two modulo the order of c. Put $\widetilde{Q} = (Q, b) \in Z$. Every successful lift in the fifteen tests must be the image of this single point. Indeed the second projection $a_H : D_H \to B_H$ sends P_H to zero and hence induces a homomorphism $J_{D_H} \to B_H$. If $[Q'_H - P_H]$ is p-primary, then $a_H(Q'_H)$ is p-primary and maps to c under λ_H. An isogeny of degree two has exactly one p-primary point over a given p-primary point: its kernel has exponent two, and $\rho_H(b)$ supplies that preimage. Thus $a_H(Q'_H) = \rho_H(b)$. In the displayed fibre product, the coordinate pair $(Q, a_H(Q'_H))$ determines Q'_H itself. Therefore

$$Q'_H = r_H(\widetilde{Q}).$$

In particular each test has at most one successful lift, and success in all fifteen tests makes their choices compatible.

Write $\delta = [\widetilde{Q} - \widetilde{P}] \in J_Z$. The exact equivalence just proved is that all fifteen tests admit a p-primary lift precisely when every $(r_H)_*\delta$ is p-primary. To see what those norm data determine, write $N_A = \sum_{a \in A} a_*$ for a subgroup $A \subset G$. In the integral group ring the hyperplane incidence identity is

$$\sum_{\substack{H \subset G \\ |G/H|=2}} N_H = 8\,\mathrm{id} + 7N_G.$$

The identity element belongs to fifteen hyperplanes and every other element to seven. Pullback and norm therefore give the following exact equality of endomorphisms of J_Z:

$$[8] = \sum_H r_H^*(r_H)_* - 7\pi^*\pi_*.$$

Let

$$K_2 = \bigcap_H \ker\big((r_H)_* : J_Z \to J_{D_H}\big).$$

Since $\pi_* = (f_H)_*(r_H)_*$, this identity proves $K_2 \subset J_Z[8]$. In particular it is finite and 2-primary. When all fifteen tests succeed, choose one power p^a killing their fifteen norm classes. For the actual point-divisor class above we obtain

$$p^a\delta \in K_2, \qquad 8p^a\delta = 0.$$

Thus the joint tests leave only a 2-primary part killed by eight in this degree-sixteen cover.

This joint kernel really is nonzero for every such genus-two curve. The character map

$$G^D = \mathrm{Hom}(G, \mathbb{G}_m) \longrightarrow \mathrm{Pic}^0(C), \qquad \chi \longmapsto Z \times^G \mathbb{A}^1_\chi$$

identifies G^D with $\ker(\pi^* : J_C \to J_Z)$. For a line bundle trivialized upstairs, the descent multipliers belong to $\Gamma(Z, \mathcal{O}_Z^*) = \overline{k}^*$ and give exactly such a character. Conversely a character gives the displayed associated line bundle. This map is injective: a trivial descended bundle would give a nonzero global function upstairs transforming by that character, whereas all such units are constant. Every character has order dividing two and its associated bundle has degree zero. There are sixteen characters; $J_C[2]$ also has sixteen geometric points. As these are finite étale groups in odd characteristic, this proves the exact equality

$$\ker \pi^* = J_C[2].$$

It follows that the actual image

$$\pi^* J_C[8] \simeq J_C[8]/J_C[2] \simeq (\mathbb{Z}/4)^4$$

is a nonzero subgroup of J_Z. For every $a \in J_C[8]$, the factorization $\pi = f_H r_H$ gives

$$(r_H)_*\pi^* a = (r_H)_* r_H^* f_H^* a = 8 f_H^* a = 0.$$

Hence $\pi^* J_C[8] \subset K_2$. In particular an a of exact order eight has $\pi^* a$ of exact order four: its double cannot vanish, since $2a$ has order four and is not in $\ker \pi^* = J_C[2]$.

The remaining point-divisor implication is precise. The fifteen conditions place the specific class $p^a[\widetilde{Q} - \widetilde{P}]$ in this nonzero finite kernel; they do not, by the norm identities alone, determine its value. We have not proved that the geometry of these two lifted points forces the residual class to vanish, or that success of all fifteen tests forces $P = Q$. Nor has a nonzero class constructed above been identified with such a point difference satisfying all the tests. The hyperplane identity is checked independently by the 15-by-16 incidence matrix in `code/AB_checks.py`.

The genus-one argument succeeds because the compactification of each $[m]^{-1}Y$ remains the same elliptic curve; the p-power order is fixed while the number of punctures grows. Higher-genus covers introduce additional Jacobian factors, and our argument does not bound the p-primary orders there. Proper covers also have no modulus points on which to force $h = 1$. These are the precise remaining obstructions to the proposed covering argument. □

B10. Marked field faithfulness in every characteristic *[Proved]*

For any field k, the contravariant functor from finitely generated field extensions of k to their profinite étale fundamental groupoids over Π_k is faithful, where an isomorphism between two functors must project to the identity natural transformation on Π_k. In characteristic $p > 0$ it remains faithful on replacing each geometric kernel by its maximal prime-to-p quotient and retaining the arithmetic base groupoid. Thus the characteristic-p margin to Theorem 2 is true in this precise marked sense, in particular for finitely generated extensions of $\mathbb{F}_p$ (149, section 1, manuscript pp. 10–11, 17, scans 22–23 and 36; F7).

Proof. Let $f, g : F \rightrightarrows L$ be k-embeddings and suppose their induced groupoid functors are isomorphic over Π_k with the stated identity condition. Base change the groupoids to a geometric point of Π_k. The components correspond to embeddings into $\overline{k}$ of the finite relative separable constant fields. The given natural isomorphism identifies the component maps. On each corresponding component it identifies the induced geometric homomorphisms up to inner conjugation. Purely inseparable extensions do not change the category of finite étale covers: a finite separable extension and its embeddings extend uniquely after such a base change, and descend by taking the separable elements. We may therefore pass to $\overline{k}$ and work with the field components of $F\overline{k}$ and $L\overline{k}$. The extended embeddings fix $\overline{k}$; restriction back to F will give the original equality.

For an integer n invertible in k, Kummer theory gives

$$H^1(G_{F\overline{k}}, \mu_n) = (F\overline{k})^*/(F\overline{k})^{*n}.$$

The coefficients are constant, and these degree-one classes factor through the geometric prime-to-p quotient when $p > 0$. Inner conjugation acts trivially on this H^1. Thus for every $a \neq 0$ we have $f(a)/g(a) \in (L\overline{k})^{*n}$ for every such n. On a normal proper model of the finitely generated field $L\overline{k}/\overline{k}$, every divisorial valuation of this ratio is divisible by arbitrarily large integers prime to the characteristic, and hence is zero. A rational function with zero divisor on a normal proper integral variety is an invertible global function, and is constant because the base is algebraically closed. Consequently $f(a) = c_a g(a)$ for some $c_a \in \overline{k}^*$.

For nonconstant a apply this also to $a - 1$. Writing $c = c_a$ and $d = c_{a-1}$ gives

$$(c - d)g(a) = 1 - d.$$

The element $g(a)$ is nonconstant, so $c = d = 1$. Constants are fixed already. Therefore the two extended field maps agree, and hence $f = g$. This proof constructs the comparison through the Kummer classes; it does not rely on an unproved reduction to a product of curves.

The arithmetic marking is material. On $\mathbb{F}_p(t)$, Frobenius $t \mapsto t^p$ acts on the Kummer class of t by multiplication by p. Choose n prime to p and not dividing $p - 1$; the t-adic valuation shows $[t^p] \neq [t]$ modulo nth powers. Thus the marked functors differ. The usual natural equivalence on the unmarked étale site coming from absolute Frobenius has component on the chosen algebraic constant field equal to arithmetic Frobenius, not to the identity. It consequently is not an allowed relative natural transformation in Theorem 2. This also explains the multiplication-by-p action on $\pi_1(\mathbb{G}_{m,\overline{\mathbb{F}}_p})^{(p')}$: it is conjugation by an arithmetic Frobenius lift, whose image in $G_{\mathbb{F}_p}$ is nontrivial. □

2.2 A prime-to-three cover resolving the small genus-two test

B14. The two ramified points in the genus-two test are separated *[Proved]*

Let $C/\mathbb{F}_3$ be the smooth projective model of

$$y^3 - y = x + x^{-1},$$

and let P, Q be its rational points over $x = 0, \infty$, respectively. Their point sections in the geometrically prime-to-three fundamental group are not geometrically conjugate. This remains true after every finite extension of $\mathbb{F}_3$. In particular their decomposition groups are not conjugate-commensurate. We construct a finite *étale* cover of $C_{\mathbb{F}_{27}}$, with geometric monodromy of order prime to three, having a rational point over P and none over Q. Thus this genus-two test is decided positively, although the general higher-genus assertion retained in B9 is not a consequence of this computation.

Passage: cote 149, §1, manuscript p. 17, scans 35–36, the proposed recovery of points by prime-to-characteristic covers (F7).

Proof. We make the covers and the section-lifting comparison explicit.

The quadratic cover and its two quotients. Put $k = \mathbb{F}_{27}$ and choose $a \in k$ with $a^3 - a = 1$; $T^3 - T - 1$ is irreducible over $\mathbb{F}_3$ since it has no root there. Set $u = y - a$ and $z = x - x^{-1}$. The original equation gives

$$z^2 = (u^3 - u + 1)^2 - 1 = u(u-1)(u+1)(u^3 - u - 1). \tag{B14.1}$$

Conversely $y = u + a$ and $x = (u^3 - u + 1 + z)/2$, so this is an isomorphism of function fields and hence of the smooth projective models. The sextic has six distinct roots; this also checks $g(C) = 2$. As in B9, $\mathrm{div}(x) = 3P - 3Q$, and $[Q - P]$ has exact order three in $J_C(\overline{k})$: order one would give a degree-one map from this genus-two curve to $\mathbb{P}^1$.

Let D be the smooth projective curve with function field

$$k(u, v, w), \qquad v^2 = u(u-1), \qquad w^2 = G(u) := (u+1)(u^3 - u - 1), \qquad z = vw. \tag{B14.2}$$

The last equality gives a degree-two morphism $f : D \to C_k$. It is connected: neither $u(u-1)$ nor its quotient by the sextic in (B14.1) is a square in $\overline{k}(u)$, as their simple zero or pole valuations show. The divisor of $u(u-1)$ on C_k is even. Indeed its two zeros are hyperelliptic branch points and have order two, while at each of P, Q it has a pole of order two. All other orders are zero. Quadratic Kummer theory in residue characteristic three therefore proves that f is *étale* everywhere.

There is also a degree-two map $g : D \to E$, where

$$E : \quad w^2 = G(u) = u^4 + u^3 + 2u^2 + u + 2. \tag{B14.3}$$

The quartic is square-free, so its projective model has genus one. It has two distinct rational points e_+, e_- at infinity, labelled by $w/u^2 = +1, -1$. The involutions of D over C and E are, respectively,

$$\tau(u, v, w) = (u, -v, -w), \qquad \sigma(u, v, w) = (u, -v, w).$$

Their product $h(u, v, w) = (u, v, -w)$ has the conic $B : v^2 = u(u-1)$ as quotient. This conic is rational, with

$$t = v/u, \qquad u = (1 - t^2)^{-1}, \qquad v = t(1 - t^2)^{-1}.$$

Writing $W = w(1 - t^2)^2$ gives the particularly simple model

$$D : \quad W^2 = 1 - t^2 - t^4 - t^8. \tag{B14.4}$$

The degree-eight polynomial is square-free. Hence $D \to B \simeq \mathbb{P}^1$ is branched at eight points and D is hyperelliptic of genus three. This agrees with the unramified formula $2g(D)-2 = 2(2g(C)-2)$.

The rational lifts and their elliptic images. At P and Q, respectively, z/u^3 tends to -1 and $+1$. Since in (B14.4) this quotient is tW, we can choose the rational lift and list the two rational lifts of Q as

$$P' = (t, W) = (1, -1), \qquad Q_+ = (1, 1), \qquad Q_- = (-1, -1). \tag{B14.5}$$

These are all defined over $\mathbb{F}_3$ on the displayed model of D; their map to the original C is defined over k because it uses a. The formula $w/u^2 = W$ gives

$$g(P') = g(Q_-) = e_-, \qquad g(Q_+) = e_+.$$

The elliptic curve (B14.3) has exactly five $\mathbb{F}_3$-points: above $u = 0$ there are none, above $u = 1$ there are two, above $u = -1$ there is one, and there are its two points at infinity. Therefore, with e_- as origin, $e_+ - e_-$ has exact order five. It follows that $[Q_+ - P'] \in J_D(\overline{k})$ is not three-primary.

The other lift has Jacobian difference of exact order six. Put $\delta = [Q_- - P']$. The two norm maps give

$$f_*\delta = [Q - P], \qquad g_*\delta = 0.$$

We verify the necessary assertion about their joint kernel instead of assuming a decomposition of the Jacobian. On divisor classes, pullback followed by norm satisfies

$$f^*f_* = 1 + \tau_*, \qquad g^*g_* = 1 + \sigma_*.$$

The hyperelliptic involution h_* is -1 on J_D: every divisor $R + hR$ is a fibre of the degree-two map to B, so applying h to a degree-zero divisor class negates it. Since $\tau\sigma = h$, we have $\tau_* = -\sigma_*$, and consequently

$$f^*f_* + g^*g_* = [2]. \tag{B14.6}$$

Thus the joint norm kernel is killed by two. Applying (B14.6) to 3δ proves $6\delta = 0$, while the exact order of $f_*\delta$ shows that the order of δ is divisible by three.

It cannot have order three. Such an order would supply a function b on $D_{\overline{k}}$ with $\operatorname{div}(b) = 3Q_- - 3P'$, hence a degree-three morphism to $\mathbb{P}^1$. Together with the degree-two hyperelliptic map, this would give a birational map onto an integral curve of bidegree $(2, 3)$ in $\mathbb{P}^1 \times \mathbb{P}^1$. To check birationality, the degree of the function-field extension over the compositum of the two rational subfields divides both two and three, so is one. An integral curve of bidegree $(2, 3)$ has arithmetic genus $(2-1)(3-1) = 2$, by adjunction; its normalization has genus at most two. This contradicts $g(D) = 3$. This argument uses function-field degrees, so does not silently assume that the degree-three map is separable. We conclude that δ has exact order six. Both lifts in (B14.5) therefore have non-three-primary difference from P', over k and over every algebraic extension of k.

The actual cover separating the fibres. Here is a finite cover constructed from the preceding classes. Write $J = J_D$ over k, let F be its 27-power Frobenius, and let $A = J(k)[3^\infty]$, viewed as a constant finite subgroup scheme. The Lang isogeny $F - 1 : J \to J$ is finite *étale* with kernel $J(k)$: its derivative is -1, and its geometric kernel is precisely the Frobenius-fixed finite group. Quotienting its source by A gives a finite *étale* isogeny

$$\lambda : J/A \longrightarrow J, \qquad \ker\lambda = J(k)/A, \tag{B14.7}$$

of degree prime to three. The comparison of rational images is exact:

$$\lambda((J/A)(k)) = A \subset J(k). \tag{B14.8}$$

Indeed, for a rational class $\bar{b} \in (J/A)(k)$, any geometric lift $b \in J(\overline{k})$ satisfies $F(b) - b \in A$, so its image under λ belongs to A. Conversely for $c \in A$ choose b with $F(b) - b = c$, using surjectivity of $F - 1$; its class modulo A is fixed by Frobenius and is rational. This proves (B14.8).

Let $\alpha : D \to J$ be $R \mapsto [R - P']$. Then

$$V = D \times_{\alpha,J,\lambda} (J/A) \longrightarrow D \xrightarrow{f} C_k \tag{B14.9}$$

is the promised cover. It has a rational point above P' supplied by $0 \in J/A$. By (B14.8) and the proved orders, it has no rational point over either Q_+ or Q_-. These are all points of D above Q, so it has none over Q. Its geometric monodromy is prime to three: the first map is a torsor under the finite group $J(k)/A$, and the second is a Galois double cover. The Galois closure of their composite embeds, over the double cover, into the product of the first torsor group and its deck conjugate; its geometric group is an extension of a subgroup of that product by a group of order two. Every order involved is prime to three. Taking a connected component containing the displayed rational point is also permissible and preserves these properties.

For numerical reproducibility, the exact script `code/F7_test.py` enumerates $\#D(\mathbb{F}_3) = 6$, $\#D(\mathbb{F}_9) = 10$, $\#D(\mathbb{F}_{27}) = 36$. Newton identities and the functional equation give its Frobenius polynomial

$$Z^6 + 2Z^5 + 2Z^4 + 4Z^3 + 6Z^2 + 18Z + 27.$$

Using the curve trace formula and the Jacobian point formula [Mil08, III, Theorem 11.1; II, Theorem 1.1] yields $\#J(\mathbb{F}_{27}) = 25920 = 2^6 \cdot 3^4 \cdot 5$. Thus (B14.7) has degree 320 and (B14.9) has degree 640 before choosing a component. The proof of separation above does not depend on these numerical counts beyond the elementary five points on E.

Comparison with point sections and constant extensions. For any finite *étale* cover whose geometric monodromy is prime to three, the rational points of its fibre over $R \in C(k)$ are exactly the fixed points of the image of $s_R(G_k)$ in the corresponding permutation representation. Geometrically conjugate sections have isomorphic such G_k-sets. The different rational fibre counts in (B14.9) therefore prove the required nonconjugacy over k, and hence over $\mathbb{F}_3$.

To obtain the asserted persistence after every finite extension, one may repeat the Lang construction with that finite field's Frobenius, after adjoining a. The two non-three-primary divisor classes retain their orders, so the same proof works. Equivalently, the precise covering argument of B8 applies to $D \to C$: after making equal sections literally equal by a geometric conjugator, the fixed coset supplied by the rational lift P' supplies a rational lift Q' with the same section in $\pi_1(D)$. This is the full geometrically prime-to-three group of D, since intersecting a prime-to-three Galois cover with its deck conjugate gives a prime-to-three Galois cover of C. Its abelianization would make $[Q' - P']$ three-primary by B7, contradicting both possibilities in (B14.5). All constant extensions used here are finite, and equality of sections restricts along their actual Galois subgroups. Finally, conjugate-commensurate decomposition groups would become equal after restricting to the open subgroup of the constant-field Galois group which is the image of their intersection. Their point sections would then be equal over a finite extension, which we have excluded. This proves every assertion of B14. □

B15. An elliptic quotient separates the genus-two quintic test *[Proved]*

Let $C/\mathbb{F}_5$ be the smooth projective curve

$$y^2 = x^5 + (x^2 + 1)^2, \qquad P = (0, 1), \quad Q = \infty.$$

Although $[P - Q]$ has exact order five in J_C, the points P, Q give distinct geometrically prime-to-five point-section classes over every finite extension of $\mathbb{F}_5$. There is a finite *étale* cover of degree 16 over $\mathbb{F}_5$, with geometric monodromy of order prime to five, having a rational point above P and none above Q.

Passage: cote 149, §1, manuscript p. 17, scans 35–36, the prime-to-characteristic covering assertion (F7).

Proof. Put

$$F(x) = x^5 + (x^2+1)^2 = (x-1)H(x), \qquad H(x) = x^4 + 2x^3 + 2x^2 - x - 1.$$

Both F and H are square-free, and $H(1) = 3$ in $\mathbb{F}_5$. Thus C has genus two and one rational point Q at infinity. The identity $(y - x^2 - 1)(y + x^2 + 1) = x^5$ shows

$$\operatorname{div}(y - x^2 - 1) = 5P - 5Q.$$

At P, the second factor is the unit 2, while at Q the orders of x, y are $-2, -5$, so that the first factor has order -5. There are no other zeros or poles. The class is nonzero because a degree-one function would make this genus-two curve rational; hence its order is exactly five.

Take the degree-two cover $f : D \to C$ given on function fields by

$$v^2 = x - 1, \qquad w = y/v, \qquad w^2 = H(x) = H(v^2 + 1). \tag{B15.1}$$

The function $x - 1$ has divisor $2R - 2Q$, where $R = (1, 0)$ is a simple hyperelliptic branch point. The quadratic Kummer extension is therefore unramified everywhere, including at Q. It is geometrically connected: neither $x - 1$ nor $(x - 1)/F(x) = 1/H(x)$ is a square in $\overline{\mathbb{F}}_5(x)$. The smooth projective model D is hyperelliptic of genus three; the degree-eight polynomial $H(v^2 + 1)$ has eight distinct roots, since H is square-free, $H(1) \neq 0$, and 2 is invertible.

The second quotient is the smooth elliptic curve

$$g : D \to E, \qquad E : w^2 = H(x).$$

Choose $P' = (v, w) = (2, 3)$, which maps to P because $x = v^2 + 1 = 0$ and $y = vw = 1$. Its elliptic image is $r = (0, 3)$. The two points Q_+, Q_- above Q are rational: the polynomial $H(v^2 + 1)$ has degree eight and leading coefficient one, so they are labelled by $w/v^4 = +1, -1$. The map g sends them to the two distinct rational infinity points e_+, e_- of E, labelled by $w/x^2 = +1, -1$. In particular neither equals r.

Direct enumeration gives

$$E(\mathbb{F}_5) = \{(0, 2), (0, 3), (3, 2), (3, 3), (4, 1), (4, 4), e_+, e_-\}.$$

With r as origin this is an elliptic group of order eight. Consequently each $e_\pm - r$ is nonzero and has order a power of two. This property persists in every extension field. The norm maps on Jacobians therefore show that neither $[Q_\pm - P'] \in J_D(\overline{\mathbb{F}}_5)$ is five-primary.

There is also a small actual separating cover. Let F_E denote the 5-power Frobenius of the elliptic curve with origin r. The map $F_E - 1 : E \to E$ is finite *étale* of degree eight and has constant kernel $E(\mathbb{F}_5)$. Form

$$V = D \times_{g, E, F_E - 1} E \longrightarrow D \longrightarrow C, \tag{B15.2}$$

where $g(P') = r$ is the identity. It has a rational point over P'. It has no rational point above Q_+ or Q_-, because $(F_E - 1)(E(\mathbb{F}_5)) = \{r\}$ whereas $g(Q_\pm) = e_\pm \neq r$. These are all points above Q. The cover (B15.2) has degree $8 \cdot 2 = 16$. Its geometric Galois closure has order prime to five by the same deck-conjugate product argument as in B14. Its rational fibre sets therefore distinguish the two point sections in the specified prime-to-five quotient.

For an arbitrary finite constant extension, equality of the point sections would, by the fixed-coset comparison in B8 and B14, give a rational lift Q' to D whose section equals that of P'. There are still exactly the two possibilities Q_+, Q_-. The geometric prime-to-five group of D is the open subgroup induced from C, since the cover has degree two. Kummer theory on J_D would make $[Q' - P']$ five-primary. Its nonzero two-primary norm $e_\pm - r$ contradicts that conclusion. This proves persistence over every finite extension, not merely separation over the original ground field. The exact polynomial, branch and rational point checks are recorded in `code/F7_test5.py` and its output; the comparison with sections uses the explicit morphisms above. □

3 C. Sections over a base and the cusp extension

We use the following notation throughout this part. A smooth pair over a scheme S is a smooth proper curve $\bar{U} \to S$ with geometrically connected fibres, together with an effective relative divisor $D \subset \bar{U}$ finite and *étale* over S; we put $U = \bar{U} \setminus D$. Thus collisions of marked points are excluded from the term "smooth pair". The type of a fibre is (g, r), with $r = \deg(D/S)$. We fix compatible geometric base points when writing homomorphisms of fundamental groups. Changing a path to a base point conjugates a displayed homomorphism by an element of the indicated geometric kernel. We write $G = \pi_1(S)$, $E = \pi_1(U)$ and $\Pi = \pi_1(U_{\bar{s}})$. These are respectively E_S, E_U and π in §3 of the manuscript, scans 64 and 79. In residue characteristic $p > 0$, a superscript (p') on a geometric kernel means its maximal pro-prime-to-p quotient; we retain the full arithmetic quotient. We write $T_{p'} = \prod_{\ell \neq p} \mathbb{Z}_\ell(1)$, and write T when $p = 0$ and all primes occur. All cohomology of profinite groups below is continuous.

C1. The Hartogs extension of a rational section *[Proved]*

[F11; §3, manuscript pp. 32, 34–35, 41; scans 65, 70–72, 83.] Let S be a smooth integral scheme of finite type over a field of characteristic zero. Let $U \to S$ be a smooth pair of constant type different from $(0, 0)$ and $(0, 1)$. For a closed subset $Z \subset S$ of codimension at least two, restriction is a bijection

$$\mathrm{Sect}_S(U) \longrightarrow \mathrm{Sect}_{S \setminus Z}(U \times_S (S \setminus Z)).$$

The same assertion holds for a finite tower of such relative curves.

Proof. Uniqueness follows from separatedness: two sections agree on the schematically dense open set $S \setminus Z$, hence their equalizer, the inverse image of the diagonal, is all of the reduced scheme S.

First suppose $g \geq 1$. We give the resolution argument behind the manuscript's reference to Weil. The question is local for the étale topology of S. A smooth proper curve has a section after an étale localization, and then a divisor of sufficiently large relative degree gives a projective embedding. We can consequently resolve the rational section $S \dashrightarrow \bar{U}$ by principalizing its base ideal. Bierstone–Milman, Theorem 1.10, supplies a finite sequence of blowups with smooth centres, disjoint from the locus where the rational map is defined, on which the ideal becomes invertible [BM97]. The resolved map is an S-morphism.

Consider the last blowup $b : S_m \to S_{m-1}$. Its geometric fibres are points or projective spaces. Every morphism from $\mathbb{P}^1$ to a smooth proper curve of genus at least one is constant: a nonconstant morphism would be finite, and the Riemann–Hurwitz equality $-2 = d(2g-2) + \deg R$ is impossible. Since any two points of a projective space lie on a projective line, the resolved map is constant on every geometric fibre of b. Let Γ be its scheme-theoretic graph image in $S_{m-1} \times_S \bar{U}$. Projection $\Gamma \to S_{m-1}$ is proper, birational, and has finite fibres, hence is finite. The target is normal, so this finite birational morphism is an isomorphism: on an affine open, its coordinate ring is an integral subalgebra of the common fraction field and is contained in the integrally closed coordinate ring of the target. The map therefore descends across b. Descending successively gives a section $\bar{f} : S \to \bar{U}$. The inverse image $\bar{f}^{-1}D$ is the zero scheme of a nonzero section of an invertible sheaf. If nonempty, it has a codimension-one component, by the principal ideal theorem in the regular local rings of S. Its support lies in Z, so it is empty.

For $g = 0$, the hypothesis gives $r \geq 2$. After an étale localization a boundary section is defined. A genus-zero smooth proper curve with this section is locally a projective-line bundle; locally on S, its complement is $\mathbb{A}^1_S$. The given section is a regular function on $S \setminus Z$. The normality equality $\mathcal{O}_S = j_* \mathcal{O}_{S \setminus Z}$ extends that function to S. For each remaining boundary section, its intersection with the extended section is again an effective Cartier divisor supported in Z, and is empty. The unique local extensions descend.

For a tower, first extend the section of the bottom curve. Pull the next curve back along that extended section, apply the one-curve assertion, and continue. These are actual Cartesian base changes, and the restriction of each pullback to $S \setminus Z$ is the pullback already carrying the given section. This proves the assertion without assuming that the entire tower embeds in a semiabelian variety.

The exclusion of $(0,0)$ matters: the section $(x,y) \mapsto [x:y]$ of $\mathbb{P}^1 \times (\mathbb{A}^2 \setminus \{0\})$ does not extend at the origin. Its restrictions to the two coordinate axes have different constant limits. The genus-zero proof also proves Hartogs extension for an affine-line bundle; that observation does not give the inertia criterion below, which fails for type $(0,1)$. □

C2. The trait criterion, including the two nonhyperbolic types *[Proved]*

[F12; §3, manuscript pp. 36–38; scans 74–78.] Let R be a discrete valuation ring, K its fraction field, and let $U = \bar{U} \setminus D$ be a smooth pair over R of type other than $(0,0), (0,1)$. Let $f_K \in U(K)$. In characteristic zero for the residue field use all geometric covers; in residue characteristic $p > 0$ use the geometrically pro-prime-to-p fundamental groups. The following are equivalent:

1. f_K extends to an R-section of U;
2. the continuous lift $G_K \to E_U^{(p')}$ induced by f_K factors through $\pi_1(\operatorname{Spec} R)$;
3. this lift is trivial on an inertia group at the closed point.

Here "factors" means that the induced map on the quotient is a section of $E_U^{(p')} \to \pi_1(R)$. For nonhenselian R, inertia is taken in a chosen decomposition group, and its closed normal closure is the kernel of $G_K \to \pi_1(R)$. The statement extends to a tower of smooth pairs whose curve factors have these types.

Proof. We can pass to the strict henselization to test the last condition. Extension of the rational point can also be tested there: properness first extends it uniquely to $\bar{U}$ over R, and whether its closed-point image lies in D is unchanged by faithfully flat base change. Thus the descent in this step is descent of this specified proper extension, not descent of an arbitrarily chosen section.

If its closed-point image is in U, functoriality gives the required factorization. Otherwise it is a cusp q. C3 below identifies the inertia map with $\kappa_q \circ [\mu]$, where $\mu > 0$ and κ_q is injective. Multiplication by a nonzero integer is injective and nonzero on $T_{p'}$. The inertia map is consequently nontrivial. This proves the one-curve assertion. The peripheral injection used here follows from the prime-to-p presentation of a punctured smooth curve in SGA 1, Exposé XIII, Corollary 2.12, and its peripheral-subgroup form recorded in Stix, §2 and Corollary 12 [GR03, Sti08a]. For a single cusp the group can be nonabelian with the cusp zero in its abelianization; C3 does not abelianize it.

For reference, the needed peripheral injection can be checked in every pro-ℓ quotient directly. The presentation of an affine curve is the free pro-ℓ group on $a_1, b_1, \ldots, a_g, b_g, c_1, \ldots, c_{r-1}$, with the last cusp the inverse of $\prod_j [a_j, b_j] \prod_i c_i$. For $r \geq 2$ each peripheral word has nonzero abelianization after choosing that cusp among the free generators. For $r = 1, g \geq 1$ its degree-two class is $\sum_j [a_j, b_j] \neq 0$ in the free degree-two Lie module. It is therefore nontrivial and has infinite order in this torsion-free free pro-ℓ group. The homomorphism $\mathbb{Z}_\ell(1)$ is injective. Taking all $\ell \neq p$ detects every coordinate of $T_{p'}$ and proves the required injection, including the one-cusp case.

For type $(0,2)$ the cusp is the whole geometric group $T_{p'}$, so the same proof applies. For $(1,0)$ properness already gives extension into $U = \bar{U}$, and smooth proper specialization gives trivial inertia in the indicated quotient. No centre-free hypothesis is needed for either case. For $(0,1)$, $R = k[[t]]$ and $f_K = t^{-1} \in \mathbb{A}^1(K)$ give a counterexample: the prime-to-p geometric group of the affine line is trivial, whereas the point has a pole.

For a tower, project the lift to the first curve. The one-curve criterion extends that projected

point. Pull back the next curve to $\operatorname{Spec} R$ along this extended section. The fundamental-group square is Cartesian: both its horizontal maps induce the same isomorphism on geometric kernels, and the right-hand quotient is $G_K \to \pi_1(R)$; its exact sequences give the Cartesian assertion by the elementary kernel-and-quotient argument. For the exceptional centre-bearing types over a trait the same exactness follows from the smooth proper or tame homotopy sequence with section, SGA 1 XIII, Proposition 4.3 and Examples 4.4. The induced lift to this pullback therefore kills inertia. Apply the one-curve criterion again and iterate. This also proves exactly which comparison maps support the manuscript's dévissage. □

C3. The nonabelian intersection formula and the five conditions *[Proved]*

[F12; §3, manuscript pp. 37–39 and 44; scans 75–79, 90.] Let R be henselian and use the notation of C2. Suppose that the proper extension of f_K meets a rational cusp q in the closed fibre. If z is a local equation of q and t a uniformizer of R, write

$$f_K^* z = t^{\mu} u, \qquad u \in R^*, \qquad \mu > 0.$$

With positive Kummer inertia defined by $\tau \mapsto \tau(z^{1/n})/z^{1/n}$, the induced continuous homomorphism on inertia is, up to the geometric conjugation arising from the chosen path,

$$I_K \longrightarrow I_K/P_K = T_{p'} \xrightarrow{[\mu]} T_{p'} \xrightarrow{\kappa_q} \Pi^{(p')}. \tag{C.1}$$

For $p = 0$, $P_K = 1$. The formula holds with one cusp. With at least two cusps it induces on the toric part of homology

$$T_\ell \longrightarrow T_\ell^{D(\bar{k})}/T_\ell, \qquad a \longmapsto \mu a e_q \pmod{T_\ell \cdot (1, \dots, 1)}. \tag{C.2}$$

In equicharacteristic zero, the five conditions a)–e) of scan 90 are equivalent for the lift $\lambda : G_K \to E_U$: injectivity of λ; the positive-multiplicity formula (C.1); nontrivial inertia; failure of descent to $\pi_1(R)$; and failure of f_K to extend into U. In that case its image has index μ in the cusp normalizer. For $p > 0$, replace the domain in the injectivity assertion by G_K/P_K, and replace the index by $\mu_{p'} = \prod_{\ell \neq p} \ell^{v_\ell(\mu)}$.

Proof. The local ring of $\bar{U}$ at $q(s)$ is regular of dimension two and $z = 0$ is a regular divisor. Over its strict henselization the prime-to-p tame covers of its complement are dominated by $z^{1/n}$, with $(n, p) = 1$. This is precisely SGA 1 XIII, Proposition 5.1 together with its normal-crossings formulation, Proposition 5.2: adjoining prime-to-residue- characteristic roots of regular parameters makes the cover étale [GR03]. These statements apply here to the one divisor $z = 0$. In the strict henselization of R, the unit u has an nth root, by Hensel's lemma. Compatible such roots exist over the inverse system (its nonempty finite levels have surjective transition maps). Pulling back $z^{1/n}$ along f_K gives $t^{\mu/n} u^{1/n}$; its inertia character is μ times the character of $t^{1/n}$. Passing through the local fundamental group and then into the global group proves (C.1). Every stage is a homomorphism of fundamental groups; no abelianization of the target has occurred. Formula (C.2) follows by applying the specified abelianization map and the single relation that the sum of peripheral classes is zero. For one cusp this latter class is zero; the injection κ_q before abelianization remains injective.

The equality of intersection multiplicities in scans 77–78 is also literal. If $A = \mathcal{O}_{\bar{U}, q(s)}$ and I_f, I_q are the ideals of the two sections, then

$$R/q^* I_f \simeq A/(I_f + I_q) \simeq R/f^* I_q.$$

The lengths of these Artinian modules are equal to μ.

For scan 90, the composite $G_K \xrightarrow{\lambda} E_U \to G_k$ is the residue quotient. Hence $\ker\lambda \subset I_K$. In characteristic zero, (C.1) is injective on I_K whenever the section meets the boundary; this proves injectivity of λ. If it does not meet the boundary, it kills $I_K \neq 1$ and is not injective. C2 proves the other equivalences. The image normalizes μL_q. In a hyperbolic curve the normalizer of any open subgroup of a peripheral group is the cusp normalizer, by the peripheral self-normalization theorem; in type $(0,2)$ the entire geometric group is peripheral and the same conclusion is immediate. Alternatively, the map from the local punctured tube constructed above lands directly in this normalizer. Its intersection with L_q is μL_q and its image in G_k is all of G_k. The coset index is therefore $[T : \mu T] = \mu$. With prime-to-p kernels the same proof applies after quotienting P_K, and gives $[T_{p'} : \mu T_{p'}] = \mu_{p'}$. Full G_K need not inject in this version: wild inertia has been killed.

For the proposed test $\mathbb{P}^1_R \setminus \{0, \infty, q\}$ with $q = t^\mu u$, the sections 0 and q collide. Thus its marked boundary is not étale and it is not a smooth pair over R. The local Kummer calculation $z \mapsto t^\mu u$ still tests the factor μ at the cusp 0. A test within the precise hypotheses is the fixed smooth pair $\mathbb{P}^1_R \setminus \{0, 1, \infty\}$ and the point $z = t^\mu u$. Both tests have trivial normal line at 0, so neither distinguishes $c_1(N)$ from $-c_1(N)$. The global transition calculation in C6 fixes that sign; assigning a nonzero Chern sign to a trivial line would not test it. □

C4. The exact sequence and the centre-free extension *[Proved]*

[F13; §3, manuscript p. 39; scan 79; also scans 64, 73.] Let S be a connected normal scheme of finite type over a field of characteristic zero, and let $U \to S$ be a hyperbolic smooth pair. Then

$$1 \longrightarrow \Pi \longrightarrow E \longrightarrow G \longrightarrow 1 \tag{C.3}$$

is exact. Its formation commutes with the base changes to a normal connected scheme used in this part, with the chosen geometric fibre identified by a path. If $\rho : G \to \mathrm{Out}(\Pi)$ is its continuous outer action, the continuous map

$$E \longrightarrow E_\rho = \{(a, g) \in \mathrm{Aut}(\Pi) \times G : [a] = \rho(g)\}, \qquad e \longmapsto (\mathrm{conj}_e, \bar{e}) \tag{C.4}$$

is an isomorphism. For a tower with factors also of types $(0,2)$ or $(1,0)$, exactness holds over an étale $K(\pi, 1)$ base, and at each stage over a trait as in C2. It is false for arbitrary bases without the hyperbolic or π_2 hypothesis.

Proof. Holzschuh–Schmidt–Stix, Theorem 5.6, applies to a geometric fibration with geometrically connected fibres over a normal quasi-compact quasi-separated scheme of characteristic zero. A smooth proper curve with a relative normal-crossings boundary, in particular our finite étale boundary, is such a fibration (their Definition 5.5). Their quasifibration theorem gives the homotopy sequence. In the segment

$$\pi_2^{\mathrm{et}}(S) \xrightarrow{\partial} \Pi \longrightarrow E \longrightarrow G \longrightarrow 1,$$

the image of ∂ is central in Π. This is the fibration boundary property: the action of a fibre loop on a spherical lift is trivial after the two boundary homotopies are composed; equivalently, the crossed-module identity gives $\partial(x) b \partial(x)^{-1} = b$ for every $b \in \Pi$. Hyperbolic geometric curve groups are centre-free. The same argument and conclusion are explicitly given in the proof of their Proposition 8.3, using Lemma 8.2 [HSS26]. Hence $\partial = 0$ and (C.3) is exact. The same theorem for a base change identifies the geometric kernel; the homomorphism to $E \times_G \pi_1(S')$ is the identity on that kernel and on the quotient, and is an isomorphism.

For (C.4), an element in its kernel lies in Π and centralizes Π, so is trivial. Given (a, g) choose $e \in E$ over g. The automorphism $a\,\mathrm{conj}_e^{-1}$ is inner, so equals conj_b for a unique $b \in \Pi$. Then be

maps to (a, g). This proves bijectivity, and compactness gives the topological isomorphism. Thus the extension reconstructed in the margin is not merely a larger object mapping onto $\pi_1(U)$ in this setting: the displayed map is an isomorphism and supplies its inverse explicitly.

If S is a $K(\pi, 1)$, its π_2 is zero, so the same homotopy sequence proves the centre-bearing cases. Over a trait the version with section is SGA 1 XIII, Proposition 4.3 and Examples 4.4 (a section exists étale locally and the claimed injection can be tested there). An actual counterexample to the deleted unrestricted margin is $S = \mathbb{P}^1_{\mathbb{C}}$ and $U = \mathcal{O}_{\mathbb{P}^1}(-1)^{\times}$. The map $(\ell, v) \mapsto v$ identifies U with $\mathbb{A}^2_{\mathbb{C}} \setminus \{0\}$; its complex analytification retracts onto S^3 and has trivial fundamental group. Riemann existence identifies its finite étale covers with finite topological covers. Consequently $\pi_1(U) = 1$, although its fibre $\mathbb{G}_m$ has group $\widehat{\mathbb{Z}}(1)$. The boundary map from $\pi_2(\mathbb{P}^1(\mathbb{C})) = \mathbb{Z}$ is multiplication by the degree -1 of the line bundle and kills the fibre group. This computes the obstruction which the margin had set aside. □

C5. Unramified rational geometric sections extend *[Proved]*

[F11; §3, manuscript pp. 32, 34–36, 41–42; scans 65, 70–74, 83–84.] Let S be smooth integral of finite type over a characteristic-zero field, with function field K, and let U/S be a tower as in C1 for which the exact geometric sequences of C4 hold. A rational section f_K extends to S if and only if its induced lift

$$\lambda_f : G_K \longrightarrow \pi_1(U)$$

kills $\ker(G_K \to \pi_1(S))$. Equivalently, it kills every divisorial inertia subgroup of G_K whose centre has codimension one in S.

Proof. The kernel in question is the closed normal subgroup generated by those inertia groups. Indeed, a finite Galois extension of K on which all these inertia groups vanish has normalization étale in codimension one over S; purity of the branch locus makes it étale everywhere. Conversely an étale cover is unramified at each such valuation. This identifies the finite quotients of the two proposed quotient groups, and therefore identifies the profinite groups themselves. The purity statement used here is the Zariski–Nagata theorem: a finite dominant map with normal source and regular target, generically étale, has branch locus either empty or of pure codimension one [The26b].

If λ_f kills the kernel, C2 applied at every height-one local ring extends the section there. Finite presentation spreads each such extension to a neighborhood of that point. The union with the original domain of definition is an open set whose complement has codimension at least two. C1 extends across that complement. The reverse implication is the functorial factorization of fundamental groups associated with the actual section. This proof specifies why the purity comparison $\pi_1(S \setminus Z) \simeq \pi_1(S)$ by itself would not prove the extension of a morphism: C1 supplies that geometric step. □

C6. The normalizer extension and its sign *[Proved]*

[F14; §3, manuscript pp. 39–41; scans 79–82.] Assume the hyperbolic case of C4 and let $q : S \to D$ be a boundary section. Put $M = I_q/I_q^2$ and $N = M^{\vee} = q^*\mathcal{O}_{\bar{U}}(q)$. For a chosen peripheral subgroup $L_q = \kappa_q(T) \subset \Pi$, the normalizer $N_E(L_q)$ fits in an exact sequence

$$1 \longrightarrow T \longrightarrow N_E(L_q) \longrightarrow G \longrightarrow 1. \tag{C.5}$$

With the positive Kummer inertia convention of C3 its extension class is $c_1(N)$, and hence is $-c_1(M)$. More precisely, its image in $H^2(G, \mu_n)$ corresponds under the comparison map to the Kummer class of N in $H^2(S, \mu_n)$; when S is an étale $K(\pi, 1)$ this comparison is an isomorphism. For type $(0, 2)$ the same conclusion holds whenever the exact sequence of C4 holds (in particular

over the arithmetic $K(\pi,1)$ bases of C7): here $L_q = \Pi = T$, its arithmetic normalizer is all of E, and the two distinguished boundary sections identify U with the punctured normal line $N^\times$. Type $(1,0)$ has no cusp, so no normalizer extension of this kind is asserted there.

Proof. The conjugacy class of L_q is stable under G, because q is defined over S. For any $g \in G$, choose $e \in E$ over g and $b \in \Pi$ with $eL_qe^{-1} = bL_qb^{-1}$. Then $b^{-1}e$ lies in the normalizer and maps to g. Its kernel is $N_\Pi(L_q) = L_q$, by peripheral self-normalization (Stix, Corollary 12). This proves surjectivity and exactness, without asserting that a surjective homomorphism has a section.

The punctured formal neighborhood of q has the same tame-cover category as the punctured normal line $N^\times$. This statement means the following concrete comparison. On an étale open where a local equation z_i for q has been chosen, all tame covers are dominated by the Kummer covers $z_i^{1/n}$; on overlaps their identifications are the identifications of the corresponding roots of the normal transition functions. SGA 1 XIII, Proposition 5.2 gives this local description; descent of finite étale covers gives the gluing. It produces a homomorphism $\pi_1(N^\times) \to E$ whose image normalizes L_q, which is the identity on the base quotient and the peripheral injection on the fibre. In particular the fibre map into $\pi_1(N^\times)$ is injective, since its composite with E is κ_q. The homotopy sequence for $N^\times \to S$ is therefore short exact. The exact sequences consequently identify the tube extension with (C.5). This is also the identification of Stix, §10.3, with the line $q^*\mathcal{O}(q)$ [Sti08a].

Here is the sign computation with all conventions fixed. Choose frames e_i of N and write $e_j = g_{ij}e_i$. For a vector $v = e_iw_i = e_jw_j$, its scalar coordinates satisfy $w_i = g_{ij}w_j$. Choose local roots $h_{ij}^n = g_{ij}$ and write $r_i^n = w_i$; the transition of the root cover is $r_i = h_{ij}r_j$. On triple overlaps the defect is $h_{ij}h_{jk}h_{ki} \in \mu_n$. This is exactly the Kummer connecting cocycle $\delta_n([N])$. The deck transformation $r_i \mapsto \zeta r_i$ is the positive inertia element specified in C3. Thus the extension cocycle, computed by multiplying three local trivializations, has sign $+\delta_n(N)$. The dual transition functions are g_{ij}^{-1}, so $\delta_n(M) = -\delta_n(N)$. Stix, §6.4, makes the line-bundle extension/Chern comparison and its sign explicit. This computation resolves the manuscript's alternatives "conormal (ou normal, le diable sait)" mathematically; it does not claim to remove the author's written doubt by palaeography.

For example $N = \mathcal{O}_{\mathbb{P}^1}(-1)$ has degree -1 and $N^\times = \mathbb{A}^2 \setminus 0$ as computed in C4. Its homotopy boundary is the degree -1, agreeing with the displayed normal-line convention. A dual convention for the generator of inertia would reverse both signs, and must also reverse the Kummer character in C3. □

C7. The arithmetic splitting and completed-unit criteria *[Proved]*

[F14–F15; §3, manuscript pp. 40–42; scans 81–82, 85.] Let S be smooth connected of finite type over a field finitely generated over $\mathbb{Q}$, and suppose that S is an étale $K(\pi,1)$. Then $\mathrm{Pic}(S)$ is finitely generated, the integral continuous comparison in degree two is an isomorphism

$$H^2(G,T) \simeq H^2(S,T),$$

and Kummer theory gives

$$H^1(G,T) \simeq H^1(S,T) \simeq \widehat{\mathcal{O}(S)^*} := \varprojlim_n \mathcal{O}(S)^*/\mathcal{O}(S)^{*n}. \tag{C.6}$$

For the boundary section of C6, (C.5) splits if and only if M, equivalently N, is a trivial line bundle. A global basis of M is sufficient over any characteristic-zero base for which (C.5) exists. Here a "uniformizer along q" means a basis of I_q/I_q^2; the statement does not require a global equation of q on an entire neighborhood of its image.

Proof. Guralnick–Jaffe–Raskind–Wiegand, Proposition 6.1, states that the Picard group of a normal scheme of finite type over a finitely generated field, or over $\mathbb{Z}$, is finitely generated [GJRW96]. Its normality and arithmetic field hypotheses hold here. They cannot be deleted, for example Pic^0 of a positive-genus curve over an algebraically closed field is divisible and usually infinite.

For every positive integer n, the Kummer sequence gives

$$0 \to \mathcal{O}(S)^*/\mathcal{O}(S)^{*n} \to H^1(S,\mu_n) \to \mathrm{Pic}(S)[n] \to 0,$$

and an injection $\mathrm{Pic}(S)/n \hookrightarrow H^2(S,\mu_n)$. If N_0 annihilates the torsion of $\mathrm{Pic}(S)$, the transition from an index divisible by nN_0 to n kills the Picard torsion term; the transition on units modulo powers is onto. Thus the images in the system $H^1(S,\mu_n)$ eventually equal the unit subgroup, and this inverse system is Mittag–Leffler. The system $H^0(S,\mu_n)$ is also Mittag–Leffler, since its terms are finite. Taking a factorial cofinal sequence of indices, the derived-limit exact sequence for continuous cohomology has zero $\varprojlim^1$ in the degrees at issue. This is the sequence (0.2) of Jannsen [Jan88], with the Mittag–Leffler hypotheses just verified. It follows that $H^i(S,T) = \varprojlim_n H^i(S,\mu_n)$ for $i = 1, 2$ here.

For a $K(\pi,1)$ the finite local-system comparison $H^i(G,\mu_n) \simeq H^i(S,\mu_n)$ holds in every degree by its definition (equivalently by the contractibility of its universal étale homotopy cover). Applying the same inverse-limit sequence on both sides proves the asserted integral comparison. The inverse limit of $\mathrm{Pic}(S)[n]$, with its multiplication transition maps, is zero because its torsion has bounded exponent. This proves (C.6). In particular we have justified the limit operation in scan 82, including the potentially missing right exactness.

If the extension splits, its continuous class vanishes, hence all $c_1(N) \bmod n$ vanish. The Kummer injections imply $[N] \in n\,\mathrm{Pic}(S)$ for every n. In a finitely generated abelian group the intersection of these subgroups is zero, so N is trivial. Conversely a nonzero normal vector along the entire section defines its tangential fibre functor, or equivalently a section of $N^\times \to S$ followed by the tube comparison in C6. It splits (C.5). This construction does not use finite generation or the $K(\pi,1)$ assumption.

Surjectivity of the normalizer map alone does not imply splitting. For a concrete correction to scan 80, let $C/\mathbb{Q}$ be a smooth proper genus-two curve and use $S = C$, $\bar{U} = C \times C \to C$ and the diagonal as its single cusp. The normal line is T_C, of degree -2. Its Kummer class modulo 3 has nonzero degree in $\mathbb{Z}/3$; hence (C.5) does not split, although it surjects onto G. The curve $C : y^2 = x^5 - x$ supplies a specified example (the polynomial has no repeated root, and the smooth proper completion has genus two). □

C8. The geometric torsor inside the completed torsor *[Proved]*

[F15; §3, manuscript p. 42; scan 85.] Under the arithmetic hypotheses of C7, $\mathcal{O}(S)^*$ contains no nonidentity element divisible by every positive integer in that group. The completion map in (C.6) is injective. When N is trivial, the set of its nowhere-zero sections is a torsor under $\mathcal{O}(S)^*$. Its tangential splittings inject into the set of T-conjugacy classes of splittings of (C.5), a torsor under $H^1(G,T)$. Under these actions the map of torsors is induced by the injective Kummer map

$$\mathcal{O}(S)^* \hookrightarrow \widehat{\mathcal{O}(S)^*}.$$

For bases of the conormal line, rather than normal vectors, multiplying the basis by a changes this Kummer class by $-\kappa(a)$.

Proof. Let F be the function field of S, finitely generated over $\mathbb{Q}$, and let k be the algebraic closure of $\mathbb{Q}$ in F; it is a number field. Suppose $a \in F^*$ is an nth power in F for every $n > 0$. On a normal projective geometrically integral model of F/k, each prime-divisor valuation of a is

divisible by every positive integer and hence is zero. Consequently a and a^{-1} are global regular functions and $a \in k^*$. Every root of a lying in F is algebraic over k, and hence lies in k. At each finite prime of k, its valuation is again zero; thus $a \in \mathcal{O}_k^*$ and all its roots in k are units. Dirichlet's unit theorem (Milne, Theorem 5.1 [Mil20b]) makes this group finitely generated; its intersection of all power subgroups is the identity. Thus $a = 1$. Applying this to $\mathcal{O}(S)^* \subset F^*$ proves the first assertions. The same argument covers transcendence degree zero by starting directly with the number field.

For completeness, the classification of the splitting torsor is explicit. Fix a splitting $s : G \to N_E(L_q)$. Every other splitting has the unique form $s_c(g) = c(g)s(g)$ for a continuous crossed homomorphism $c : G \to T$. The homomorphism identity for s_c is $c(gh) = c(g) + gc(h)$, and conjugation by $b \in T$ adds the coboundary $b - gb$. Thus the classes form the $H^1(G, T)$-torsor asserted. If a normal frame is multiplied by a, the corresponding root frames differ by compatible roots of a; the cocycle is exactly $g \mapsto g(a^{1/n})/a^{1/n}$. Therefore its class is $\kappa(a)$. The kernel is the intersection of all power subgroups, already proved trivial. Dualizing a conormal frame replaces a by a^{-1} and changes the sign. It is the ordinary unit group, not its completion, that acts on actual algebraic frames. This is the mathematical correction supported by scan 85, where a possible circumflex remains a palaeographic uncertainty. □

C9. Divisorial failure of extension and tangential specialization *[Proved]*

[F16; §3, manuscript pp. 46–47; scans 93–95.] Let S be smooth integral in characteristic zero and let f_K be a rational section of a smooth pair as in C1. Let S' be its maximal open domain as a section of U. Every irreducible component of $S \setminus S'$ has codimension one. At the generic point of such a component D_α, the proper extension meets a unique cusp, and

$$\mu_\alpha = \operatorname{ord}_{D_\alpha}(f_K^* z) > 0$$

is its intersection multiplicity, independent of a unit multiple of the local cusp equation z. The sum $\sum_\alpha \mu_\alpha D_\alpha$ is an effective Cartier divisor because S is regular.

On an open subset W of the regular locus of D_α where the compactified section and the cusp are defined, and where a uniformizer t of D_α has been fixed, write $z \circ f = t^{\mu_\alpha} u$ with u regular and invertible along W. The resulting conormal isomorphism is

$$f_W^*(I_q/I_q^2) \xrightarrow{\sim} (I_{D_\alpha}/I_{D_\alpha}^2)^{\otimes \mu_\alpha}, \qquad z \longmapsto u\, t^{\mu_\alpha}. \tag{C.7}$$

It produces tangential specialization of the geometric section at every point of W, compatible with specialization inside W. Changing t to at replaces the leading coefficient by $a^{-\mu_\alpha} u$, and changes the associated Kummer class by $-\mu_\alpha \kappa(a)$ in the scalar-coordinate convention.

Proof. If an irreducible component of the complement had codimension at least two, remove the other components and work near its generic point. The section is then defined outside a codimension-at-least-two closed set, and C1 extends it, contradicting maximality. At a codimension-one generic point, properness extends f_K to $\bar{U}$. Its image cannot lie in U, again by maximality and finite presentation. The finite étale boundary has disjoint sections after an étale localization, so exactly one is met, and C3 gives the stated multiplicity. The divisorial sum is locally finite because the complement is a closed subset of the noetherian scheme S; on a regular scheme Weil divisors are Cartier. We do not identify it with the pullback of a global divisor by an everywhere-defined compactified rational map: such a map can still have a codimension-two indeterminacy in genus zero.

Where $z \circ f = t^\mu u$ with u a unit, reduction modulo the next power of I_D gives precisely (C.7). On the punctured normal lines the corresponding map is, in scalar coordinates, $v \mapsto uv^\mu$. It sends the source tangent vector with $t(v) = 1$ to the target tangent vector with $z(w) = u$. For every n,

pulling the Kummer cover of the target back by this map gives the μth-power Kummer cover multiplied by the unit u. These compatible maps of finite covers define the map of tangential fibre functors, hence the specialization of sections of the exact fundamental-group extensions. Evaluation at a point of W commutes with these pullbacks of covers, proving the asserted compatibility. The replacement $t' = at$ gives $u' = a^{-\mu}u$ directly, and Kummer functoriality gives the final formula. At a zero or pole of u, this construction no longer gives a nonzero tangent vector; this is exactly the regular-and-invertible restriction in the manuscript. □

4 D. Valuations and the birational quotient

Throughout this part, L/K is a finitely generated field extension of characteristic zero. We fix an algebraic closure $\overline{L}$, put $\overline{K} = \overline{L} \cap K^{\mathrm{alg}}$, and write $G_L = \mathrm{Gal}(\overline{L}/L)$ and $P_{L/K} = \mathrm{Gal}(\overline{L}/L\overline{K})$. A valuation v is trivial on K, and a chosen prolongation $\overline{v}$ to $\overline{L}$ fixes its decomposition group D_v and inertia group I_v as actual closed subgroups. Changing the prolongation conjugates both by G_L. We use additive value groups $\Gamma_v = v(L^\times)$ and residue fields $\kappa(v)$. The notation $T = \widehat{\mathbb{Z}}(1)$ means $\varprojlim_n \mu_n(\overline{K})$, with power transition maps.

The distinction between the initially discrete valuations of scans 117–120 and the arbitrary valuations of scans 121–127 changes several answers. We give explicit comparison homomorphisms throughout; no inference about a valuation is made merely from its rational rank.

D1. The model-independent quotient (F18) *[Known]*

Let X and Y be proper, regular, integral finite-type K-schemes with function field L. Their finite *étale* extensions of L form the same full subcategory of finite separable L-algebras. Consequently their quotients $G_L \twoheadrightarrow \pi_1(X)$ and $G_L \twoheadrightarrow \pi_1(Y)$ have the same kernel, with the generic geometric point fixed. The assertion also holds in positive characteristic whenever these proper regular models exist.

Passage: cote 149, §4, manuscript p. 57, scans 115–116, “ne dépend pas du modèle X choisi”.

Proof. We use purity in the form of Stacks, Lemma 58.21.4, Tag 0BMB [The]: for a quasi-finite morphism of locally noetherian schemes, a normal source local ring and a regular target local ring of the same positive dimension, unramifiedness at the generizations of source codimension one implies *étaleness* at the point under consideration. These are all the hypotheses used below. Let M/L be a connected finite *étale* cover of X. At a codimension-one point y of Y, properness of X extends the generic map $\operatorname{Spec} L \to X$ uniquely to $\operatorname{Spec} \mathcal{O}_{Y,y} \to X$. Pullback of the given cover is finite *étale*; its generic algebra is M. The normalization Y_M of Y in M is finite, because finite-type schemes over fields are excellent. Thus $Y_M \to Y$ is unramified at every codimension-one point. Purity makes it *étale* everywhere. Exchanging X, Y proves equality of the two categories. Their fiber functors are both the set of L-embeddings into $\overline{L}$, so their corresponding kernels in G_L agree. This proves equality as quotients, including the specified generic base point. □

D2. Riemann–Zariski points and the noetherian correction (F21) *[Proved]*

The inverse limit of all proper integral K-models of L, with its colimit local rings, has points exactly the valuation rings V of L containing K. This identification uses no resolution of singularities. Its noetherian local rings are exactly L and the discrete valuation rings of L/K. A discrete valuation ring need not be divisorial: there exists one for $\mathbb{Q}(x, y)/\mathbb{Q}$ with residue field $\mathbb{Q}$ and hence manuscript codimension two.

Passage: cote 149, §4, manuscript pp. 60–61, scans 121–123.

Proof. Products followed by the closure of the common generic point give a directed family of models. For a compatible point $a = (a_X)_X$, the union $V_a = \varinjlim_X \mathcal{O}_{X,a_X} \subset L$ is local and contains K. For $f \in L^\times$, take the closure of the graph of the rational map $f : X \dashrightarrow \mathbb{P}^1_K$. This is a proper model with a morphism to $\mathbb{P}^1_K$. At its center, either f or f^{-1} is regular. Therefore V_a is a valuation ring, and its fraction field is L. Conversely properness gives a unique center on each model to $\operatorname{Spec} V \to \operatorname{Spec} K$ and its generic identification with $\operatorname{Spec} L$. The centers are compatible. To prove that their colimit ring is exactly V, take $f \in V$ and use the same graph model. The map $\operatorname{Spec} V \to \mathbb{P}^1$ defined by f has its closed point in the finite affine chart, so f belongs to the local ring at the center on that graph. This proves both inverse identifications.

Normalizing models gives the same limit and does not change the proof. In a valuation ring a finitely generated ideal is principal, since any two principal ideals are comparable. A noetherian valuation domain is therefore a local principal ideal domain. For a nonfield let $\mathfrak{m} = (\varpi)$. Every nonzero element is a unit times a nonnegative power of ϖ: otherwise repeatedly dividing that element by ϖ produces a strictly increasing chain of principal ideals. Hence its value group is $\mathbb{Z}$ and it is a DVR. Conversely a DVR and a field are noetherian. For the counterexample let $h(t) = \sum_{j\geq 2} t^{j!} \in \mathbb{Q}[[t]]$, and put $x = t$, $y = t^2 + t^3 h(t)$. The series h is transcendental over $\mathbb{Q}(t)$: viewed on the complex unit disc, for every root of unity ζ the terms $\zeta^{j!}$ are eventually 1, so $h(r\zeta)$ tends to infinity as $r \uparrow 1$. An algebraic function has only finitely many branch points and poles on the complex projective line, whereas this gives infinitely many singularities. Thus $\mathbb{Q}(x, y)$ embeds in $\mathbb{Q}((t))$. Its restricted t-adic valuation has value group $\mathbb{Z}$, because $t = x$ belongs to the field, and residue field $\mathbb{Q}$, because every Laurent coefficient is rational. Its valuation ring is a DVR. Since $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(x, y) = 2$ and $\operatorname{trdeg}_{\mathbb{Q}} \kappa(v) = 0$, the codimension defined at scan 121 is two. This refutes the asserted exhaustion by the generic and divisorial points. □

D3. Explicit valuations with divisible value groups *[Proved]*

For K of characteristic zero and $L = K(x, y)$, the following valuations trivial on K exist:

$$(\Gamma_v, \kappa(v)) = (\mathbb{Q}, K), \qquad (\Gamma_v, \kappa(v)) = (\mathbb{Z}[1/2], K).$$

They have centers on the boundary of $(\mathbb{P}^1 \setminus \{0, 1, \infty\})^2$, and their centers can lie on only one boundary component. This construction will supply the corrections to F19, F20, F22 and F25.

Passages: cote 149, §4, manuscript pp. 59–65, scans 119–131.

Proof. Work in the Hahn field $K((t^{\mathbb{Q}}))$. Take $N_j = j!$ for $j \geq 1$, or take $N_j = 2^j$ for $j \geq 1$, respectively. Put

$$a_j = j + N_j^{-1}, \quad s_m = \sum_{j=1}^{m} t^{a_j}, \quad x = t, \quad y = 2 + \sum_{j\geq 1} t^{a_j}.$$

The support is well ordered and the sums belong to the Hahn field. Only the subfield of series whose supports are finite below each real bound is needed here: sums and products are defined coefficientwise, and the inverse of a series is obtained by factoring out its least monomial and using the geometric series, whose remaining terms have a positive least exponent. This verifies the required field operations directly. We prove directly that the value group of $K(t, y)$ is $\bigcup_m N_m^{-1}\mathbb{Z}$. For a fixed m, form

$$P_m(X, Y) = \operatorname{Norm}_{K(X^{1/N_m})/K(X)} \left(Y - 2 - \sum_{j=1}^{m} X^{a_j} \right) \in K(X)[Y].$$

In a splitting field this is a product indexed by N_m roots of unity. The partial sum s_m has N_m distinct conjugates: its last exponent has numerator coprime to N_m, and equality of Hahn series forces equality coefficient by coefficient. At $(X, Y) = (t, y)$ the identity factor has value a_{m+1}, while every other factor has value in $N_m^{-1}\mathbb{Z}$, since its first nonzero term occurs in the difference of two partial sums. Hence

$$v(P_m(t, y)) = a_{m+1} + c_m, \qquad c_m \in N_m^{-1}\mathbb{Z}.$$

Starting with $v(x) = 1$ and $v(y - 2) = a_1$, induction gives $1/N_m$ in the value group and then $1/N_{m+1}$ by subtracting $c_m + (m + 1)$. The reverse containment follows because every term of a rational expression in t, y has exponent in the union group. If y were algebraic over $K(t)$,

the fundamental inequality would bound the index of its value group over $\mathbb{Z}$ by $[K(t,y):K(t)]$, contradicting this unbounded index. Thus the indicated field is a rational function field in two variables. Its residue field is K, since all leading coefficients of its Hahn series belong to K. On the product compactification its center is $(0,2)$. The element x has positive value, while y and $y-1$ are units. Thus this center lies on precisely the boundary branch $x=0$. The same construction with y instead of $y-2$ gives a center at $(0,0)$. □

D4. Inertia, its ranks, and a splitting (F22) *[Known]*

For every valuation v of L/K there are a continuous exact sequence and a canonical pairing

$$1 \longrightarrow I_v \longrightarrow D_v \longrightarrow G_{\kappa(v)} \longrightarrow 1, \qquad I_v \simeq \operatorname{Hom}\bigl(\Gamma_v \otimes \mathbb{Q}/\Gamma_v, \mu_\infty\bigr) \simeq \operatorname{Hom}(\Gamma_v, T).$$

The sequence has a continuous splitting, with no canonical choice. The conjugation action on inertia is the cyclotomic action on the values in T. More precisely,

$$I_v \simeq \prod_\ell \mathbb{Z}_\ell(1)^{r_\ell}, \qquad r_\ell = \dim_{\mathbb{F}_\ell}(\Gamma_v/\ell\Gamma_v) \le \dim_{\mathbb{Q}}(\Gamma_v \otimes \mathbb{Q}) \le \operatorname{trdeg}_K L - \operatorname{trdeg}_K \kappa(v).$$

The product decomposition is noncanonical. Neither the manuscript codimension nor, in general, the rational rank determines the inertia group. In particular $\Gamma_v = \mathbb{Q}$ gives $I_v = 1$, and $\Gamma_v = \mathbb{Z}[1/2]$ gives $I_v = \prod_{\ell \neq 2} \mathbb{Z}_\ell(1)$.

Passage: cote 149, §4, manuscript p. 61, scan 124, “des T^α_∞ essentiellement” and “scinder”.

Proof. The fixed field of D_v is the chosen henselization L_v^h. Reduction of finite *étale* extensions of its valuation ring gives the quotient $G_{\kappa(v)}$ and its kernel I_v. The fixed field $F = \overline{L}^{I_v}$ is strictly henselian, has residue field $\kappa(v)^{\mathrm{sep}}$, and has value group Γ_v. The precise tame input is Temkin, Theorem 3.4.4 and Corollary 3.5.3 [Tem]: over a strictly henselian field, a value-group enlargement with quotient of order prime to the residue characteristic determines the corresponding tame extension and its character group; wild inertia is a pro-p group in residue characteristic p, and is trivial in residue characteristic zero. Thus all algebraic extensions of F are obtained by adjoining the roots of elements representing the divisible hull of Γ_v. For $a \in F^\times$, $a^{1/n} \in \overline{L}$, and $\sigma \in I_v$, the value

$$\langle \sigma, v(a)/n \bmod \Gamma_v \rangle = \sigma(a^{1/n})/a^{1/n} \in \mu_n$$

is independent of a and the root: units in F have all nth roots in F by the simple-root form of Hensel’s lemma. These finite Kummer pairings give the first character-group identification. Sending a character c to the compatible functions $\gamma \mapsto c(\gamma/n \bmod \Gamma_v)$ gives the second one. This explicitly types the comparison with T. The torsion group $\Gamma_v \otimes \mathbb{Q}/\Gamma_v$ is divisible, and its ℓ-torsion is $\Gamma_v/\ell\Gamma_v$ via $\gamma \mapsto \gamma/\ell$. The classification of divisible torsion abelian groups, obtained by successively choosing compatible ℓth roots of a basis of the ℓ-torsion, therefore gives the displayed ℓ-adic factors. Linear independence modulo ℓ implies rational linear independence after clearing denominators and canceling a maximal common power of ℓ, proving the first bound. For the last bound, choose elements with rationally independent values and units whose residues are algebraically independent. Any polynomial relation in them has a unique least-value collection of monomials, and the coefficient of this least value remains a nonzero polynomial in the independent residues. This proves their algebraic independence and the bound. We include a direct construction of a complement. Among algebraic extensions of L_v^h in $\overline{L}$ having unchanged residue field, choose a maximal one M by unions of chains. Its value group is divisible. Indeed, a value $\gamma \notin \ell\Gamma_M$ gives a degree-ℓ extension by an ℓth root of an element of value γ; its ramification index is already ℓ, so its residue extension is trivial, contradicting maximality. The field M is henselian. Every finite extension of a henselian field of residue characteristic zero has degree equal to its ramification index times residue degree; this is the characteristic-zero case of the

defect formula, also recorded as Corollary 2.9 of [Kuh14]. Its maximal unramified extension has divisible value group and algebraically closed residue field, and hence admits no further finite extension. Therefore $G_M \to G_{\kappa(v)}$ is an isomorphism. Its inverse, followed by $G_M \hookrightarrow D_v$, is the required section. This is also the characteristic-zero instance of Haran–Jarden–Pop, Proposition 7.3(a) [HJP02]. Every element of D_v fixes the value of every element of L; applying it to the Kummer quotient above changes only the root of unity. This proves the cyclotomic action and also proves $I_v \subset P_{L/K}$. □

D5. The exact inertia normalizer for arbitrary valuations (F19) *[Proved]*

Let $I_v \neq 1$. Let Δ_v be the largest divisible convex subgroup of the ordered group Γ_v, and let w be the coarsening with value group Γ_v/Δ_v, with its prolongation induced by $\overline{v}$. Then

$$I_w = I_v, \qquad N_{G_L}(I_v) = D_w, \qquad N_{P_{L/K}}(I_v) = D_w \cap P_{L/K}.$$

There is an exact sequence

$$1 \longrightarrow I_v \longrightarrow N_{P_{L/K}}(I_v) \longrightarrow G_{\kappa(w)\overline{K}} \longrightarrow 1.$$

Here $\kappa(w)\overline{K}$ is the compositum inside the residue algebraic closure induced by $\overline{v}$. Consequently geometric self-normalization holds precisely when that compositum is separably closed. The arithmetic identity $N_{G_L}(I_v) = D_v$ holds precisely when the induced residual valuation v/w is henselian on $\kappa(w)$. It holds in particular for every rank-one valuation with nontrivial inertia, including every DVR. For $I_v = 1$ the inertia normalizer is all of G_L.

Passages: cote 149, §4, manuscript pp. 59 and 63, scans 119–120 and 127.

Proof. Convex subgroups form a chain, and a union of divisible subgroups in that chain is divisible, so Δ_v exists. For a convex subgroup $\Delta \subset \Gamma_v$ the quotient is torsion free. The resulting exact sequence of divisible torsion groups, followed by character duality, gives

$$1 \longrightarrow I_{v/\Delta} \longrightarrow I_v \longrightarrow \mathrm{Hom}(\Delta, T) \longrightarrow 1.$$

The first arrow is the actual subgroup inclusion of the inertia of the coarsening: in the Kummer pairing it is pullback along $\Gamma_v \to \Gamma_v/\Delta$. In particular it is an equality precisely when Δ is divisible. This proves $I_w = I_v$. Set $F = \overline{L}^{I_v}$. It is henselian for the induced valuation, has separably closed residue field, and value group Γ_v. A coarsening of that valuation has separably closed residue field precisely when its convex kernel is divisible. To verify this assertion, the residual refinement is henselian of characteristic zero, has separably closed final residue field and value group that convex kernel; D4 identifies its absolute Galois group with the character group of that kernel. Thus $w|F$ is the coarsest henselian valuation on F with separably closed residue field. We justify "coarsest" among all such valuations, not merely those already in the displayed chain. The exact input is the F. K. Schmidt theorem in Kuhlmann, Theorem 9.44 and Corollary 9.45 [Kuh]: two independent nontrivial henselian valuations force a field to be separably closed; for two incomparable henselian valuations the residue field of their finest common coarsening is separably closed. Given another henselian valuation u of F with separably closed residue field, incomparability with $w|F$ would give a strictly coarser such valuation, contrary to the construction. Comparability then shows that u refines $w|F$. Since $I_v \neq 1$, F is not separably closed, and this coarsest valuation is nontrivial. An element $g \in N_{G_L}(I_v)$ acts on F as an L-automorphism. The just-proved intrinsic description makes it preserve $w|F$. Henselianity gives a unique extension of $w|F$ to $\overline{L}$, so $g \in D_w$. Conversely D_w normalizes $I_w = I_v$ by the definition of inertia. This proves the arithmetic formula, and intersection with the explicitly specified subgroup $P_{L/K}$ proves the geometric formula. Reduction identifies the kernel of $D_w \to G_K$ with the inverse image of $G_{\kappa(w)\overline{K}}$; its kernel in that sequence is I_v. Finally D_v is the inverse image

in D_w of the decomposition group of v/w on $\kappa(w)$. That subgroup is the whole residue Galois group exactly when the residual valuation is henselian. Rank one has only the convex subgroups $0, \Gamma_v$; nontrivial inertia excludes the divisibility of the latter, so then $w = v$. □

D6. Both unqualified self-normalization assertions (F19) *[Refuted]*

The geometric assertion $N_{P_{L/K}}(I_v) = I_v$ is false for the x-adic valuation of $L = \mathbb{Q}(x, y)$ over $K = \mathbb{Q}$. The arithmetic assertion $N_{G_L}(I_v) = D_v$ is false for general valuations with $I_v \neq 1$, already for $L = \mathbb{Q}(x, y, z)$. D5 is the exact replacement.

Passages: cote 149, §4, manuscript pp. 59 and 63, scans 119–120 and 127.

Proof. For the first example $\Gamma_v = \mathbb{Z}$ and $\kappa(v) = \mathbb{Q}(y)$. The last quotient in D5 is $G_{\overline{\mathbb{Q}}(y)}$, which has the nontrivial quadratic extension obtained by adjoining $\sqrt{y}$. Thus the normalizer strictly contains I_v. For the second let w be the x-adic valuation of $\mathbb{Q}(x, y, z)$, and on $\kappa(w) = \mathbb{Q}(y, z)$ take the valuation r of D3 with $\Gamma_r = \mathbb{Q}$, $\kappa(r) = \mathbb{Q}$, $r(y) > 0$. Compose w and r to obtain v. Its value group is $\mathbb{Z} \oplus_{\text{lex}} \mathbb{Q}$; the divisible convex subgroup is the second factor. Therefore $I_v = I_w = T$ and $N_{G_L}(I_v) = D_w$. The residual decomposition group D_r is proper. Indeed $Z^2 - (1 + y)$ is irreducible over $\mathbb{Q}(y, z)$, and its two roots have residues $+1, -1$ under the two prolongations of r. They give different valuations on this quadratic field: the root minus 1 has positive value for the first and value zero for the second. Thus D_r has trivial image in this quadratic quotient. Since D_v is the inverse image of D_r in D_w, it is a proper subgroup. These are actual unequal subgroups for compatible choices of prolongations, with the same nonzero inertia subgroup. □

D7. Detecting a center: the DVR assertion and its arbitrary-valuation extension (F20) *[Refuted]*

For a DVR V of L/K and an elementary tower U/K of smooth curves with good relative compactifications and no fiber of type $(0, 0)$ or $(0, 1)$, the homomorphism $I_v \to \pi_1(U)$ is trivial precisely when V has a center on U. For an arbitrary valuation, having a center still implies that this homomorphism is trivial, but its converse is false even for $U = (\mathbb{P}^1 \setminus \{0, 1, \infty\})^2$. Outside the indicated curve-type hypothesis the converse fails already for $\mathbb{A}^1$ and a DVR.

Passage: cote 149, §4, manuscript p. 59, scan 120, “A t-on une réciproque?”.

Proof. For every valuation with a center, the morphism from its generic point to U extends to $\operatorname{Spec} V$. After strict henselization its inertia homomorphism factors through $\pi_1(\operatorname{Spec} V^{\text{sh}}) = 1$. This proves the implication without a discreteness hypothesis. For a DVR, make this same strict henselian base change and apply the trait criterion of C2 successively along the tower. If the composite generic map has trivial fundamental-group image, its projection to the next stage has trivial image, so it extends there by induction. Relative to this extended base map, C2 extends the generic map to the last curve. For hyperbolic fibers this is exactly Hoshi’s Lemma 2.8 [Hos14]; the $(0, 2)$ case uses Kummer theory, and the proper $(1, 0)$ case uses the valuative criterion of properness, as proved in C2. Uniqueness by separatedness descends the extended map from the strict henselization to V: the two pullbacks agree at the generic point and therefore agree everywhere. All the homomorphisms in this induction are those induced by the displayed tower projections and base changes. Now use the valuation with group $\mathbb{Q}$ in D3. Its center on $(\mathbb{P}^1)^2$ is $(0, 2)$, outside U, and uniqueness of centers on the proper separated compactification rules out a center on U. But D4 gives $I_v = 1$, so the homomorphism is trivial. Finally the valuation at infinity of $K(t)$ has no center on $\mathbb{A}^1_K$, while its geometric inertia dies in $\pi_1(\mathbb{A}^1_K) = G_K$ in characteristic zero. This is the promised counterexample without the curve-type restriction. □

D8. Restriction of valuation rings and of local Galois groups (F23) *[Proved]*

Let $K \subset L_1 \subset L$ and $V_1 = V \cap L_1$. Then V_1 is a valuation ring with fraction field L_1, and reduction embeds $\kappa(V_1)$ into $\kappa(V)$. With compatible algebraic closures and prolongations, restriction gives continuous homomorphisms

$$D_V \longrightarrow D_{V_1}, \qquad I_V \longrightarrow I_{V_1}.$$

Their existence does not assert that either homomorphism is injective or surjective. If L_1/K has transcendence degree one and $\kappa(V)$ is algebraic over K, V_1 is the DVR of a closed point on the proper smooth curve with function field L_1.

Passage: cote 149, §4, manuscript p. 62, scans 125–126.

Proof. For $a \in L_1^\times$, either $a \in V$ or $a^{-1} \in V$; these alternatives take place in V_1. Moreover either $a = a/1$ or $a = 1/a^{-1}$ is a quotient of elements of V_1, proving its fraction-field assertion directly. The maximal ideal is $\mathfrak{m}_V \cap V_1$, so the residue map is injective. Inside $\overline{L}$, let $\overline{L_1}$ be the algebraic closure of L_1. An automorphism preserving the prolonged valuation on $\overline{L}$ preserves its restriction. An inertia automorphism acts identically on the larger residue field and hence on the residue subfield of $\overline{L_1}$. This proves the two homomorphisms, and their continuity is continuity of restriction in the Krull topology. In transcendence degree one, a nontrivial valuation trivial on K has its center at a closed point of the proper smooth curve. The local ring there is a DVR, and any valuation ring of the same function field dominating a DVR is either that DVR or the field; the latter has no closed-point center. The valuation cannot be trivial because its residue field would then be L_1, which does not embed over K in the algebraic field $\kappa(V)$. This proves the final assertion. □

D9. The written function-field formula at scan 127 (F24) *[Refuted]*

Assume here that K is finitely generated over $\mathbb{Q}$. The two assertions (32)–(33) with subscript L have the same truth value, but their common assertion of triviality is false. For $L = \mathbb{Q}(t)$, the t-adic place, and a decomposition section $s : G_{\mathbb{Q}} \to D_v \subset G_L$, both fixed groups are $I_v = T \neq 1$. Under the quotient to $\pi_1(U)$ for $U = \mathbb{P}^1 \setminus \{1, 2, \infty\}$, this section becomes the point section at 0 and both fixed groups become trivial. For point sections of elementary anabelian U the corresponding U-formulas hold.

Passage: cote 149, §4, manuscript p. 63, scans 127–128, (32)–(33).

Proof. We use the attached return [GPT26a, Statements B6–B8 and C6], whose maps and arguments were reread here. In B6 the quotient $G_L \to G_K$ sends any element centralizing $s(G_K^\circ)$ into $C_{G_K}(G_K^\circ) = 1$, so the arithmetic centralizer equals its intersection with $P_{L/K}$. This proves the equivalence using the actual inclusion $P_{L/K} \hookrightarrow G_L$. In B7 the compatible inertia groups on the hyperbolic open curves in the function field inverse system identify the centralizer of a decomposition section with I_v; B8 applies it to the explicit point above. Nontriviality is witnessed already by $\sqrt{t}$, whose inertia quotient is $\mathbb{Z}/2$. The continuous quotient $q_U : G_L \twoheadrightarrow \pi_1(U)$ kills I_v, because 0 is an interior point. Its image section is the point section at 0, whose centralizer is trivial by C6, proved there by induction through the fiber inclusions and tower projections. We do not infer the reverse centralizer implication under q_U. The scan was also visually inspected: it supports the written L in both formulas. The mathematics corrects the assertion to its model quotient in this context; it does not justify changing the transcription silently. □

D10. Normalizers of decomposition groups (completion of F19) *[Proved]*

Put $H = L_v^h$. If $\kappa(v)$ is not separably closed, then

$$N_{G_L}(D_v) = D_v.$$

If $\kappa(v)$ is separably closed, then $D_v = I_v$; its normalizer is D_w as in D5 when $I_v \neq 1$, and is G_L when $I_v = 1$. In all cases $N_{P_{L/K}}(D_v) = N_{G_L}(D_v) \cap P_{L/K}$. For the geometric decomposition subgroup $D_v^{\mathrm{geom}} = D_v \cap P_{L/K}$, the same alternatives hold with $G_L, L, \kappa(v)$ replaced by $P_{L/K}, L\overline{K}, \kappa(v)\overline{K}$: it is self-normalizing when this last residue field is not separably closed; otherwise it equals I_v and D5 gives its normalizer.

Passages: cote 149, §4, manuscript pp. 59 and 63, scans 119–120 and 127.

Proof. An element of $N_{G_L}(D_v)$ restricts to an L-automorphism of H. The given valuation on H is henselian and immediate over L: its value group and residue field are Γ_v and $\kappa(v)$. The transported valuation under that automorphism is another henselian valuation, with the same restriction to L. If the two valuations were incomparable, Corollary 9.45 of [Kuh] would make the residue field of their finest common coarsening separably closed. The residue of every valuation of a separably closed field in characteristic zero is algebraically closed: a monic polynomial over the residue field lifts to a monic polynomial over the valuation ring, splits in the algebraically closed field, and its integral roots reduce to a splitting. The residue field $\kappa(v)$ of the refinement would therefore be separably closed, a contradiction. The two valuations are thus comparable. A proper coarsening of the valuation on H kills a nonzero convex subgroup of Γ_v; because H/L is immediate, that subgroup already contains values of elements of L. It cannot have the same restriction to L. Therefore the two valuations coincide. Uniqueness of prolongation of a henselian valuation to $\overline{L}$ puts the normalizer element in D_v. When $\kappa(v)$ is separably closed the decomposition–inertia exact sequence in D4 gives $D_v = I_v$, proving the other alternatives by D5. Restriction of the chosen valuation to $L\overline{K}$ has unchanged value group and residue field $\kappa(v)\overline{K}$. Its decomposition group in $\mathrm{Gal}(\overline{L}/L\overline{K})$ is exactly $D_v \cap P_{L/K}$ by the definition as a stabilizer. Applying the proof just given to that field proves the geometric assertion, with all subgroup comparisons explicit. □

5 E. Boundary meridians and the homotopy obstruction

We write X for a smooth proper compactification of a smooth U in characteristic zero, and $D = X \setminus U$ for a simple normal crossings divisor. At a geometric point $\bar{x}$ on exactly r branches, choose regular parameters $z_1, \dots, z_r$ defining them and a compatible geometric generic point. All meridian homomorphisms below are genuine continuous homomorphisms with these choices; changing the connecting path conjugates their images in $\pi_1(U)$.

E1. The local group and the two local injectivity assertions (F25, 1°) *[Proved]*

For the strict henselization $R = \mathcal{O}_{X,\bar{x}}^{\mathrm{sh}}$ there is a parameter-labelled isomorphism

$$\pi_1\big(\operatorname{Spec} R[1/(z_1 \cdots z_r)]\big) = T^r.$$

Its ith factor is the meridian of D_i, characterized by its action on compatible roots of z_i. For an ordinary point x, the henselian local group fits into

$$1 \longrightarrow T^r \longrightarrow \pi_1\big(\operatorname{Spec} \mathcal{O}_{X,x}^h \cap U\big) \longrightarrow G_{\kappa(x)} \longrightarrow 1.$$

When $\kappa(x)/K$ is finite, injectivity of the entire local homomorphism into $\pi_1(U)$ is equivalent to injectivity of its restriction to T^r. This equivalence does not hold for general nonclosed points. If

U is a hyperbolic elementary polycurve, each individual branch meridian $T \to \pi_1(U)$ is injective, so the geometric local homomorphism is injective whenever $r = 1$.

Passage: cote 149, §4, manuscript pp. 64–65, scans 130–131, the two versions of 1°.

Proof. Adjoining all Nth roots of the z_i gives a finite cover of the punctured local scheme with group $(\mu_N)^r$. The corresponding ring over R is regular local and strictly henselian: the adjoined roots replace the first r members of a regular system of parameters, and the residue field is unchanged. Conversely, take a connected finite *étale* cover of the punctured scheme and normalize R in its function field. This normalization is finite because R is excellent. The possible codimension-one ramification is confined to the $z_i = 0$. Choose N divisible by the inertia exponents of its finite Galois closure at all those divisors. D4, applied to their henselian DVRs, shows that adjoining the indicated roots kills that ramification: its character pairing becomes restriction along $\mathbb{Z} \hookrightarrow N^{-1}\mathbb{Z}$, and the finite inertia quotient becomes trivial. The normalization after this base change is unramified in codimension one. Purity, with the regular target and normal finite source of D1, makes it *étale* everywhere, and strict henselianity makes it a disjoint union of copies of the base. Thus these Kummer covers are cofinal, proving the displayed group and its specified generators. Descent from the strict henselization to the henselization gives the second exact sequence. The composite of the local homomorphism with $\pi_1(U) \to G_K$ is reduction $G_{\kappa(x)} \to G_K$. For a finite residue extension this last map is the injective restriction map associated to the chosen embedding $\kappa(x) \subset \overline{K}$. A kernel element of the entire local map consequently lies in T^r, proving the equivalence. For the one-branch assertion, take the DVR at the generic point of that branch. Follow its generic map through the hyperbolic tower and let $U_i \to U_{i-1}$ be the first stage to which the map does not extend, although its projection extends. Its good relative compactification gives a boundary section after an unramified residue extension. C3 identifies the inertia homomorphism, followed by $\pi_1(U) \to \pi_1(U_i)$, with multiplication by a positive intersection multiplicity μ followed by that cusp's inertia embedding into the fiber group. The latter embedding is injective by the hyperbolic loop theorem of A, and the fiber group embeds into $\pi_1(U_i)$: Hoshi, Proposition 2.4(i), gives this exact sequence for a hyperbolic curve over a connected normal locally noetherian base [Hos14]. Multiplication by μ is injective on T. Thus the original divisorial inertia is injective. The ith local coordinate factor above identifies with this inertia by the common Kummer character of z_i, proving the assertion at every $\bar{x}$. Finally the need for the finite-residue hypothesis is explicit on a product $U = C_1 \times C_2$ of affine hyperbolic curves. Take x to be the generic point of the divisor $\{c\} \times \overline{C}_2$, where c is a K-rational cusp of C_1. The residue field is $K(C_2)$. Product coordinates and a tangential parameter at c give a section of the local group over $G_{K(C_2)}$ whose global image in $\pi_1(U)$ is the tangential section in the first factor and $G_{K(C_2)} \to \pi_1(C_2)$ in the second. At an interior rational point of C_2, the latter homomorphism kills its nonzero inertia group, and its image in G_K is also trivial. Its image under the displayed local section is therefore a nontrivial kernel subgroup of the whole local homomorphism. The local meridian T still embeds in the first curve factor. Taking $C_1 = C_2 = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ supplies all rational points and parameters used here. □

E2. The blow-up map, and the product compactification (F25, 1°) *[Refuted]*

For $U = (\mathbb{P}^1 \setminus \{0, 1, \infty\})^2$, blow up $(0, 2)$ in $(\mathbb{P}^1)^2$. At the intersection of the exceptional divisor with the strict transform of $x = 0$, the local homomorphism is

$$T^2 \longrightarrow T \hookrightarrow \pi_1(U), \qquad (a, b) \longmapsto a + b.$$

Its kernel is $\{(a, -a) : a \in T\}$. Thus 1° is false even for an elementary hyperbolic surface. On the original product compactification of any two smooth hyperbolic curves, every geometric local meridian homomorphism is injective. That product compactification is terminal among

smooth proper compactifications of the same U with simple normal crossings boundary: every such compactification has a unique morphism to it inducing the identity on U.

Passage: cote 149, §4, manuscript pp. 64–65, scans 130–131, the blow-up margin and "Minimalité".

Proof. Use coordinates x and $y-2$ at the center. In the chart $x = uv$, $y-2 = v$, the strict transform is $u = 0$ and the exceptional divisor is $v = 0$. A positive loop in either u or v gives one positive loop in x; the coordinate $y = 2+v$ lies in a disc about the interior point 2 and has zero image in $\pi_1(C_2)$. Thus the actual homomorphism on Kummer characters is addition in the first cusp group and zero in the second. The latter cusp group is embedded, so the kernel is exactly the stated antidiagonal. The same computation over $\mathbb{C}$ uses ordinary positive meridians, and its compatible finite cyclic quotients are the displayed profinite formula. On $\overline{C}_1 \times \overline{C}_2$, at most one boundary branch comes from each factor. The product isomorphism $\pi_1(U_{\overline{K}}) \simeq \pi_1((C_1)_{\overline{K}}) \times \pi_1((C_2)_{\overline{K}})$ sends each local meridian into the corresponding cusp factor and the other coordinate to the identity. Since each cusp factor embeds, their product embeds. For terminality let Y be another such compactification. Each projection $U \to C_i \subset \overline{C}_i$ defines a rational map $Y \dashrightarrow \overline{C}_i$. For $g(\overline{C}_i) \geq 1$, the extension theorem C1 applies; its proof resolves the rational map by blow-ups of points and contracts each exceptional rational tree because a positive-genus curve admits no nonconstant map from $\mathbb{P}^1$. Thus this rational map extends to Y. For genus zero, make a finite extension of K splitting its at least three omitted points and identify the target with $\mathbb{P}^1$. At an alleged base point of the rational map, write it as $[a : b]$ in the two-dimensional regular local ring of Y, with a, b coprime and both in its maximal ideal. For three distinct omitted values, the three local equations $a - cb$ (and b for the value at infinity) are nonunits with pairwise disjoint sets of irreducible factors. Their zeros all lie in $Y \setminus U$, because the map on U avoids the three values. They give at least three distinct boundary branches through that point. Simple normal crossings on a smooth surface permit at most two. This contradiction eliminates every base point. The extension is unique by separatedness, and therefore descends from the finite field extension. The two extended projections define $Y \to \overline{C}_1 \times \overline{C}_2$, a proper birational morphism equal to the identity on U. Uniqueness follows because U is dense and the target is separated. This is the precise minimality property proved here. □

E3. The image of an arbitrary valuation at a crossing (F25, 2°) *[Proved]*

Let a valuation v have center x on X, let the branches through x have parameters $z_1, \dots, z_r$, and put $\gamma_i = v(z_i)$. The homomorphism from valuation inertia to the local geometric group is

$$\rho_v : \mathrm{Hom}(\Gamma_v, T) \longrightarrow T^r, \qquad \chi \longmapsto (\chi(\gamma_1), \dots, \chi(\gamma_r)).$$

It is dual to the homomorphism of discrete torsion groups

$$\psi_v : \mathbb{Q}^r/\mathbb{Z}^r \longrightarrow (\Gamma_v \otimes \mathbb{Q})/\Gamma_v, \qquad (q_i) \longmapsto \sum_i q_i\gamma_i \pmod{\Gamma_v}.$$

Consequently $\operatorname{im}\rho_v$ is open exactly when $\ker\psi_v$ is finite, and it is all of T^r exactly when $\ker\psi_v = 0$. More explicitly, openness requires rational independence of the γ_i. Under that independence, with $\Lambda = \sum_i \mathbb{Z}\gamma_i$,

$$\ker\psi_v = (\Gamma_v \cap \operatorname{span}_{\mathbb{Q}}\{\gamma_i\})/\Lambda.$$

Thus openness is exactly finite saturation index, and surjectivity is exactly equality of this saturation with Λ. Equality of rational rank and number of branches alone is insufficient.

Passage: cote 149, §4, manuscript pp. 64–65, scans 130–131, 2° and the monomial example (36).

Proof. The root $z_i^{1/n}$ defines the ith cyclic local Kummer cover. On its pullback to the valuation field, the action of $\sigma \in I_v$ is $\sigma(z_i^{1/n})/z_i^{1/n} = \langle \sigma, \gamma_i/n \bmod \Gamma_v \rangle$. This proves the formula in every finite quotient and hence in the inverse limit. Replacing a parameter by a unit times that parameter does not change the formula: its unit has value zero and an nth root in the strictly henselian valuation ring. This also proves independence of the chosen regular equations, with only the specified branch labels retained. The character group μ_∞ is divisible, so a character of a subgroup of a discrete torsion abelian group extends to the whole group. Taking such characters is therefore an exact contravariant operation. It identifies the image of ρ_v with the annihilator of $\ker \psi_v$ and identifies $T^r / \operatorname{im} \rho_v$ with its character dual. The image is compact and closed. Its quotient is finite precisely when that kernel is finite, and trivial precisely when that kernel is zero, proving the first criteria. If the values have a nontrivial integer relation, its rational span maps to zero and supplies an infinite divisible subgroup of $\ker \psi_v$; the image cannot be open. Under rational independence the rational map $\mathbb{Q}^r \to \Gamma_v \otimes \mathbb{Q}$ is injective. Taking the preimage of Γ_v and quotienting by $\mathbb{Z}^r$ gives exactly the displayed saturation quotient. For a DVR, choose the positive generator of $\Gamma_v = \mathbb{Z}$ and write $\gamma_i = m_i > 0$. The formula is $a \mapsto (m_1 a, \dots, m_r a)$. For $r = 1$ its index is m_1 and it is surjective exactly when $m_1 = 1$; for $r > 1$ it is never open. This includes nondivisorial DVRs: D2 at $(0,0)$ has $(m_1, m_2) = (1, 2)$. For a monomial valuation whose independent parameter weights generate its value group, the selected boundary weights are part of a basis, and ρ_v is surjective. More generally, for a finitely generated monomial value lattice the full-rank integer weight matrix gives an open image; its finite index is the product of its Smith invariants, and surjectivity says all those invariants are one. Finally take the valuation with $\Gamma_v = \mathbb{Z}[1/2]$ of D3 at $(0,2)$. Here $r = 1$, $\gamma_1 = 1$, and $\ker \psi_v = \mathbb{Z}[1/2]/\mathbb{Z}$ is infinite. The inertia image in T is exactly $\prod_{\ell \neq 2} \mathbb{Z}_\ell(1)$, by D4 and evaluation at 1; it is closed and not open. The rational rank and branch count are both one. Taking $\Gamma_v = \mathbb{Q}$ instead makes the same boundary inertia map zero. These examples decide the general-valuation assertion, including its rational-rank shortcut. □

E4. The precise criterion for the stratum extension (F26) *[Proved]*

Work first over an algebraically closed field of characteristic zero. For a nonempty connected open stratum

$$S = D_J^\circ = \Big(\bigcap_{j \in J} D_j \Big) \setminus \Big(\bigcup_{i \notin J} D_i \Big), \qquad L_j = N_{D_j/X}|_S,$$

let $P = \prod_{j \in J,S} L_j^\times$ be the product of the punctured normal line bundles. Its fundamental groups have the exact homotopy sequence

$$\pi_2^{\mathrm{et}}(S) \xrightarrow{\partial} T^J \longrightarrow \pi_1(P) \longrightarrow \pi_1(S) \longrightarrow 1.$$

The same sequence computes the punctured tubular neighborhood of this stratum. With positive fiber meridians and the clutching convention specified below,

$$\partial(a) = \big(\langle c_1(L_j), a \rangle \big)_{j \in J}.$$

In particular (37) is left exact precisely when these Chern evaluations vanish. An equivalent criterion, involving only finite covers and line bundles, is: for every positive integer n there is a connected finite *étale* cover $S' \to S$ such that every $L_j|_{S'}$ is an nth tensor power in $\operatorname{Pic}(S')$. Vanishing of $\pi_2^{\mathrm{et}}(S)$ suffices, but is not necessary. Over a nonclosed field the same assertion descends with the cyclotomic action and the permutation of geometric branch labels.

Passage: cote 149, §4, manuscript pp. 66–67, scans 133–135, (37) and its qualification.

Proof. The bundle $P \to S$ is a Zariski locally trivial torsor under $\mathbb{G}_m^J$. The exact sequence displayed above is the geometric-fibration sequence explicitly used in the proof of Borne–Emsalem–Stix, Proposition 8 [BES15]; its hypotheses are characteristic zero, a geometrically connected geometrically unibranch base, and a torsor under a torus. Our smooth stratum satisfies them. Their proof identifies the fiber fundamental group with T^J. Their Proposition 16 and its proof identify the transgression with the Kummer first Chern class, including its finite-coefficient versions; the vanishing assumption of that proposition is used there for the cohomology isomorphism, whereas the transgression computation by transition functions does not use it. We give the transition computation and an independent finite-cover proof of the criterion. Choose trivializations of L_j on a cover of S. Their transition functions $g_{ab,j}$ define the line bundles. For a sphere mapped into the complex analytic stratum, trivialize each pullback on its northern and southern discs. We orient the equator as the boundary used when comparing the northern lift with the southern lift; the clutching function has winding number $\langle c_1(L_j), [S^2]\rangle$. The difference of the lifted boundaries is precisely that winding number of the positive fiber loop. This fixes the sign convention and proves the displayed formula. Algebraically, choose local nth roots $h_{ab,j}$ of $g_{ab,j}$. The triple-overlap products $h_{ab,j}h_{bc,j}h_{ca,j} \in \mu_n$ are both the obstruction to lifting transition functions through $[n] : \mathbb{G}_m \to \mathbb{G}_m$ and the Kummer cocycle $c_1(L_j)$ mod n. Evaluation on the universal *étale* cover gives the same boundary modulo n. Since all n are used, this proves the formula for T^J, with no replacement by a single prime or by an uncompleted group. Here is a proof of the finite-cover criterion that also checks its necessity. Suppose $A = T^J$ injects into $G = \pi_1(P)$. For a given n, compactness and separation in the profinite group give an open normal subgroup N of G with $N \cap A \subset nA$. Put $Q = G/N$. The quotient $Q/\operatorname{im}(A)$ is a finite quotient of $\pi_1(S)$. The associated connected cover $S' \to S$ makes the image of $\pi_1(P_{S'})$ in Q lie in $\operatorname{im}(A)$. Compose with its quotient A/nA. We obtain a finite *étale* $(\mu_n)^J$-torsor on $P_{S'}$ restricting to the standard nth-power cover on every torus fiber. On an *étale* cover of S' where the line bundles and this cover trivialize, its coordinates are $w_{a,j}^n = z_{a,j}$. On overlaps $w_{a,j} = h_{ab,j}w_{b,j}$ with $h_{ab,j}^n = g_{ab,j}$; these functions satisfy the cocycle condition because they are the transition functions of the actual cover. Thus they define line bundles M_j with $M_j^{\otimes n} \simeq L_j|_{S'}$. Conversely these roots give a finite *étale* cover $\prod M_j^\times \to P_{S'} \to P$ whose restriction to each torus fiber is the nth-power map. The image of A in its finite permutation representation has kernel contained in nA. Doing this for every n separates every nonidentity element of A, proving injectivity. This proves the exact equivalence without any assumption that profinite completion preserves an arbitrary discrete exact sequence. To identify the stratum neighborhood, use local equations $z_{a,j}$ along S. Their quotients on overlaps restrict to the same $g_{ab,j}$ for the normal bundle. Any higher-order difference is a unit congruent to 1 along S, and has a unique compatible nth root with residue 1 in the henselian neighborhood, by Hensel's lemma. E1 shows that all local finite covers are Kummer covers of the $z_{a,j}$. Their gluing therefore depends exactly on the above transition functions modulo n and has the same finite-cover category as the normal-bundle torsor, for every n. This identifies the required fundamental-group sequence. Finally finite *étale* descent records permutation of the branches, while conjugation acts on their Kummer roots through the cyclotomic character, giving the stated arithmetic form. □

E5. Nonzero Chern boundary for an elementary hyperbolic variety (F26) *[Refuted]*

The homomorphism $\pi_2(D_J^\circ) \to T^J$ need not vanish for an elementary anabelian U, even when U is a product of three copies of the thrice-punctured line. Thus the proposed extension (37) is not left exact for every simple normal crossings compactification of an elementary variety.

Passage: cote 149, §4, manuscript p. 67, scan 135, "On va donc admettre que (37) définit bien une structure d'extension".

Proof. Take $U = (\mathbb{P}^1 \setminus \{0, 1, \infty\})^3$ over $\mathbb{Q}$, $X_0 = (\mathbb{P}^1)^3$, and blow up $p = (0, 2, 2)$. Only the

boundary component $D_0 = \{x = 0\}$ contains p. Let $E \simeq \mathbb{P}^2$ be the exceptional divisor and D' the strict transform of D_0. They meet in the full line $S = D' \cap E \simeq \mathbb{P}^1$, and no other boundary component meets E. Hence $S = D^{\circ}_{\{D',E\}}$ is indeed an open stratum, not merely a closed intersection from which further points must be removed. The total transform relation is $D' = b^*D_0 - E$. The restriction of $b^*\mathcal{O}(D_0)$ to S is trivial, since S maps to the point p. The exceptional normal bundle is $\mathcal{O}_E(E) = \mathcal{O}_{\mathbb{P}^2}(-1)$. Therefore

$$N_{D'/X}|_S = \mathcal{O}_{\mathbb{P}^1}(1), \qquad N_{E/X}|_S = \mathcal{O}_{\mathbb{P}^1}(-1).$$

E4 gives, with its convention,

$$\partial : T = \pi_2^{\mathrm{et}}(\mathbb{P}^1) \longrightarrow T^2, \qquad a \longmapsto (a, -a).$$

The same nonzero map on ordinary complex homotopy groups is $\mathbb{Z} \to \mathbb{Z}^2$, $1 \mapsto (1, -1)$. In particular its quotient modulo every $n > 1$ is nonzero, so no issue about completion can erase this counterexample. The neighborhood torsor is $\mathcal{O}(1)^{\times} \times_S \mathcal{O}(-1)^{\times}$; its fundamental group is the quotient $T^2/\{(a, -a)\} = T$. The map to $\pi_1(U)$ sends both meridians to the original x-cusp, by exactly the coordinate calculation of E2. Thus the Chern relation is killed by the actual global homomorphism as required. This also explains why nonzero diagonal inertia, established in the prior return, cannot imply (37). □

E6. Algebraic residues need not be finite; the arithmetic openness criterion *[Refuted]*

A valuation of a finitely generated extension L/K can have residue field algebraic and infinite over K, even when it is a DVR. For a valuation with algebraic residue field centered at x on X, the entire homomorphism $D_v \to \pi_1(\operatorname{Spec} \mathcal{O}^h_{X,x} \cap U)$ has open image precisely when both $[\kappa(v) : \kappa(x)]$ and $|\ker \psi_v|$ of E3 are finite. Its image is the entire local group precisely when $\kappa(v) = \kappa(x)$ and $\ker \psi_v = 0$. Thus the inserted "fini" on scans 124 and 130 is an additional hypothesis, not a consequence of algebraicity.

Passages: cote 149, §4, manuscript pp. 61 and 64–65, scans 124 and 130–131.

Proof. Let $p_0, p_1, \ldots$ be distinct rational primes, $a_j = \sqrt{p_j} \in \overline{\mathbb{Q}}$, $F_m = \mathbb{Q}(a_0, \ldots, a_m)$, and put

$$x = t, \qquad y = \sum_{j \geq 0} a_j t^{j+1} \in \overline{\mathbb{Q}}[[t]].$$

The series y is transcendental over $\mathbb{Q}(t)$. Otherwise its orbit under coefficientwise $G_{\mathbb{Q}}$ would be contained in the finite root set of its minimal polynomial over $\mathbb{Q}(t)$. Its coefficient fields F_m have unbounded degrees 2^{m+1}, so that orbit is infinite. Thus $L = \mathbb{Q}(t, y)$ is a rational function field of transcendence degree two. Its restricted t-adic valuation has group $\mathbb{Z}$. We prove that its residue field is exactly $F_\infty = \bigcup_m F_m$. First a_0 is the residue of y/t. Suppose F_m has already been obtained. Put $s_m = \sum_{j=0}^{m} a_j t^{j+1}$ and take

$$P_m(X, Y) = \operatorname{Norm}_{F_m(X)/\mathbb{Q}(X)}(Y - s_m(X)) \in \mathbb{Q}[X, Y].$$

At (t, y) the identity factor has leading term $a_{m+1}t^{m+2}$. Every other factor has a leading term determined by the first changed sign among $a_0, \ldots, a_m$. Therefore the product has leading coefficient $c_m a_{m+1}$ for an explicit $c_m \in F_m^{\times}$, and integral order M_m. The residue of $P_m(t, y)/t^{M_m}$ is $c_m a_{m+1}$; dividing in the residue field by c_m recovers a_{m+1}. Induction proves $F_\infty \subset \kappa(v)$. Conversely the leading coefficient of any rational expression in t, y uses only finitely many of the coefficients a_j, so belongs to F_∞. This proves equality. The valuation is a DVR by D2, and its residue extension over $\mathbb{Q}$ is algebraic and infinite. For the criterion, algebraicity of $\kappa(v)/K$ makes x a closed point, so $\kappa(x)/K$ is finite. In characteristic zero the inclusion $\kappa(x) \subset \kappa(v)$

identifies $G_{\kappa(v)}$ with a closed subgroup of $G_{\kappa(x)}$ for compatible separable closures. Reduction of the homomorphism $D_v \to N_x$ is exactly this inclusion, by D8. Therefore, writing H for its image and $A = T^r$ for local inertia, we have

$$H \cap A = \operatorname{im}(I_v \to A), \qquad [N_x : H] = [G_{\kappa(x)} : G_{\kappa(v)}]\,[A : H \cap A]$$

with the usual cardinal interpretation when an index is infinite. The first index is the field degree and the second is $|\ker \psi_v|$ by character duality in E3. This proves both assertions, including the infinite-residue case excluded by the manuscript's inserted assumption. □

6 F. Arithmetic centralizers, topoi, and reduction

6.1 The two rows referring to the attached cyclotomic return

F8a. The starred arithmetic centralizer assertion *[Known]*

Fix $\overline{\mathbf{Q}}$ and write $G_{\mathbf{Q}}$ for its full profinite Galois group. For every open subgroup $\Gamma' \subset G_{\mathbf{Q}}$, let Γ'° be the kernel of its cyclotomic character. Then

$$C_{G_{\mathbf{Q}}}(\Gamma'^{\circ}) = 1.$$

Manuscript passage: Cote 149, §2, manuscript p. 22, scan 45, starred "à vérifier !".

Proof. Let $k = \overline{\mathbf{Q}}^{\Gamma'}$. Galois correspondence gives the literal equality of closed subgroups $\Gamma'^{\circ} = G_{k(\mu_\infty)}$ in $G_{\mathbf{Q}}$. Statement B1 of the supplied `CYCLOTOMIC_KERNEL` return [AI-26, Statement B1 and its proof] proves $C_{G_{\mathbf{Q}}}(G_{k(\mu_\infty)}) = 1$ for every number field k. The prompt explicitly permits this citation. We checked that proof, including the following two points essential to the proposed R3 argument. After a finite Galois enlargement of k, choose a prime p unramified in that original coefficient field and split in the finite field whose elements a proposed centralizer element moves. The value group in $k(\mu_\infty)$, normalized by $v(p) = 1$, is $(p-1)^{-1}\mathbf{Z}[1/p]$. A CRT element of value one therefore has nonzero Kummer class for a prime $\ell \nmid p(p-1)$. Equality of stabilizers forces $\sigma(a) = c^{\ell}a^{j}$, $1 \le j < \ell$; at a moved prime it gives $0 = \ell v(c) + j$, a contradiction in that value group. Repeating over every finite Galois number field containing k forces $\sigma = 1$. Thus the cited statement has exactly the subgroup, ambient group, and quantifier required here. □

F10a. The local extension and the normalizer correction *[Proved]*

Let K be finitely generated over $\mathbf{Q}$, let $U = X \setminus D$ be a smooth geometrically connected hyperbolic curve, and let $c \in D(K)$. Choose a geometric base point and a lift of c to the normalized universal covering. Its inertia I_c and decomposition group D_c satisfy

$$D_c = N_{E_U}(I_c), \qquad 1 \longrightarrow I_c = \widehat{\mathbf{Z}}(1) \longrightarrow D_c \longrightarrow G_K \longrightarrow 1, \qquad C_{E_U}(I_c) = D_c \cap p^{-1}(\ker \chi_K).$$

A local parameter supplies a splitting. The I_c-conjugacy classes of splittings form a torsor under $H^1(K, \widehat{\mathbf{Z}}(1)) = \varprojlim_n K^*/K^{*n}$.

Manuscript passage: Cote 149, §3, manuscript p. 26 and continuation, scans 53–54, formulas (2)–(3).

Proof. Stix [Sti08b, Proposition 11; Corollary 12] applies to a hyperbolic curve over a characteristic-zero field: distinct lifted cusps have disjoint nontrivial inertia, the geometric normalizer of cusp inertia is the inertia itself, and its arithmetic normalizer is the cusp decomposition group. The tame local covers are $z^n = t$; conjugating $z \mapsto \zeta_n z$ by an arithmetic lift sends ζ_n to $\zeta_n^{\chi_n}$. This computes the conjugation action and the displayed centralizer. A coefficient action fixing compatible roots of a parameter t splits the local group. Two splittings differ by a continuous 1-cocycle in I_c; I_c-conjugacy changes it by a coboundary. The Kummer sequence and Hilbert 90 give $H^1(K, \mu_n) = K^*/K^{*n}$. Passing to the limit introduces no $\varprojlim^1 H^0$ term, since the finite groups $\mu_n(K)$ satisfy the Mittag–Leffler condition. This proves the torsor assertion with its precise acting group.

The algebraic closure of $\mathbf{Q}$ in K is a number field k, and K/k is regular. Restriction $G_K \twoheadrightarrow G_k$ therefore makes χ_K have the same nontrivial open image as χ_k. The centralizer has image $\ker \chi_K$, whereas D_c surjects onto G_K. The word *centralisateur*, retained by our inspection of scan 54, consequently requires a mathematical correction to *normalisateur*; this is not a silent change of transcription. □

F10b. Embedded, disjoint cusp packets, including type $(0,3)$ *[Known]*

Under the hypotheses of F10a the map from local splitting classes to $\pi_1(U_{\overline{K}})$-conjugacy classes of sections of $E_U \to G_K$ is injective. Its image is disjoint from the classes of points of $U(K)$ and from every other rational cusp packet. These assertions hold after each finite extension of K.

Manuscript passage: Cote 149, §3, manuscript pp. 26–27, scans 54–55, Proposition and its margin.

Proof. The exact input is Stix [Sti08b, Theorem 14(1)–(2)]. It assumes injectivity of the point-section map for smooth proper curves of genus at least two over every finite extension of the characteristic-zero base field, and gives the packet assertions just stated for a hyperbolic open curve. We verify its arithmetic hypothesis. Distinct points a, b of such a proper curve C/L have nonzero class $[b-a] \in J_C(L)$: a rational function with divisor $b-a$ would give a degree-one map $C \to \mathbf{P}^1$. Equality of point-section classes pulls back the finite *étale* multiplication covers $[n]$ of J_C to the same fibre torsors; their fibre at $[a-a]=0$ is trivial. Consequently $[b-a] \in \bigcap_n nJ_C(L)$, which is zero by the finite generation proved and sourced in Part B. Every finite L/K is again finitely generated over $\mathbf{Q}$, so all the hypotheses are satisfied. This supplies the derivation for $(0,3)$ as well as every other hyperbolic type. The theorem is used for section classes with geometric conjugation, exactly as in the manuscript. □

F10c. The first assertion of scan 54 in type $(0,2)$ *[Refuted]*

For $U = \mathbf{G}_{m,K}$ each of the two rational cusp packets equals the entire set of section classes, and in particular contains every point section. The first assertion of scan 54 is therefore false in this type. For every affine characteristic-zero type with nontrivial geometric fundamental group, the disjointness conclusion holds exactly in the hyperbolic cases covered by F10b.

Manuscript passage: Cote 149, §3, continuation of manuscript p. 26, scan 54.

Proof. The covers $z^n = t$ identify the geometric group of $\mathbf{G}_m$ with $T = \widehat{\mathbf{Z}}(1)$. The point 1 identifies its arithmetic group with $T \rtimes G_K$. The local group at 0 is this entire semidirect product; at ∞ the parameter t^{-1} gives the same subgroup with inertia generator inverted. Thus both local-to-global maps on splitting classes are bijections with $H^1(K,T)$. A point $a \in K^*$ maps to its Kummer class in that same set. The cases $(0,0)$ and $(0,1)$ have trivial geometric group; proper curves have no cusp packet. These computations also locate the failure of the filling-a-hole proof in scan 55: filling one hole of $\mathbf{G}_m$ gives $\mathbf{A}^1$, where the point-section map has only one value. □

6.2 F9: what the entire étale topos reconstructs

In this subsection a morphism *over* S_{et} includes a specified invertible 2-cell to the structure morphism. Isomorphisms between such morphisms must respect that 2-cell. This is essential, including over a finite field. A geometric morphism is called *pinned* when its map on underlying spaces sends closed points to closed points, and *chivalrous* when it sends constructible subsets to constructible subsets. These are conditions on a geometric morphism, not additional structure attached to a scheme.

F9.1. The 1-faithfulness assertion over the stated base *[Proved]*

[F9; §2, manuscript p. 25; scan 52.] For a locally noetherian scheme S, a reduced S-scheme X locally of finite presentation, and an S-scheme Y locally of finite presentation, the map

$$\mathrm{Hom}_S(X,Y) \longrightarrow \pi_0 \,\mathrm{Hom}_{S_{\mathrm{et}}}(X_{\mathrm{et}}, Y_{\mathrm{et}}) \tag{F.1}$$

is injective. Thus the manuscript's 1-faithfulness statement does not need its tentative characteristic-zero restriction.

Proof. The finite-presentation, quasi-compact quasi-separated-base assertion is Carlson–Stix, Lemma 5.4, in version 2 (20 September 2026) [CS26]. Its source hypothesis is that X is reduced; no assumption that its fibres are reduced is present in that lemma. Scheme morphisms of finite presentation induce chivalrous geometric morphisms, and the lemma gives their injectivity into the indicated groupoid of such morphisms. Inclusion of this full subgroupoid in all geometric morphisms does not change whether its two objects are isomorphic.

Here is the localization giving precisely the manuscript's scope. If $f_{\mathrm{et}} \simeq g_{\mathrm{et}}$, their maps on underlying spaces coincide, because open subsets are the subterminal objects of the étale topos. Choose an affine open in S, an affine open in Y over it, and affine opens in its common inverse image under f, g. On each such source open, both restricted maps have affine target and are between finite-presentation schemes over the chosen affine base. Apply the cited lemma there. These source opens cover X, so $f = g$. This proves the reduction even when S, X, Y are not quasi-compact.

Absolute Frobenius is not a counterexample to (F.1). Over $\mathbb{F}_p$, the natural isomorphism on étale covers induced by relative Frobenius restricts on the base to the arithmetic Frobenius automorphism of the identity functor of $BG_{\mathbb{F}_p}$. That automorphism is nonidentity. Thus it does not give the required isomorphism over S_{et} with its fixed 2-cell. □

F9.2. The precise full-faithfulness theorems *[Known]*

[F9; §2, manuscript pp. 25–26; scans 51–52.] Let UH denote the universal homeomorphisms. The functor

$$(\mathrm{Sch}_k^{\mathrm{ft}})[\mathrm{UH}^{-1}] \longrightarrow \mathrm{RTop}_{k_{\mathrm{et}}}^{\mathrm{pin}}, \qquad X \longmapsto X_{\mathrm{et}}, \tag{F.2}$$

is fully faithful for every infinite field k finitely generated over its prime field, and for every sub-p-adic field k (a subfield of a finitely generated extension of some $\mathbb{Q}_p$). The first assertion is Carlson–Haine–Wolf, Theorem 0.1; the second is Carlson–Stix, Theorem A (=Theorem 4.1). In characteristic zero, the localization is represented by seminormalization; in positive characteristic it is represented by perfection.

There is also the following relative assertion. Let S be noetherian, Nagata, geometrically unibranch, and suppose each generic residue field of S is sub-p-adic. Restrict to finite-type S-schemes each of whose irreducible components is universally open over S. After inverting UH, the topos functor into chivalrous geometric morphisms over S_{et} is fully faithful. This is Carlson–Stix, Theorem D (=Theorem 6.12 and Proposition 6.11).

Proof. These are the exact statements of the opened primary sources [CHW24, CS26]. To derive the assertions about the manuscript's original objects, observe first that a universal homeomorphism induces an equivalence of étale topoi, by topological invariance of the étale site. Thus the functor factors through the displayed localization. In characteristic zero the universal homeomorphism $X^{\mathrm{sn}} \to X$ represents the localization: maps in it can be calculated from the seminormalization of the source. Accordingly, for a seminormal finite-type source over the fields in (F.2), the comparison on actual scheme Hom-sets is already a bijection; for two seminormal schemes, topos equivalences are exactly scheme isomorphisms. Over the relative bases just stated, normal sources are absolutely weakly normal and the analogous actual-Hom comparison holds; this is Corollary E of Carlson–Stix. These conclusions follow by applying the displayed fully faithful functors to their localized objects, not by identifying the original schemes without their normality hypothesis.

For a general base, chivalrous cannot be replaced by pinned in the relative statement. Take a DVR R, an element a of positive valuation, and the closed point $(aT - 1)$ of $\mathbb{A}^1_R$. The projection

$\mathbb{A}^1_R \to \operatorname{Spec} R$ sends it to the generic point. Its topos morphism is therefore not pinned, although it is induced by a scheme morphism in the stated category. Its image of any constructible set is constructible by Chevalley's theorem. This also gives the precise reason for the different target of the relative theorem. □

F9.3. Fullness into all topoi, and fullness over an algebraically closed field *[Refuted]*

[F9; §2, manuscript p. 25; scan 52.] Without a restriction on geometric morphisms, the proposed fullness fails already over $\mathbb{Q}$. With pinned morphisms retained, it fails over $\mathbb{C}$, even for smooth proper genus-one curves and even after universal homeomorphisms are inverted.

Proof. For the first assertion take the t-adic henselian valuation of $\mathbb{Q}(t)$. The coefficient action, fixing compatible roots of t, gives a continuous section $G_{\mathbb{Q}} \to G_{\mathbb{Q}(t)}$ through its decomposition group. One can construct it in the Puiseux field $\bigcup_n \overline{\mathbb{Q}}((t^{1/n}))$: coefficient automorphisms preserve the algebraic closure of $\mathbb{Q}(t)$ inside this field and fix t. This section gives a geometric morphism

$$\mathbb{Q}_{\mathrm{et}} \longrightarrow \mathbb{Q}(t)_{\mathrm{et}} \longrightarrow (\mathbb{P}^1_{\mathbb{Q}})_{\mathrm{et}}$$

over $\mathbb{Q}_{\mathrm{et}}$, whose point maps to the generic point. An actual $\mathbb{Q}$-morphism $\operatorname{Spec}\mathbb{Q} \to \mathbb{P}^1$ has closed-point image. Universal-homeomorphism localization does not change this Hom-set, since source and target are seminormal. This is the concrete counterexample of Carlson–Haine–Wolf, Example 2.30, with its section constructed above.

For the second assertion choose $\lambda \in \mathbb{C}$ transcendental over $\mathbb{Q}$ and an automorphism σ of $\mathbb{C}$ with $\sigma(\lambda) = 2\lambda$. Such an automorphism is obtained by replacing λ by 2λ in a transcendence basis and extending to an algebraic closure. Set

$$E_\lambda : y^2 = x(x-1)(x-\lambda), \qquad E_{2\lambda} : y^2 = x(x-1)(x-2\lambda).$$

Applying σ to the coefficients gives an isomorphism of the underlying schemes over $\operatorname{Spec}(\sigma)$, hence an equivalence of their étale topoi. Since $\operatorname{Spec}\mathbb{C}_{\mathrm{et}} = \mathrm{Set}$, this is a pinned equivalence over the specified base topos. A $\mathbb{C}$-scheme isomorphism between these curves would force equality of their j-invariants

$$j(\lambda) = \frac{256(1-\lambda+\lambda^2)^3}{\lambda^2(1-\lambda)^2}, \qquad j(2\lambda).$$

These rational functions differ, as the polynomial calculation in `code/CF_checks.py` verifies. Transcendence of λ therefore excludes equality. A morphism inducing this equivalence would induce a bijection of underlying topological spaces. It would therefore be nonconstant, hence finite between these smooth proper curves. In characteristic zero its generic degree equals the number of geometric points of a general fibre, so that bijectivity forces degree one. A finite degree-one map onto a normal curve is an isomorphism, which the displayed j-invariants exclude. Both curves are normal, so the characteristic-zero universal-homeomorphism localization cannot produce an additional isomorphism. This constructs the comparison map between the two topoi and proves explicitly why it does not come from a morphism of the required schemes. It does not contradict the injectivity in F9.1. □

F9.4. A finite-field obstruction to pinned fullness *[Refuted]*

[F9 and the characteristic question in its margin; §2, manuscript p. 25; scan 52.] Pinned fullness after universal-homeomorphism localization fails for $\mathbb{A}^1_{\mathbb{F}_3}$ over $\mathbb{F}_3$. Thus the positive transcendence-degree hypothesis in the positive-characteristic case of F9.2 has an actual obstruction.

Proof. Put $X = \mathbb{A}^1_{\mathbb{F}_3}$ and $p : X_{\mathrm{et}} \to B\widehat{\mathbb{Z}}$ for its structure map. For any $\gamma \in \widehat{\mathbb{Z}}$ the central action on $\widehat{\mathbb{Z}}$-sets defines a natural automorphism of the identity of $B\widehat{\mathbb{Z}}$, and hence an automorphism α_γ of p. The pair $(\mathrm{id}_{X_{\mathrm{et}}}, \alpha_\gamma)$ is a pinned self-equivalence in the relative 2-category. Choose $\gamma \notin \mathbb{Z}$; for example use 2-adic coordinate 0 and every odd-adic coordinate 1.

Suppose it came from a morphism in the localization. This is a morphism $X^{\mathrm{perf}} \to X$, hence is given by $f \in \mathbb{F}_3[x^{1/3^\infty}]$. Since the underlying topological map is the identity, f fixes every closed point as a point. Choose r with $P = f^{3^r} \in \mathbb{F}_3[x]$, of degree D. For each $a \in \mathbb{F}_{3^N}$, preservation of its closed point gives $P(a) = a^{3^i}$ for some $0 \le i < N$. Unless one polynomial $P(T) - T^{3^i}$ is identically zero, their union of roots has size at most

$$\sum_{i=0}^{N-1} \max(D, 3^i) \le ND + \frac{3^N - 1}{2} < 3^N$$

for sufficiently large N. This contradicts the preceding assertion for all 3^N elements. Therefore $f = x^{3^m}$ for an integer m, where negative exponents mean inverse Frobenius on the perfection.

We check the entire 2-isomorphism ambiguity. A natural automorphism of $\mathrm{id}_{X_{\mathrm{et}}}$ restricts to each closed subtopos $\operatorname{Spec}\mathbb{F}_{3^n,\mathrm{et}}$ because the direct image of a closed immersion is fully faithful. Naturality with respect to the adjunction unit $A \to i_* i^* A$ makes its base component agree with the restriction of the original component. On that subtopos its induced base automorphism lies in $G_{\mathbb{F}_{3^n}} = n\widehat{\mathbb{Z}}$: the automorphisms of the identity of BH are $Z(H)$, here $H = n\widehat{\mathbb{Z}}$. There is a closed point of every degree n on $\mathbb{A}^1_{\mathbb{F}_3}$. Thus its base component belongs to $\bigcap_n n\widehat{\mathbb{Z}} = 0$. All isomorphisms $(\mathrm{Frob}^m)_{\mathrm{et}} \simeq \mathrm{id}$ consequently have the same base component. The canonical one, from relative Frobenius of étale covers, has component m; with the relative-pair convention above this represents $(\mathrm{id}, \alpha_{-m})$. It cannot represent our $\gamma \notin \mathbb{Z}$. This proves the counterexample, including the possible noncanonical natural transformations. □

F9.5. The proposed function-field automorphism test *[Known]*

[F9; §2, manuscript p. 25; scan 52.] Let k be a number field, C/k a smooth proper geometrically connected curve of any genus, $L = k(C)$, and $\Delta = \ker(G_L \to G_k)$. Then

$$\mathrm{Aut}_k(C) \longrightarrow \mathrm{Isom}_{G_k}(G_L, G_L)/\operatorname{Inn}(\Delta) \tag{F.3}$$

is bijective, with the usual contravariance on function fields. In particular the additional requirement that the automorphism preserve all loop subgroups is unnecessary for this conclusion; the automorphism obtained from (F.3) preserves them by functoriality of valuations.

Proof. We use Tsujimura, Corollary 4.10, in the opened RIMS-1957 version [Tsu23]. It applies to function fields in one variable over a subfield K of a finitely generated extension of the field obtained from a number field F by adjoining all roots of a fixed prime number. Take $F = K = k$; it is a subfield of that field, so every hypothesis is met. The corollary gives the bijection between k-field isomorphisms and the right-hand side of (F.3), with conjugation exactly by the geometric kernel Δ.

A k-automorphism of L gives a rational map $C \dashrightarrow C$. Every local ring of the smooth proper curve at a closed point is a DVR. The valuative criterion for properness extends the rational map there; finite presentation and separatedness glue these extensions to a morphism. The same argument for the inverse field automorphism gives its inverse morphism, proving the claimed bijection for C. It carries the local DVR at x to that at its image, and hence carries the corresponding inertia and decomposition groups to the groups at that image, modulo the stipulated conjugation.

On the image, scan 52 appears to abbreviate “structure à lacets” as “à lcts”, rather than the transcription’s alternatives “Qcts”/“Dcts”. We record this as evidence, not a settled reading.

The proved statement covers both the version with peripheral structure required and the stronger version where it is recovered. □

6.3 F17: good reduction and the quotients of monodromy

F17.1. The nonabelian criterion and the valid forward implications *[Known]*

[F17; §3, manuscript pp. 49–51; scans 100–103.] Let R be a DVR with fraction field K and residue characteristic $p \geq 0$. Let $U = \bar{U} \setminus D$ be a smooth geometrically connected hyperbolic curve over K, with $\bar{U}$ smooth proper and D reduced and geometrically étale. Good reduction means a smooth proper model of $\bar{U}$ over R on which the boundary extends to a finite étale divisor. Good reduction is equivalent to trivial inertia in $\mathrm{Out}(\Pi^{(p')})$; it is also equivalent to trivial inertia in $\mathrm{Out}(\Pi^{(\ell)})$ for one, equivalently every, prime $\ell \neq p$. For $p = 0$ the pro-prime-to-p group is the full geometric group, so this proves exactly a)⇔b).

Good reduction implies c) and d) with prime-to-p coefficients, through the characteristic quotient maps $\Pi^{(p')} \to \Pi^{(p'),\mathrm{ab}} \to H_1(U_{\bar{K}}, \mathbb{Z}_\ell)$. When the residue field is finitely generated over $\mathbb{Q}$, a) and b) imply e) for every splitting of the residue exact sequence: the resulting outer action is that of the actual smooth special fibre.

Proof. The precise source is Tamagawa, Theorem (5.3), printed p. 177, equivalently Theorem (0.8), with the definitions immediately before it [Tam97]. Its hypotheses are a DVR, a smooth proper geometrically connected curve and relatively étale effective boundary over its fraction field, and $2 - 2g - r < 0$. It makes the four conditions just stated equivalent; no semistability assumption is added. Thus its proper-curve case includes Oda's criterion and its marked case covers colliding cusps as well.

Inner automorphisms act trivially on abelianization. Therefore an outer inertia action equal to the identity induces the identity on each indicated abelian quotient, giving the actual comparison maps for a)⇒b)⇒c)⇒d). In good reduction, specialization identifies the geometric groups of generic and special fibres, and identifies their arithmetic extensions after pullback along $G_K \to G_k$. The outer action therefore factors through G_k. Composing this factorization with any continuous splitting $G_k \to G_K$ yields that same special-fibre action. Since Π is centre-free, C4 identifies the extension reconstructed from the action with the extension of the special fibre. This proves e), without an additional field extension.

In mixed characteristic one cannot include the residue prime in the homological conclusion. For example the smooth marked curve $\mathbb{P}^1_{\mathbb{Z}_p} \setminus \{0, 1, \infty\}$ has good reduction, but its generic H_1 contains $\mathbb{Z}_p(1)^2$ and p-adic cyclotomic inertia is nontrivial. This is why the criterion above specifies its quotient rather than silently using the full group at residue characteristic p. □

F17.2. A compact-type genus-two counterexample, including class two *[Refuted]*

[F17; §3, manuscript pp. 49–51; scans 100, 102–103.] Over $R = \mathbb{Q}[[t]]$, the smooth projective completion of

$$C_t : \quad y^2 = (x^3 - t^6)(x^3 - 1) \tag{F.4}$$

has genus two, bad reduction, and trivial inertia on all of its abelianizations. Its inertia action is even trivial on every class-two nilpotent geometric quotient. Its full outer inertia action is nontrivial. Thus c)⇒a) and d)⇒a) are false for proper curves.

Proof. The sextic discriminant is $729t^{12}(1 - t^6)^6$, as checked in the attached program. It is nonzero in $\mathbb{Q}((t))$, so the double cover has six distinct branch points and genus $(6 - 2)/2 = 2$.

We describe its stable model explicitly. Separate the two sets of three branch points using the model of the base projective line with local plumbing equation $xz = t^2$. On the inner chart put $x = t^2X, y = t^3Y$; the equation becomes

$$Y^2 = (X^3 - 1)(t^6X^3 - 1),$$

with special fibre $E_{\text{in}} : Y^2 = 1 - X^3$. On the outer component the normalization is $E_{\text{out}} : V^2 = x(x^3 - 1)$, with $V = y/x$. Both are smooth genus-one curves. On the outer curve the substitutions $X = 1/x$ and $Y = V/x^2$ give $Y^2 = 1 - X^3$, so the two smooth proper components are isomorphic over $\mathbb{Q}$. Over $\mathbb{Q}(\zeta_3)$ the automorphism $(X, Y) \mapsto (\zeta_3X, Y)$ fixes the origin and has order three, giving the CM action by $\mathbb{Z}[\zeta_3]$. At the joining point set $A = (1 - z^3)(x^3 - 1)$, a unit. The equations are

$$V^2 = xA, \qquad W^2 = z/A, \qquad VW = t, \qquad W = t/V.$$

They follow by substitution in (F.4); conversely they recover the original equation on the generic fibre. Since $A(0, 0) = -1$, the implicit-function theorem for complete local rings identifies the completed local equation with $VW = t$. The model is smooth away from this node, since the remaining branch sections are disjoint and simple. Normalizing the proper separated-branch base model therefore gives a proper stable curve whose special fibre consists of these two elliptic curves meeting transversely at one point. Both components are stable, and the fibre is not smooth. Uniqueness of the stable model (the input also used in Tamagawa, Remark (5.2)) excludes a smooth model.

The single vanishing circle separates the two genus-one pieces. Choose a geometric topological presentation

$$\pi_1(C_t(\mathbb{C})) = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1\rangle, \qquad c = [a_1, b_1].$$

The plumbing $VW = t$ twists one boundary circle once as t circles zero. Cutting at this circle shows directly that the induced automorphism conjugates a_1, b_1 by c and fixes a_2, b_2, or its inverse with the opposite loop orientation. This description can also be obtained by identifying V-annulus and W-annulus coordinates through $W = t/V$: a full turn of t adds one turn to the gluing coordinate. Finite topological covers and finite étale covers are identified by SGA 1 XII, Theorem 5.1; the same formula therefore describes the profinite and pro-ℓ tame inertia actions. The family is defined algebraically near $t = 0$, so the topological loop maps to the positive generator of its tame inertia, with this identification.

The element c belongs to the second lower-central subgroup Γ_2. In the class-two quotient Π/Γ_3 it is central, so the displayed automorphism is the identity there, and in particular on Π^{ab}. This is an identity in the quotient itself, not merely an outer identity. The full outer action is nontrivial by F17.1 and the proved failure of good reduction. The special fibre is of compact type and its Jacobian is the product of the two elliptic component Jacobians. More precisely, the relative Pic^0 of a compact-type model is an abelian scheme, and its special fibre is the product of the component Jacobians; the construction and properness argument are given in [DR17, §1, p. 1]. That text assumes algebraically closed residue field. Apply it after the faithfully flat strict-henselian base extension; formation of the relative Picard functor commutes with that base change, and smoothness and properness descend. This proves the same assertion over $\mathbb{Q}[[t]]$ and explains the absence of a homological vanishing cycle. □

F17.3. Colliding cusps detected by class-two monodromy *[Refuted]*

[F17; §3, manuscript pp. 50–51; scans 102–103.] Let

$$U_t = \mathbb{P}^1_{\mathbb{Q}((t))} \setminus \{0, t, 1, \infty\}. \tag{F.5}$$

It has bad marked reduction and trivial inertia on $\Pi^{\mathrm{ab}} = \widehat{\mathbb{Z}}(1)^3$, while inertia acts nontrivially in the outer automorphism group of its class-two pro-ℓ quotient for every prime ℓ.

Proof. All four punctures are rational. The homology is the quotient of the free Tate module on the four punctures by its diagonal. The t-adic inertia fixes all roots of unity (they are constants), so its action on this quotient is trivial. The cross ratio of the ordered four points is t up to one of its six cross-ratio transforms. In a smooth marked projective line over a strictly henselian trait, carrying three disjoint sections to $0, 1, \infty$ makes the fourth coordinate and its difference from 1 units. Our valuation $v(t) > 0$ precludes this. Alternatively, blowing up the collision at $(x, t) = (0, 0)$ gives two three-pointed projective lines joined at a node, the stable model. This proves failure of good reduction independently of the monodromy computation.

Choose free geometric generators a, b, c around $0, t, 1$; the loop at infinity is $(abc)^{-1}$. A loop of t around zero is the full twist of the first two punctures. Cutting out a disc containing exactly these two points shows its action to be

$$a \longmapsto (ab)a(ab)^{-1}, \qquad b \longmapsto (ab)b(ab)^{-1}, \qquad c \longmapsto c, \tag{F.6}$$

up to reversing the generator of inertia. This is the same local plumbing calculation as in F17.2, now for a disc with two punctures. Let $[x, y] = xyx^{-1}y^{-1}$ and pass to the free class-two pro-ℓ group. Its commutator subgroup is freely generated as a $\mathbb{Z}_\ell$-module by $c_{ab} = [a, b], c_{ac} = [a, c], c_{bc} = [b, c]$. Formula (F.6) becomes

$$a \mapsto a c_{ab}^{-1}, \qquad b \mapsto b c_{ab}, \qquad c \mapsto c. \tag{F.7}$$

Conjugation by an element with abelian coordinates (u, v, w) gives

$$a \mapsto a c_{ab}^{-v} c_{ac}^{-w}, \quad b \mapsto b c_{ab}^{u} c_{bc}^{-w}, \quad c \mapsto c c_{ac}^{u} c_{bc}^{v}.$$

Matching the first two images forces $u = v = 1, w = 0$; the third then has the nonzero commutator coordinates $(1, 1)$. Thus (F.7) is not inner. This calculation holds over every $\mathbb{Z}_\ell$, including $\ell = 2$. The program verifies it separately in the finite class-two exponent-five quotient by examining all 5^3 possible abelian coordinates of an inner automorphism. This finite check supplements the integral proof and does not replace its all-primes conclusion. □

F17.4. Homological unramifiedness need not give a geometric residue action *[Refuted]*

[F17; §3, manuscript pp. 50–51; scans 101–103, condition e).] For the curve (F.5), choose the coefficient splitting $G_{\mathbb{Q}} \to G_{\mathbb{Q}((t))}$ which fixes compatible roots of t. The resulting outer action on $\widehat{F}_3$ is not realized by a smooth hyperbolic curve over any finite extension of $\mathbb{Q}$. Thus neither c) nor d) implies e), even allowing the finite extension in e).

Proof. We use the nonlinear inertia criterion of Nakamura, Theorem (2.1.1) [Nak94b]. Its hypotheses are a smooth geometrically connected affine hyperbolic curve over a number field and a full class of finite groups; we take the class of all finite groups. It states that a nonidentity geometric element belongs to an inertia group precisely when it has a closed cyclotomic normalizer whose image in the arithmetic Galois group is open. "Cyclotomic" here means that the action on its closed procyclic subgroup factors through that arithmetic image and is the cyclotomic character.

Let E_0 be the extension of $G_{\mathbb{Q}}$ by $\Pi = \widehat{F}_3$ obtained by the chosen coefficient splitting. Each of the four cusps supplies a decomposition subgroup in E_0 and a tangential section normalizing its loop through the cyclotomic character. There is a fifth such subgroup for the vanishing loop $d = ab$. We justify this extra subgroup locally. At the node of the stable marked model the regular equation is $xz = t$. The punctured formal neighborhood, with the two vertical divisors removed, has tame geometric group T^2, by the two-parameter form of SGA 1 XIII, Proposition 5.2. Its map to the tame inertia of the base is

$$T^2 \longrightarrow T, \qquad (r, s) \longmapsto r + s,$$

because $t = xz$. Coefficient automorphisms split its arithmetic extension and act cyclotomically on both factors. Choose the roots so that $x^{1/n} z^{1/n} = t^{1/n}$ for every n; coefficient automorphisms fix both factors and hence their product. They therefore map to our specified coefficient splitting of the base. The kernel $\{(r, -r)\}$ maps to the vanishing loop subgroup $\langle ab \rangle$ of the generic geometric curve. The map on this kernel is injective: ab has primitive abelianization in the free geometric group, and the local annulus identifies the two orientations as opposite. Consequently its image together with the coefficient section is a closed subgroup of E_0 projecting onto $G_{\mathbb{Q}}$ and acting on $\langle ab \rangle$ cyclotomically. This construction persists after restriction to every finite number field.

Suppose E_0 became the arithmetic extension of a smooth hyperbolic curve V/k after such a restriction. Its geometric group has abelian rank three, so V is affine and $2g(V) + r(V) - 1 = 3$, whence $r(V) \leq 4$. Transfer the five cyclotomic normalizers through the asserted extension isomorphism. Nakamura's criterion puts the five elements

$$a,\ b,\ c,\ (abc)^{-1},\ ab$$

in inertia groups of V. In $H_1(\Pi, \mathbb{Q}_\ell)$ their lines are generated by

$$e_1, \quad e_2, \quad e_3, \quad -e_1 - e_2 - e_3, \quad e_1 + e_2.$$

They are five distinct lines. Elements belonging to conjugate inertia groups at one cusp have proportional abelian classes. Hence these five elements require at least five distinct cusps of V, contradicting $r(V) \leq 4$. The attached program checks the pairwise rank-two test for these five lines. This proves the failure of e) with its full finite-extension allowance. □

F17.5. A geometric residue action can coexist with bad reduction *[Refuted]*

[F17 and an additional flagged step; §3, manuscript pp. 50–51; scans 101, 103.] Over $K = \mathbb{Q}((t))$ let

$$E_t : y^2 = x^3 - t^2 x, \qquad U_t = E_t \setminus \{O\}. \tag{F.8}$$

It has bad reduction. For every continuous splitting $s : G_{\mathbb{Q}} \to G_K$, however, its restricted outer action is realized by a once-punctured elliptic curve over $\mathbb{Q}$ itself. Thus e)⇒a) is false even with "every splitting", and bad reduction alone does not force the proposed residue action to be nongeometric.

Proof. Write $E_0 : Y^2 = X^3 - X$. Over $K(\sqrt{t})$ the substitution $x = tX$, $y = t\sqrt{t}\,Y$ gives an isomorphism $E_0 \simeq E_t$, as checked by direct substitution. The nonidentity automorphism of this quadratic extension acts through $[-1]$ on E_0. Therefore E_t is the twist by the ramified quadratic character $\epsilon_t : G_K \to \{\pm 1\}$ associated to t. Its inertia action on $H_1(E_{\bar{K}}, \mathbb{Z}_\ell)$ is -1, so good reduction is impossible by the forward implication in F17.1 (or smooth proper base change).

For any splitting s, the composite $\epsilon_t \circ s$ is a quadratic character of $G_{\mathbb{Q}}$, hence is the Kummer character of some $d \in \mathbb{Q}^*/\mathbb{Q}^{*2}$. Restriction of the twisting cocycle is precisely the cocycle defining

the quadratic twist $E_0^{(d)}$. Functorial descent of finite étale covers then identifies the pulled-back extension with that of $E_0^{(d)} \setminus \{O\}$. Explicitly, the geometric action is $g \mapsto \iota(\epsilon_t(s(g)))\rho_0(g)$, where $\iota(+1) = \mathrm{id}$ and $\iota(-1) = [-1]$, and this is the descent action for that twist. This proves e) without enlarging the residue field. The point $(0, 0)$ is fixed by $[-1]$, so the same construction can be made with actual point sections, rather than only outer actions. □

F17.6. Stable bad reduction obstructs every geometric residue action *[Proved]*

[F17; §3, manuscript pp. 49–51; scans 100–103; supplied R4 and H2.] Let R be a henselian discrete valuation ring with fraction field K and residue field k finitely generated over $\mathbb{Q}$. Let U/K be a hyperbolic curve whose smooth proper compactification and reduced boundary have a stable marked model $(\mathcal{C}, \mathcal{D})$ over R. Put $\Pi = \pi_1(U_{\bar{K}})$, with its full profinite topology. For a continuous section $s : G_k \to G_K$ of the residue map, let

$$E_s = \pi_1(U) \times_{G_K, s} G_k.$$

Every geometric node of $\mathcal{C}_k$ gives an injected subgroup $I_e \simeq \widehat{\mathbb{Z}}(1)$ of Π, well defined up to Π-conjugacy. In E_s it is essentially cyclotomically normalized: after a finite extension of k fixing that node and its two branches, a closed subgroup of its normalizer maps onto the residue Galois group and acts on I_e by the cyclotomic character. This holds for every s. Distinct geometric nodes and cusps give distinct conjugacy classes of maximal procyclic subgroups.

If $\mathcal{C}_k$ has a node, the extension E_s is not isomorphic, over G_k, to the arithmetic fundamental-group extension of any smooth hyperbolic curve over k, and it does not become so after any finite extension of k. If $r > 0$ is the number of geometric cusps, the number of conjugacy classes of essentially cyclotomically normalized maximal procyclic subgroups of Π equals r exactly when this stable marked model is smooth. In the singular case it is at least $r + n$, where $n > 0$ is the number of geometric nodes.

Proof. We give the full comparison which was missing in the supplied prior return E4. All logarithmic fundamental groups below classify finite Kummer étale covers; all sets of primes are the set of *all* primes. This choice is legitimate because k has characteristic zero.

First pass from R to its completion. The restriction map $G_{\widehat{K}} \to G_K$ is compatible with the residue maps to G_k. The two inertia groups are $T = \widehat{\mathbb{Z}}(1)$: over the maximal unramified extension every finite extension is killed by adjoining a root of a uniformizer, by the tame local calculation of SGA 1 XIII, Propositions 5.1–5.2. The restriction map induces the identity on T, as is seen on the same compatible roots of that uniformizer, and the identity on G_k. It is therefore an isomorphism of the two extensions. Geometric fundamental groups of a smooth curve are invariant under an extension of algebraically closed fields in characteristic zero: this follows from SGA 1 XIII, Proposition 4.3, Examples 4.4, and §4.5.3, applied to the constant smooth proper pair with a chosen geometric point section. The base-change map $\pi_1(U_{\widehat{K}}) \to \pi_1(U)$ consequently induces isomorphisms on both geometric kernel and arithmetic quotient, and hence is an isomorphism. Thus it transports each specified s, its pullback extension, and the stable special fibre without changing the question.

We now work over the complete ring. Give $S = \operatorname{Spec} R$ its divisorial log structure, and $\mathcal{C}$ the log structure from $\mathcal{C}_k \cup \mathcal{D}$. These are fs log schemes; $\mathcal{C}^{\log}$ is log regular and proper over $S^{\log}$. For example a split node of thickness m has completed equation $xy = t^m$ after absorbing a unit into y, and monoid

$$P = \langle X, Y, T_0 \mid X + Y = mT_0 \rangle.$$

Its group completion has rank two; the closed log stratum is regular of dimension zero, and the local ring has dimension two. These are precisely the log-regularity dimension and regularity

conditions. At a smooth point and at a marked section the charts are respectively the one-parameter and two-parameter normal-crossing charts. The monoid P is saturated: in the basis (X, T_0) its integral cone is generated as a monoid by $(1,0),(0,1),(-1,m)$.

We use the following two opened, numbered comparison results. Illusie, Theorem 7.6 [Ill02], states that for a log-regular fs scheme, restriction to its trivial-log open identifies finite Kummer étale covers with finite étale covers tame along the boundary at its codimension-one points. Here every such residue characteristic is zero, so all covers are tame. Hoshi, Corollary 3.9 [Hos09], states that a proper log-regular fs scheme over the spectrum of a complete local ring has the same finite Kummer étale cover category as its strict closed log fibre. Applying these results also to the base gives a commuting square of isomorphisms

$$\begin{array}{ccc} \pi_1(\mathcal{C}_k^{\log}) & \xrightarrow{\sim} & \pi_1(U) \\ \downarrow & & \downarrow \\ \pi_1(k^{\log}) & \xrightarrow{\sim} & G_K. \end{array} \tag{F.9}$$

The upper arrow is the strict-closed-fibre comparison followed by the inverse of restriction to the interior. These functors commute with finite extension of the residue field, with log base change, and with the morphism from a strict point at a node. The resulting maps of fundamental groups have only the usual base-point path, hence inner-conjugacy, ambiguity.

For clarity, the geometric kernel in (F.9) is also identified in an opened source. Mochizuki, Example 2.5 [Moc07], treats a stable log curve over the standard log point $\mathbb{N} \to k$, $1 \mapsto 0$. Its log Galois group is an extension of G_k by T, and the geometric kernel is the fundamental group $\Pi_{\mathcal{G}}$ of the semi-graph of groups of the stable curve. The identification is natural and the arithmetic action preserves that graph. The finite covers in this description are obtained by matching the tame branch covers at each node, with inverse branch orientations; the chart $X + Y = mT_0$ gives exactly this matching. Hoshi–Mochizuki, Definition 2.10(iv), Lemma 2.11(ii)–(iii), and Remark 2.14.1 [HM11b], make the log-group injection and its change under the smoothing-exponent map explicit. In particular, with all primes, (F.9) identifies

$$\Pi_{\mathcal{G}} = \ker[\pi_1(\mathcal{C}_k^{\log}) \to \pi_1(k^{\log})] \xrightarrow{\sim} \ker[\pi_1(U) \to G_K] = \Pi. \tag{F.10}$$

Thus (F.10) is an isomorphism of the full geometric groups, with the arithmetic action inherited from (F.9); it is not only an isomorphism of their abelianizations or pro-ℓ quotients.

Fix a geometric node. Enlarge k by a finite extension which defines the node and its two branches, and make the corresponding unramified extension of the trait. This replaces G_k by its specified open subgroup and s by its restriction. The log point at that node has arithmetic group

$$D_e = T_P \rtimes G_k, \qquad T_P = \{(a,b,c) \in T^3 : a + b = mc\},$$

where G_k acts cyclotomically on all three coordinates. Its map to the base log Galois group, using the same uniformizer splitting, is

$$(a,b,c,g) \longmapsto (c,g) \in T \rtimes G_k = G_K. \tag{F.11}$$

This map is surjective, since $(mc,0,c) \in T_P$ for every c. Its kernel is $\{(a,-a,0)\} \simeq T$. Compatible Kummer roots may be chosen with $x^{1/n}y^{1/n} = (t^{1/n})^m$, which proves (F.11) and its compatibility with the specified base action. The map from this kernel into (F.10) is the injected edge group of Lemma 2.11. Stability is essential here: these are the edges of the stable curve, not contractible circles attached to an unstable rational tail. Resolving a node of thickness m merely subdivides its edge and gives the same essential circle.

Let H_s be the inverse image of $s(G_k)$ under (F.11). It is closed, maps onto $s(G_k)$, and its geometric kernel is the displayed antidiagonal T. Its image in E_s therefore is a closed normalizing subgroup mapping onto G_k. Conjugation on that kernel is $a \mapsto \chi(g)a$, since T_P is abelian and

its arithmetic action is cyclotomic. No complement in H_s has been chosen or assumed. This proves the assertion for every splitting, including one different from the uniformizer splitting used to write (F.11).

Mochizuki, Proposition 1.2(i)–(ii) [Moc07], applies to every semi-graph of profinite pointed-stable-curve type: its edge subgroups are commensurably terminal, and open intersection of two edge subgroups forces the same edge. Thus each edge group is maximal procyclic, since an abelian group containing it normalizes it and is contained in its commensurator. A conjugacy between groups belonging to two geometric edges would identify those edges by part (i); distinct nodes and cusps therefore give distinct Π-conjugacy classes. Marked sections give the same cyclotomic-normalizer construction by their one-branch Kummer charts. This proves the bound $r+n$. If $n=0$, (F.9) is the good-reduction comparison, and Stix, Theorem 20 and Corollary 21, give exactly the r cusp classes. The required hypothesis that k distinguishes tori from abelian varieties follows from the checked finitely generated characteristic-zero result AE1 of [AI-26]: a Tate-module map from a split torus would have image fixed by the cyclotomic kernel. This also proves the asserted counting criterion.

We finally exclude realization by an arbitrary smooth curve, whose number of cusps has not been fixed in advance. This needs an additional argument in the affine case. The comparison (F.10) gives the graphic quotient and the homology filtration described in Mochizuki, Definition 1.1(ii) and Remarks 1.1.3–1.1.4. If the dual graph has first Betti number $b>0$, killing its vertex groups gives a surjection

$$\Pi \twoheadrightarrow \widehat{\pi_1(\mathcal{G})}, \qquad H_1(\Pi,\mathbb{Q}_\ell) \twoheadrightarrow \mathbb{Q}_\ell^{b}. \tag{F.12}$$

After a finite extension fixing the graph, the right-hand module has trivial G_k-action. This statement is independent of s: logarithmic inertia fixes the underlying graph, and the entire action on this quotient factors through the finite action of G_k on that graph.

No smooth curve V/k has a nonzero trivial quotient of its $\mathbb{Q}_\ell$-homology, after any finite extension of k. Indeed, after its cusps split, the actual homology sequence is

$$0 \longrightarrow \mathbb{Q}_\ell(1)^{r_V-1} \longrightarrow H_1(V_{\bar{k}},\mathbb{Q}_\ell) \longrightarrow V_\ell \operatorname{Jac}(\bar{V}) \longrightarrow 0 \tag{F.13}$$

for $r_V>0$; in the proper case the first term is zero. A map from the first term to the trivial representation is zero, because the ℓ-adic cyclotomic character has infinite image over every field finitely generated over $\mathbb{Q}$. For the last term put $A=\operatorname{Jac}(\bar{V})$. The Weil pairing $T_\ell A \times T_\ell(A^\vee)\to\mathbb{Z}_\ell(1)$ identifies $(V_\ell A)^*$ with $V_\ell(A^\vee)(-1)$. Its invariants under the cyclotomic kernel vanish by the same AE1 theorem, applied to the dual abelian variety. Hence its G_k-invariants vanish as well. Applying $\operatorname{Hom}_{G_k}(-,\mathbb{Q}_\ell)$ to (F.13) proves the assertion and contradicts (F.12).

It remains to treat a tree. For a proper curve, the existence of one node subgroup already contradicts Stix's Theorem 20 for a proper realizing curve. Such a realizing curve must indeed be proper: the continuous group $H^2(\Pi,\mathbb{Z}/\ell)$ is $\mathbb{Z}/\ell$ for a proper hyperbolic curve and zero for an affine one, by the curve $K(\pi,1)$ comparison and the surface/free group presentations. An isomorphism of geometric groups induces an isomorphism on this cohomology.

For an affine curve with $r>0$ and tree dual graph, the same graph presentation gives an exact sequence of G_k-modules

$$0 \longrightarrow \mathbb{Q}_\ell(1)^{r-1} \longrightarrow H_1(\Pi,\mathbb{Q}_\ell) \longrightarrow \bigoplus_v V_\ell \operatorname{Jac}(C_v) \longrightarrow 0, \tag{F.14}$$

after a finite extension defining the smooth normalized components C_v, nodes, branches, and marks. Here is the rank computation: at each vertex the sum of its boundary classes is zero; eliminating an end vertex of the tree expresses its unique incident node class as a sum of its cusp classes. Successively removing end vertices expresses every node class in terms of the r cusp classes, with only their single total-sum relation. Quotienting by these classes fills every

puncture on every component and gives precisely the last term of (F.14). The action there is the ordinary arithmetic action of the actual curves C_v/k; log inertia is trivial on this quotient. These facts follow from the same functorial vertex maps in (F.10), so they hold after every restriction along s.

The torus/abelian-variety vanishing used above gives

$$\dim_{\mathbb{Q}_\ell} \mathrm{Hom}_{G_k}\big(\mathbb{Q}_\ell(1), H_1(\Pi, \mathbb{Q}_\ell)\big) = r - 1. \tag{F.15}$$

A smooth realizing curve must be affine by H^2 as above, including the distinction between $r = 1$ and $r = 0$. Applying (F.13) to it shows that the same dimension is $r_V - 1$; hence $r_V = r$. Stix's criterion then gives it exactly r maximal essentially cyclotomic classes, contradicting the $r + n$ already constructed. Every argument survives a further finite extension of k. This proves the nonrealization assertion in its full finite-extension form and, in particular, settles the two isomorphic CM elliptic components of F17.2. □

F17.7. Condition e detects potential good reduction *[Proved]*

[F17; §3, manuscript pp. 50–51; scans 101–103; the correction proposed in the reader's notes.] Let R, K, k, U have the hypotheses of F17.6 except that a stable model over R is not assumed. Fix any continuous residue splitting $s : G_k \to G_K$. The following three assertions are equivalent: U has good marked reduction after a finite extension of K; the outer action obtained through s is realized by a smooth hyperbolic curve over k; that action is realized after restriction to $G_{k'}$ for some finite extension k'/k. Consequently "some splitting" and "every splitting" give the same condition e), and allowing a finite residue extension does not enlarge it in these hypotheses.

Proof. We first make the finite-extension comparison explicit. If K_1/K is finite inside the chosen separable closure, put

$$H = s^{-1}(G_{K_1}) \subset G_k, \qquad G_L = G_{K_1} \cap p^{-1}(H), \qquad p : G_K \to G_k. \tag{F.16}$$

The subgroup H is open, so $H = G_{k_2}$ for a finite extension k_2/k. It is contained in the image of G_{K_1} in G_k. The fixed field L of the displayed open subgroup is finite over K, contains K_1, and has residue field k_2: the restriction $p : G_L \to H$ is surjective, because every $h \in H$ has a lift in G_{K_1}. Moreover $s(H) \subset G_L$, and $s|_H$ is a section of that residue map. The arithmetic fundamental-group base-change square for a field extension gives

$$\pi_1(U_L) = \pi_1(U) \times_{G_K} G_L, \qquad \pi_1(U_L) \times_{G_L, s|_H} H = E_s \times_{G_k} H. \tag{F.17}$$

These are actual continuous extension isomorphisms, up to geometric inner conjugation from base points. Residue splittings exist in this characteristic-zero setting: write a separable closure as the maximal unramified extension with all compatible roots of a uniformizer adjoined, and extend each residue automorphism by fixing those roots. The Kummer description used in F17.6 proves the resulting section is continuous.

Suppose U_{K_1} has good marked reduction. We prove realization over k itself, including its descent map. Enlarge K_1/K to a finite Galois extension, which preserves good reduction, and write $G = \mathrm{Gal}(K_1/K)$ and k_1 for its residue field. Uniqueness of the stable marked model extends the natural semilinear G-action on the generic pair to its smooth marked model: for each $\sigma \in G$ the generic isomorphism $\sigma^*(\bar{U}_{K_1}, D_{K_1}) \to (\bar{U}_{K_1}, D_{K_1})$ extends uniquely between the two stable models. The cocycle identities hold because they hold on the generic fibre and the extension is unique. The marked divisor is unordered here; permuting labels merely relabels the ordered stable model and does not obstruct this uniqueness argument. Reducing gives isomorphisms $\lambda_\sigma : \sigma^*(C_1, D_1) \to (C_1, D_1)$ of the special pair over k_1, where σ acts on k_1 by its residue automorphism.

The continuous homomorphism $\alpha : G_k \xrightarrow{s} G_K \to G$ has finite image and projects to the ordinary restriction map on k_1. Its kernel is G_{k_2} for a finite Galois extension k_2/k containing k_1. For $\gamma \in \mathrm{Gal}(k_2/k)$, the base change of $\lambda_{\alpha(\gamma)}$, combined with the scalar action of γ on k_2, defines an isomorphism

$$\gamma^*(C_1, D_1)_{k_2} \longrightarrow (C_1, D_1)_{k_2}. \tag{F.18}$$

Here $\alpha(\gamma)$ means the value of α on any lift; the definition is independent of that lift by the choice of k_2. The identities for the λ_σ make (F.18) an actual finite Galois descent datum.

We also verify its effectiveness. The line bundle $\omega_{C_1}(D_1)^{\otimes 3}$ is very ample on every smooth hyperbolic marked curve: its degree is at least $2g+1$ when $g \geq 1$, and positive when $g = 0$. Its formation is functorial in the pair, so (F.18) gives a compatible linear action on its space of sections. For a finite Galois extension L/k and a finite-dimensional semilinear L-space M, the natural map $L \otimes_k M^{\mathrm{Gal}(L/k)} \to M$ is an isomorphism. Explicitly, trace-dual bases a_i, b_i of L/k give invariant vectors $v_i = \sum_\gamma \gamma(b_i v)$ and $v = \sum_i a_i v_i$; invertibility of the embedding matrix $(\gamma(a_i))$ gives injectivity. Apply this to the section space and to each graded piece of the stable homogeneous ideals of the embedded curve and its marked divisor. They descend to a projective marked pair (C_s, D_s) over k, whose base change to k_2 is the displayed pair. Smoothness and the finite-étale property of the marks descend by this faithfully flat field extension. Put $V_s = C_s \setminus D_s$.

Here is the comparison of the full arithmetic actions, including the action of the infinite subgroup G_{k_2}. Put

$$Q = G \times_{\mathrm{Gal}(k_1/k)} G_k, \qquad q : G_K \longrightarrow Q, \quad \delta \longmapsto (\delta|_{K_1}, p(\delta)).$$

The map q is surjective: lift the first coordinate to G_K and adjust its residue by an element of G_{K_1}, whose residue map surjects onto G_{k_1}. Its kernel is $G_{K_1} \cap I_K = I_{K_1}$. Good reduction over K_1 makes this kernel act trivially in $\mathrm{Out}(\Pi)$, so ρ_U factors through Q. On the geometric special pair, (σ, g) acts through λ_σ together with g on the scalars $\bar{k}$; equality of their restrictions to k_1 makes this action well defined. The good-reduction comparison for covers is functorial in isomorphisms of marked models. It therefore identifies this Q-action on the special geometric group with the factored generic outer action. Finally $q \circ s(g) = (\alpha(g), g)$, exactly the action obtained from the descent datum (F.18). Thus the descent action of G_k on $\pi_1((V_s)_{\bar{k}})$ is identified with $\rho_U \circ s$ on Π. In particular, on $G_{k_2} = \ker \alpha$ the second coordinate retains the ordinary arithmetic action on the curve over k_2; no factorization of that action through a finite group is being asserted. Since Π is centre-free, the extension reconstruction in C4 identifies their full arithmetic extensions over G_k as well. This proves realization over k for the arbitrary chosen splitting. Realization after a finite residue extension follows by base change.

For the converse choose K_1/K over which the marked curve has stable reduction. The precise source is Knudsen, Theorem 2.7, printed p. 179 [Knu83]: for every $2g - 2 + r > 0$ the stack of stable r-pointed genus-g curves is proper and smooth over $\mathbb{Z}$. After first splitting and ordering the finitely many geometric cusps, its properness, applied to the classifying generic point, supplies the stable marked model after a finite trait extension; separatedness gives uniqueness and compatibility with further base change. The unmarked semistable reduction statement is also Stacks, Theorem 55.18.1 [The26c]. If the geometric stable special fibre has a node, it retains that node after every finite base change: base change of a stable marked curve is stable, and its special fibre is the corresponding field extension of that nodal fibre. Use (F.16)–(F.17), and intersect H with any open residue subgroup over which the proposed realizing curve is defined. F17.6 then contradicts that realization. Thus the stable special fibre is smooth, which says exactly that U_{K_1} has good marked reduction. □

F17.8. Individual node invariance needs the finite-extension qualification *[Refuted]*

[F17; §3, manuscript pp. 50–51; scans 101–103; the literal individual-node assertion in supplied H2(b).] The full residue Galois group can permute the node groups. The strongest uniform conclusion used in F17.6 is invariance under an open subgroup fixing the node and its branches, with cyclotomic action there.

Proof. Over $\mathbb{Q}[[t]]$ take the projective hyperelliptic family

$$y^2 = (x^2+1)^2(x^2-2) + t = x^6 - 3x^2 - 2 + t. \tag{F.19}$$

Its generic sextic has discriminant $-46656\,t^2(t-2)(t-4)^2$, which is nonzero in $\mathbb{Q}((t))$; the program checks this formula. Its special fibre is integral with precisely the two ordinary nodes $(x,y) = (i,0), (-i,0)$. The coefficient of $(x-i)^2$ in the sextic at i is $12 \neq 0$, and the same holds at $-i$. At infinity the two points are smooth. Normalizing by $w = y/(x^2+1)$ gives $w^2 = x^2 - 2$, a smooth genus-zero curve; the four inverse images of the nodes make this an irreducible stable genus-two fibre. The total space is regular at each node because the derivative of the defining equation with respect to t is -1. Thus this is a stable degeneration. An element of $G_{\mathbb{Q}}$ taking i to $-i$ interchanges its two geometric nodes. By the graphic comparison (F.10), every residue splitting gives this same permutation of their group classes. Proposition 1.2 of [Moc07] makes the two classes distinct. Hence neither class is invariant under the whole $G_{\mathbb{Q}}$. A finite extension fixes both nodes and their branches and recovers exactly the conclusion of F17.6. □

Combining F17.1–F17.8, in residue characteristic zero the justified arrows are a)⇔b)⇒c)⇒d) and a),b)⇒e). The converses c),d)⇒a), the arrows c),d)⇒e), and e)⇒a) all have the explicit counterexamples just given. Condition e), with the finite extension permitted in the manuscript, is equivalent to potential good reduction under the finitely generated residue-field hypotheses of F17.7. The predicate in f), "pas trop loufoque (à définir convenablement)", has no definition in the manuscript, so it supplies no additional mathematical condition to put in an equivalence table.

7 G. Cote 142: surfaces and categorical completion

Throughout this section a surface is connected, oriented, and obtained by removing a finite set D from a compact oriented surface $\overline{S}$. Its type is (g,ν), with $\nu = |D|$, and it is hyperbolic when $2g - 2 + \nu > 0$. Homeomorphism groups have the compact-open topology. The arrows under consideration are finite covering maps; the superscript + requires preservation of orientation. We write $A(S) = \mathrm{Homeo}^+(S)$, $A_0(S)$ for its identity component, and $M(S) = A(S)/A_0(S)$. Punctures are preserved as a set, unless explicitly labelled. This fixes the conventions needed in scans 3–14.

G1. Local form and invariant complex structure *[Known]*

A nonidentity orientation-preserving finite-order homeomorphism of a connected surface has isolated fixed points, each with fixed-point index +1. A cyclic finite group acting by orientation-preserving homeomorphisms on a surface of finite type preserves a complex structure of a smooth algebraic curve of that type.

Manuscript passage: Cote 142, opening, continuation of manuscript p. 1 (scan 6), and p. 3 (scan 7). The reading order is 4, 6, 5, 7.

Proof. We use the attached paper of Constantin–Kolev [CK03, Lemma 2.5; Theorem 3.1; Proposition 3.3(1)]. Lemma 2.5 supplies arbitrarily small invariant closed disc neighbourhoods of a fixed point of a periodic homeomorphism of an arbitrary surface. Theorem 3.1 conjugates

a periodic homeomorphism of a closed disc to an orthogonal linear map. Proposition 3.3(1) says that a nonidentity orientation-preserving such map has just one fixed point, in the interior. These hypotheses apply to the restriction to each invariant disc. A restriction that is the identity on an open set forces the global map to be the identity: the set of points having an identity neighbourhood is open, and an invariant disc at any point of its boundary would be a rotation with an open fixed subset, hence the identity. This set is therefore also closed. Connectedness proves the assertion. The remaining local rotations are nontrivial, so their unique fixed points are isolated. The degree of $z \mapsto z - e^{i\theta}z$ on a small circle is $+1$. The orientation hypothesis in this local assertion is necessary: reflection of a sphere fixes its equator. A deck transformation between oriented connected surfaces preserves orientation, since the local orientation signs in $p\sigma = p$ cancel. Thus the hypothesis holds in the covering argument of scan 6. In G2 below it follows directly from the homology hypothesis.

The finite-end topology extends every homeomorphism of S uniquely to its compactification $\overline{S}$: neighbourhoods of an added end are the punctured discs defining that end. Apply the preceding local description to the cyclic action on $\overline{S}$. Its quotient B is a compact oriented topological surface, and the quotient map is locally $z \mapsto z^e$, where e is the stabilizer order. There are only finitely many branch points, because each nonidentity group element has a closed discrete fixed set on the compact surface. Choose a complex algebraic curve of the genus of B and a homeomorphism from B to its analytic surface. Such curves are supplied explicitly by $\mathbf{P}^1$ in genus zero and by the smooth projective model of $y^2 = \prod_{j=1}^{2h+2}(x - a_j)$, with distinct complex a_j, in genus $h \geq 1$; cutting the branched double covering computes that genus. Remove the branch values. The finite topological covering of this open complex algebraic curve is algebraic by SGA 1 XII, Theorem 5.1 [G+71]: for a complex scheme locally of finite type, analytification equates finite *étale* algebraic covers with finite analytic covers. Normalizing the compact curve in the resulting function field fills each puncture by the unique disc covering $z \mapsto z^e$. The local topological conjugacies identify its analytic surface with $\overline{S}$. The deck transformations are algebraic, and deleting the given invariant finite set D proves the assertion for S. □

G2. Periodic homological rigidity, with the level-two exception *[Proved]*

For a hyperbolic surface S, a finite-order homeomorphism acting trivially on $H_1(S, \mathbf{Z})$ is the identity. The same conclusion follows from trivial action on $H_1(S, \mathbf{Z}/n)$ for any $n \geq 3$. In particular, a periodic homeomorphism isotopic to the identity is the identity. The assertion with $n = 2$ is false: $z \mapsto -z$ on $(\mathbf{R}^2/\mathbf{Z}^2) \setminus \{0\}$ is a counterexample of type $(1, 1)$.

Manuscript passage: Cote 142, opening, manuscript p. 2, scan 5; continuation p. 3, scan 7.

Proof. We first give the entirely topological proof. The exact sequence

$$0 \longrightarrow \mathbf{Z}^D/\mathbf{Z}\Big(\sum_{d\in D} e_d\Big) \longrightarrow H_1(S, \mathbf{Z}) \longrightarrow H_1(\overline{S}, \mathbf{Z}) \longrightarrow 0$$

holds for $D \neq \varnothing$ by cutting out disjoint discs and applying the cellular homology sequence; the e_d are positively oriented small loops. For $D = \varnothing$ the right map is the identity. An orientation-preserving map acts on the left term by the permutation of D, and an orientation-reversing map acts by minus that permutation.

Trivial integral action forces preservation of orientation. For $g > 0$ this follows from the intersection pairing on the quotient $H_1(\overline{S}, \mathbf{Z})$: an orientation reversal changes its sign, while the pairing takes the value 1. For $g = 0$ we have $\nu \geq 3$; the trace of a permutation on $\mathbf{Q}^D/\mathbf{Q}(\sum e_d)$ is the number of its fixed punctures minus 1, which is at least -1. Thus that permutation cannot be -1 on a space of dimension $\nu - 1 \geq 2$. Orientation reversal again cannot act trivially. Moreover a permutation inducing the identity on this quotient fixes every puncture for $\nu \geq 2$:

the equality $e_{\sigma(d)} - e_d \in \mathbf{Z}\sum e_d$ forces its left side to be zero. For $\nu = 1$ its one puncture is fixed automatically.

Suppose the map is nonidentity. Its extension to $\overline{S}$ has finitely many fixed points and each has index $+1$, by G1. The cellular Lefschetz formula gives

$$\#\operatorname{Fix}(\overline{\sigma}) = 1 - \operatorname{tr}(\sigma_* \mid H_1(\overline{S}, \mathbf{Q})) + 1 = 2 - 2g.$$

One may compute this formula by triangulating the quotient surface with the branch values as vertices, lifting that triangulation, and taking a barycentric subdivision: a nonfixed simplex contributes zero to the alternating trace, and each fixed vertex contributes one. Thus this use of the formula requires neither a conformal structure nor an algebraic Jacobian. For $g \geq 2$ the displayed right side is negative. For $g = 1$ it is zero, whereas hyperbolicity supplies a fixed puncture. For $g = 0$ it is two, whereas at least three punctures are fixed. Each case is a contradiction.

For the finite-coefficient assertion we prove the integral congruence lemma being used. A finite-order matrix $A \in \mathrm{GL}_r(\mathbf{Z})$ congruent to 1 modulo $n \geq 3$ is 1. Otherwise take a prime-order power B of A, of order q. Choose either an odd prime $p \mid n$, or $p = 2$ when $4 \mid n$. Write $B = 1 + p^s C$, with $C \not\equiv 0 \pmod{p}$; then $s \geq 1$ and, for $p = 2$, $s \geq 2$. The expansion $B^q = 1$, divided by p^s and reduced modulo p, rules out $q \neq p$. For $q = p$, divide instead by p^{s+1}. The first term is C; all subsequent terms are divisible by p, since $s(p-1) - 1 \geq 1$. This contradicts the choice of C. Every $n \geq 3$ has one of the two indicated divisors. Apply the lemma to the finite-order integral action of the homeomorphism and then apply the preceding proof. Finally the displayed involution of a punctured torus has matrix $-I_2$, which is I_2 modulo two, and is nonidentity. The matrix checks are reproduced in `code/G_checks.py`. □

G3. The complete-curve translation assertion used in scan 7 *[Proved]*

For a complex hyperbolic curve Y in its generalized Jacobian torsor, an equality $Y + u = Y$ implies $u = 0$. The assertion involving a nonempty open subset $W \subset Y$ and only $W + u \subset Y$ requires the correction given in Part B; that assertion fails for type $(1, 1)$.

Manuscript passage: Cote 142, opening, manuscript p. 3, scan 7; cote 149, §1, manuscript p. 18, scan 37.

Proof. Write $j : Y \hookrightarrow J^1$ for the Abel map. A translation preserving its whole image defines an automorphism σ of Y. The Abel map identifies $H_1(Y, \mathbf{Z})$ with $H_1(J^1, \mathbf{Z})$: its torus extension has kernel $(\mathbf{C}^*)^{\nu-1}$, the small puncture loops map to its standard lattice modulo their one sum relation, and its quotient map is the usual Abel map to the compact Jacobian, whose period lattice is $H_1(\overline{Y}, \mathbf{Z})$. The two short exact lattice sequences and these explicit maps give the claimed isomorphism; for $\nu = 0$ it is the compact Jacobian statement itself. Thus the translation gives the identity on $H_1(Y, \mathbf{Z})$. The automorphism group of a hyperbolic complex algebraic curve is finite. Here is the relevant verification: for $g \geq 2$ the faithful action on a pluricanonical embedding is an algebraic group whose tangent space is $H^0(Y^{\mathrm{cpt}}, T_{Y^{\mathrm{cpt}}}) = 0$ because this line bundle has degree $2 - 2g < 0$; hence it is zero-dimensional and finite. For $g = 1$, preserving the nonempty puncture set bounds the translation part to finitely many choices. An origin-fixing automorphism lifts to $z \mapsto az$ with $a\Lambda = \Lambda$ for the rank-two period lattice. Covolume gives $|a| = 1$, and a fixed nonzero lattice vector has only finitely many lattice images of the same length. Thus there are finitely many such automorphisms. For $g = 0$, preservation of a set of at least three points gives an injection into its finite permutation group. Thus G2 applies and gives $\sigma = 1$. At one point of $j(Y)$ the equality $j(y) + u = j(y)$ then gives $u = 0$. This argument uses full stability of the curve. For the open-subset form take $Y = E \setminus \{0\}$, $u \neq 0$ and $W = E \setminus \{0, -u\}$ inside $J^1_Y \simeq E$. □

7.1 Covering spaces and their components

G4. Components, stabilizers, and the finiteness of covering orbits *[Known]*

Let X, Y be connected hyperbolic finite-type surfaces. In the space of orientation-preserving finite covering maps $X \to Y$, each connected component is homeomorphic to $A_0(X)$ and is contractible. Its isotopy class has stabilizer, under precomposition by $M(X)$, exactly the finite deck group of that covering, embedded by its mapping classes. There are finitely many $M(X)$-orbits of isotopy classes. As the target Y varies, a finite subgroup of $M(X)$ occurs as such a stabilizer exactly when it has a free realization on X by orientation-preserving homeomorphisms.

Manuscript passage: Cote 142, opening, manuscript pp. 1–3, scans 3,4,6–8.

Proof. The homotopy lifting property for a finite covering $p : X \to Y$ gives the precise isotopy assertion. A path p_t of covering maps has a unique lift $h_t : X \to X$ of p_t, starting at the identity. Lifting locally in an evenly covered disc shows that h_t is a local homeomorphism. It is proper because $h_t^{-1}(K)$ is closed in the compact set $p_t^{-1}(p(K))$ for every compact $K \subset X$. The degree of p_t is the fixed ratio $\chi(X)/\chi(Y)$; hence h_t has covering degree one. Thus h_t is a homeomorphism and $p_t = ph_t$. Conversely ph_t is a path of covers for every isotopy h_t. If $ph = ph'$ with $h, h' \in A_0(X)$, the homeomorphism hh'^{-1} is a deck transformation, of finite order, and G2 makes it the identity. Thus $h \mapsto ph$ is a continuous bijection from $A_0(X)$ to the component of p.

For the inverse topology, fix a base point and a sheet above a small evenly covered disc. The branch of the lift at that point varies continuously with the map. On a compact subset of X, use finitely many evenly covered neighbourhoods and finitely many paths joining them to the base point; uniqueness of lifted paths then gives uniform continuity of the inverse lifting operation on that compact set. These are exactly the subbasic neighbourhoods of the compact-open topology. The bijection is a homeomorphism. We also verify that path components are open, so that they are the connected components asserted in scan 8. Choose a compact core carrying a finite set of based generators of $\pi_1(X)$. A sufficiently small change of p on this core preserves its based homomorphism to $\pi_1(Y)$, after identifying the nearby base points along a small disc. The changed cover p' therefore lifts through p to a local homeomorphism $h : X \to X$. It is proper, because $h^{-1}(K)$ is closed in the compact set $(p')^{-1}(p(K))$ for every compact K. Both covers have the same degree $\chi(X)/\chi(Y)$, so h has degree one and is a homeomorphism. The injectivity of p_* gives $[h_*] = 1$ in $\mathrm{Out}(\pi_1 X)$. The Dehn–Nielsen–Baer injection for a hyperbolic surface, including punctured genus zero [FM, Theorem 8.8], makes h isotopic to the identity. Thus the neighbourhood just constructed belongs to the path component of p. The open path components are also closed, and the asserted identity of connected and path components follows.

For noncompact X the contractibility is Yagasaki [Yag00, Theorem 1.1(ii)]: the identity component of the compact-open homeomorphism group of a connected separable PL 2-manifold is contractible outside the explicitly listed plane, annulus and punctured-projective-plane exceptions. None is hyperbolic and orientable. For compact X of genus at least two, the same source gives the compact case in Lemma 2.1(i), with $N = X$ and the fixed subpolyhedron empty. Its hypotheses exclude only the sphere, torus and the other displayed elementary exceptions. The supplied Earle–Eells announcement [EE67, Theorem 1(c)] proves the corresponding result for smooth diffeomorphisms; the Yagasaki statement is the one used here for homeomorphisms.

An isotopy $pa \simeq p$ gives $pa = ph$ for some $h \in A_0(X)$; therefore ah^{-1} is a deck transformation representing $[a]$. Its mapping class is unique by G2. This proves the asserted stabilizer. The orbits are the isomorphism classes of finite covers of Y with total surface homeomorphic to X. Euler characteristic fixes their degree to $d = \chi(X)/\chi(Y)$. A group generated by r elements has at most $(d!)^r$ homomorphisms to S_d, so has only finitely many subgroups of index d. This proves finiteness, including the case where the required degree is not a positive integer and there are no arrows. Every deck action is free. Conversely a free finite orientation-preserving action gives a

finite unramified quotient cover $X \to X/F$, whose full deck group is F: every automorphism over the quotient is determined by the image of one point, and the fibre is exactly its F-orbit. The quotient has Euler characteristic $\chi(X)/|F| < 0$, as required. □

G5. The finite-subgroup margin of scan 8 *[Refuted]*

The assertion that every finite subgroup of $M(X)$ occurs as a deck stabilizer for the fixed source X is false. A hyperelliptic involution on a closed genus-two surface gives a subgroup of order two which cannot occur.

Manuscript passage: Cote 142, opening, manuscript p. 3, scan 8, first NB.

Proof. The double branched cover of the sphere at six distinct points is a closed genus-two surface. Its deck involution has six fixed points; it has nontrivial mapping class by G2 and generates a subgroup of order two. A free orientation-preserving involution on this surface would have closed orientable quotient of Euler characteristic $-2/2 = -1$, whereas every closed orientable surface has even Euler characteristic. Thus this particular subgroup cannot have a free realization. The realization theorem of Kerckhoff [Ker80, Theorem 1] provides isometric realizations of finite mapping-class subgroups; it allows fixed points. The exact replacement for the margin is G4. □

7.2 A precise profinite envelope

For this subsection C_0 is a small category, $A_X = \mathrm{Aut}_{C_0}(X)$, and all A_X initially have the discrete topology. For $u : X \to Y$ set

$$S_u = \{a \in A_X : ua = u\}, \qquad G_u = \{(a, b) \in A_X \times A_Y : bu = ua\}, \qquad H_u = \mathrm{pr}_Y(G_u).$$

The condition of scan 11 is $[A_Y : H_u] < \infty$ for every u. A bar on a subgroup denotes its closure in the explicitly displayed profinite ambient group. The symbol $\widehat{A}$ denotes the full profinite completion of the discrete group A.

G6. Hausdorff profinite envelope under the finite-index condition *[Proved]*

For every small C_0 satisfying the preceding finite-index condition there is a category $\widehat{C}_0$ with the same objects and

$$\mathrm{Hom}_{\widehat{C}_0}(X, Y) = \coprod_{uA_X \in \mathrm{Hom}_{C_0}(X,Y)/A_X} \overline{S_u}\backslash\widehat{A}_X. \tag{G.1}$$

Its Hom spaces are locally compact Hausdorff and totally disconnected; composition is continuous and associative, and extends composition in C_0. Each double orbit also equals the quotient of $\widehat{A}_X \times \widehat{A}_Y$ by the closure of G_u, with the opposite-action convention dictated by $bu = ua$. The functor $C_0 \to \widehat{C}_0$ has dense image in each Hom space. It is conservative, and

$$\mathrm{Isom}_{\widehat{C}_0}(X, Y) = \mathrm{Isom}_{C_0}(X, Y) \overset{A_X}{\times} \widehat{A}_X, \qquad \mathrm{Aut}_{\widehat{C}_0}(X) = \widehat{A}_X. \tag{G.2}$$

The displayed contracted product for isomorphisms is a torsor product, so no closure correction is needed there. Among Hausdorff topological categories with continuous composition and profinite automorphism groups, every functor from C_0 extends uniquely to a continuous functor from this envelope whenever the target Hom topology induces its stated profinite automorphism-group topology. This characterizes the envelope up to a unique isomorphism fixing C_0.

Manuscript passage: Cote 142, opening, manuscript pp. 4–6, scans 9–12 and 14; formulas (12)–(17), NB of scan 11, and conservativity NB of scan 14.

Proof. We construct the actions and composition; their existence is essential to the uniqueness assertion. A right orbit of u is $S_u\backslash A_X$. Its completion in (G.1) is the inverse limit of the finite sets $S_uN\backslash A_X$, where N ranges over the finite-index normal subgroups of A_X. Indeed the compatible finite cosets are exactly $\overline{S}_u\backslash\widehat{A}_X$, by compactness.

For $b \in H_u$ choose $a \in A_X$ with $bu = ua$. Such an a normalizes S_u: applying b and b^{-1} to $us = u$ gives both inclusions. Its class in $N_{A_X}(S_u)/S_u$ is independent of the choice, and $b \mapsto aS_u$ is a homomorphism. On each finite set $S_uN\backslash A_X$, left multiplication by a is well-defined. It factors through a finite quotient of H_u, because its kernel is the inverse image of $S_u(N \cap N_{A_X}(S_u))/S_u$. The homomorphism therefore extends continuously from H_u to $\widehat{H}_u$ on every finite set, and the extensions are compatible. The finite-index inclusion $H_u \subset A_Y$ induces $\widehat{H}_u \simeq \overline{H}_u \subset \widehat{A}_Y$: every finite-index subgroup of H_u has finite index in A_Y, and its A_Y-normal core gives the required cofinal neighbourhoods.

Choose finitely many representatives b_i for A_Y/H_u. The double orbit of u is the disjoint union of the right orbits of b_iu. The preceding actions, together with this finite permutation of orbits, extend its left A_Y-action to a continuous left $\widehat{A}_Y$-action. They commute with the right $\widehat{A}_X$-action. The stabilizer of u is exactly $\overline{G}_u$ in the product. To check the nonautomatic inclusion, take a stabilizing pair $(\widehat{a}, \widehat{b})$ and prescribed open normal neighbourhoods N_X, N_Y in the completed endpoint groups. The component permutation puts $\widehat{b}$ in $\overline{H}_u$. Choose $b \in H_u$ close to $\widehat{b}$ modulo N_Y, and close enough that its action on $S_u(N_X \cap A_X)\backslash A_X$ has the same value as $\widehat{b}$. A lift a_b with $bu = ua_b$ then belongs to the left S_u-coset of $\widehat{a}$ modulo N_X. Multiplication of a_b on the left by a suitable element of S_u preserves $bu = ua_b$ and puts it in the prescribed neighbourhood of $\widehat{a}$. Every product neighbourhood therefore meets G_u. Consequently the double-orbit topology is the profinite quotient topology. There are $[A_Y : H_u]$ right orbits in it, so every right orbit is open. This proves the NB of scan 11 and the equivalence of the two *Hausdorff* versions of (17).

For an original arrow $v : Y \to Z$, define a continuous map

$$L_v : \mathrm{Hom}_{\widehat{C}_0}(X, Y) \longrightarrow \mathrm{Hom}_{\widehat{C}_0}(X, Z), \qquad u\alpha \longmapsto (vu)\alpha.$$

It is well-defined because $S_u \subset S_{vu}$, and hence the same inclusion holds for their closures in $\widehat{A}_X$. For completed arrows $f = u\alpha$ and $g = v\beta$ define

$$g \circ f = L_v(\beta \cdot u)\, \alpha. \tag{G.3}$$

Changing α within its left $\overline{S}_u$-coset does not change this expression, by equivariance. Changing β within its left $\overline{S}_v$-coset does not change it either: the identity $L_v(sh) = L_v(h)$ holds first for $s \in S_v$ and original h, and then for $s \in \overline{S}_v$ and completed h by continuity and density. Changing the representative u or v of an original right orbit merely performs the corresponding translation, so (G.3) is independent of those choices too. On a product of two completed right orbits it is the quotient of a continuous map from $\widehat{A}_X \times \widehat{A}_Y$. Those orbit pieces are compact and the quotients Hausdorff, so this proves joint continuity. The disjoint union topology gives it globally. The two triple compositions are continuous maps to a Hausdorff Hom space and agree on the dense set of original triples. They agree everywhere. The same argument supplies the identity laws.

The orbit of the identity arrow is precisely $\widehat{A}_X$ and is open. Suppose f has inverse g in the completed category. Choose neighbourhoods of f and g so that both products lie in the respective identity orbits, and approximate both arrows by original ones f_0, g_0 in these neighbourhoods. Their original composites g_0f_0 and f_0g_0 lie in identity right orbits, hence are original automorphisms. Thus f_0 is an original isomorphism. Moreover f_0 lies in the same right orbit as the chosen original representative of f, by taking that orbit as its neighbourhood. That representative is therefore an isomorphism. This proves (G.2), conservativity, and density of original isomorphisms among completed ones.

Finally let $F : C_0 \to D$ be a functor to a target as in the statement. The group homomorphism $A_X \to \mathrm{Aut}_D(FX)$ extends to $\widehat{A}_X$, since every finite quotient of its profinite target has finite-index

kernel. The continuous orbit map $\alpha \mapsto F(u)F(\alpha)$ has closed stabilizer in the Hausdorff target; it kills $\overline{S}_u$. It therefore defines the desired map on each component of (G.1). The maps commute with composition by density, and continuity on disjoint components is continuity. Any other extension agrees on a dense subset, hence agrees everywhere. Applying this property in both directions proves the uniqueness claimed in the statement. □

G7. Two missing qualifications in the general categorical formulas *[Refuted]*

The set-theoretic contracted products in scan 12 need not be Hausdorff under the finite-index condition alone. Also the abstract completion $\widehat{G}_u$ need not embed in the product of the completed endpoint automorphism groups. The corrected subgroup in G6 is $\overline{G}_u$.

Manuscript passage: Cote 142, opening, manuscript pp. 5–6, scans 12 and 14, (17) and the final NB.

Proof. Take two objects X, Y, with $A_X = \mathbf{Z}$, $A_Y = 1$, a single arrow $u : X \to Y$ satisfying $ua = u$ for all a, and no arrows $Y \to X$. All endomorphisms are the stipulated automorphisms. The condition on H_u is automatic. The raw contracted product for this orbit is $\mathbf{Z}\backslash\widehat{\mathbf{Z}}$; it is not Hausdorff because $\mathbf{Z}$ is dense and proper in $\widehat{\mathbf{Z}}$. Its separated quotient is one point, exactly (G.1). Properness follows, for example, from the uncountability of $\mathbf{Z}_2$, a quotient of $\widehat{\mathbf{Z}}$, whereas $\mathbf{Z}$ is countable.

For the second assertion take $A_X = (\mathbf{Q}, +)$, $A_Y = 1$ and $\mathrm{Hom}(X, Y) = \mathbf{Z}\backslash\mathbf{Q}$, with the translation action and again no reverse arrows. The stabilizer of the zero coset is $G_u = \mathbf{Z}$. The group $\mathbf{Q}$ has no nontrivial finite quotient: multiplication by the order of such a quotient is surjective on $\mathbf{Q}$ and zero on the quotient. Thus $\widehat{A}_X = 1$, whereas $\widehat{G}_u = \widehat{\mathbf{Z}} \neq 1$. The canonical map to the completed product has this entire group as kernel. This proves the need for the closure in the general statement. □

G8. The surface category and the marked-factorization assertion *[Proved]*

The isotopy category of orientation-preserving finite covers of connected hyperbolic surfaces satisfies the condition of G6. Its finite deck groups embed in the profinite completions of the source mapping-class groups. For this category, $\widehat{G}_u \to \widehat{M}(X) \times \widehat{M}(Y)$ is injective and its image is the arrow stabilizer. Every completed arrow has a factorization $u\widehat{f}_0$, where u is an original finite cover and $\widehat{f}_0$ a completed isomorphism; the resulting *marked factorization* is determined up to a unique original isomorphism over the target.

Manuscript passage: Cote 142, opening, manuscript pp. 5–6, scans 11 and 14.

Proof. An original degree-d cover is represented by a conjugacy class of an index-d subgroup of $\pi_1(Y)$. There are finitely many such subgroups by the permutation-representation argument in G4. The group $M(Y)$ acts on this finite set. Its stabilizer is exactly H_u: a homeomorphism of Y lifts to the cover precisely when its induced outer automorphism preserves the conjugacy class of that subgroup, by the covering-space lifting criterion. A lift is a homeomorphism, and changing representatives through an isotopy changes its lift through an isotopy. This proves the finite-index condition with the specified comparison map.

By G2 the finite representation $M(X) \to \mathrm{GL}(H_1(X, \mathbf{Z}/3))$ is injective on the deck group: its kernel contains no nonidentity finite-order homeomorphism. Hence $S_u \to \widehat{M}(X)$ is injective, and its image is finite and closed. We also prove the requested residual finiteness for every hyperbolic type. Put $P = \pi_1(X)$, a finitely generated free group when X is punctured and a closed surface group otherwise. Paris [Par07, Theorem 1.4] proves that the kernel of $\mathrm{Out}(P) \to \mathrm{GL}(H_1(P, \mathbf{F}_p))$ is residually p. The Dehn–Nielsen–Baer injection [FM, Theorem 8.8] embeds $M(X)$ into $\mathrm{Out}(P)$, including the punctured genus-zero case. Its intersection K with this kernel is normal of finite

index and residually finite. An element outside K is detected by $M(X)/K$. For $x \in K \setminus \{1\}$ choose a finite-index normal subgroup N of K missing x. The intersection of its conjugates by a finite set of coset representatives for $M(X)/K$ is normal of finite index in $M(X)$ and still misses x. These finite quotients prove residual finiteness, with the comparison maps explicitly specified.

The sequence $1 \to S_u \to G_u \to H_u \to 1$ gives $\widehat{G}_u/\overline{S}_u = \widehat{H}_u$: both sides classify exactly the finite quotients killing S_u. The kernel of the map $\widehat{G}_u \to \widehat{M}(X) \times \widehat{M}(Y)$ therefore lies in the finite subgroup S_u, since $\widehat{H}_u \hookrightarrow \widehat{M}(Y)$. Its first projection is injective on that subgroup, so the kernel is trivial. Its compact image is the closure $\overline{G}_u$, identified in G6.

Every point of a component of (G.1) is $u\alpha$, which proves existence of the factorization. Compare two such expressions. Their equality puts the original covers in the same source-automorphism orbit, hence supplies an original isomorphism between the covers. After using one such isomorphism to identify them, the discrepancy between the two markings is an element of $\overline{S}_u = S_u$. There is exactly one such deck transformation, because $S_u \to \widehat{M}(X)$ is injective. Correcting by that deck transformation gives the unique isomorphism respecting both markings. This names the ambiguity suppressed in the unmarked wording. □

G9. Injectivity alone does not give the general uniqueness claim *[Refuted]*

For an arbitrary category satisfying the finite-index condition, the injectivity of every $S_u \to \widehat{A}_X$ does not by itself imply the marked-factorization uniqueness asserted in scan 14. Closedness of its image is also needed. Finite stabilizers, as in G8, have this property.

Manuscript passage: Cote 142, opening, manuscript p. 6, scan 14.

Proof. Use the first two-object category in G7. Here $S_u = \mathbf{Z}$ embeds in $\widehat{\mathbf{Z}}$. Choose $\alpha \in \widehat{\mathbf{Z}} \setminus \mathbf{Z}$. The unique completed arrow $X \to Y$ has the two factorizations $u \circ 1$ and $u \circ \alpha$. An original comparison of these marked factorizations would be an integer whose completed image is α, which does not exist. Replacing injectivity by a closed embedding makes the proof of G8 apply without the finiteness assumption: the discrepancy in $\overline{S}_u$ is then a unique element of S_u. □

G10. The category with already profinite automorphism groups *[Proved]*

Let C be a small category with profinite automorphism groups A_X. For every arrow $u : X \to Y$, suppose that $G_u = \{(a, b) : bu = ua\}$ is closed in $A_X \times A_Y$ and its image H_u in A_Y has finite index. Give each double orbit its compact quotient topology, and give each Hom set the disjoint union topology. Composition is continuous. The image H_u is closed and open, and the right A_X-orbits are open in each double orbit.

Manuscript passage: Cote 142, manuscript pp. 4–5, scans 10–11, especially (14)–(16).

Proof. The closed subgroup G_u is compact. Its projection to the Hausdorff group A_Y is compact and therefore closed. A closed subgroup of finite index is open: its complement is a finite union of closed cosets. There are $[A_Y : H_u]$ right orbits in the double orbit of u; each is compact, and its complement is a finite union of compact subsets. Thus each is open and closed. For fixed $u : X \to Y$ and $v : Y \to Z$, the map $b \mapsto vbu$ is continuous on H_u: its composite with the compact quotient map $G_u \to H_u$ is $(a, b) \mapsto (vu)a$, continuous by the quotient definition of the target orbit. On a coset cH_u the same argument uses vc in place of v. There are finitely many open cosets, so continuity holds on A_Y. On a product of double orbits, composition is consequently the quotient of the continuous map

$$(a, b, c, d) \longmapsto c\, v\, (d^{-1}b)\, u\, a^{-1}.$$

The quotient maps from compact groups onto the Hausdorff orbits, and their products, are quotient maps. This proves continuity on each product of open pieces and hence on the entire product of Hom spaces. No new assertion of associativity is needed here: the multiplication is the original composition in C. □

G11. The doubtful homology qualification in scan 5 *[Proved]*

For orientation-preserving periodic homeomorphisms of a hyperbolic finite-type surface, trivial action on ordinary H_1 or on Borel–Moore H_1 with integral coefficients implies the identity. The same is true with coefficients $\mathbf{Z}/n$, $n \geq 3$. Without the orientation-preserving hypothesis, still for periodic homeomorphisms, the Borel–Moore version is false, whereas the ordinary version remains G2. Replacing either by the homology of the compactification alone is also false.

Manuscript passage: Cote 142, manuscript p. 2, scan 5: the doubtful word following H_1, transcribed as "propre".

Proof. Put $\overline{S} = S \cup D$ and $R = \mathbf{Z}$ or $\mathbf{Z}/n$. Relative chains identify $H_1^{\mathrm{BM}}(S, R)$ with $H_1(\overline{S}, D; R)$: a proper arc ending at D becomes a relative chain, and truncating it in successively smaller end discs gives the inverse locally finite class. Intersection defines a pairing between this group and ordinary $H_1(S, R)$. It is perfect, as follows from an explicit basis. Use the usual $2g$ handle cycles on a compact core, together with $\nu - 1$ small puncture loops in ordinary homology; use the handle cycles and $\nu - 1$ arcs from the first $\nu - 1$ ends to the last end in relative homology. The cellular sequences of the punctured polygon and the pair $(\overline{S}, D)$ show these are bases. The intersection matrix has a symplectic handle block and an identity end block, and determinant ± 1. For $D = \varnothing$ only the handle block occurs. Orientation-preserving maps preserve intersection. Thus identity on Borel–Moore homology forces identity on ordinary homology by this perfect pairing. G2 applies. This gives the exact comparison required for that reading of "propre".

For the orientation issue take $S = \mathbf{P}^1(\mathbf{C}) \setminus \{0, 1, \infty\}$ and complex conjugation. It has order two, reverses orientation, and fixes all three ends. The relative-chain description gives

$$H_1^{\mathrm{BM}}(S, \mathbf{Z}) \simeq H_1(\mathbf{P}^1(\mathbf{C}), \{0, 1, \infty\}; \mathbf{Z}) \simeq \ker(\mathbf{Z}^3 \xrightarrow{\mathrm{sum}} \mathbf{Z}).$$

The action is the identity, because the three ends are individually fixed, whereas the map itself is nonidentity. Its action on ordinary H_1 is -1, as seen on positive small loops, consistently with the orientation sign in cap product. Finally $z \mapsto 1 - z$ on the same surface is an orientation-preserving involution and acts trivially on $H_1(\mathbf{P}^1(\mathbf{C}), \mathbf{Z}) = 0$; it is not the identity. These examples keep the alternatives distinct by explicit homology maps. They provide mathematical evidence for the reading ledger; they do not determine the handwritten word. □

8 Z. Additional steps in cote 149, scans 1–40

B11. An affine hyperbolic polycurve need not immerse in a semiabelian variety *[Refuted]*

The semiabelian-immersion assertion on scan 26, repeated parenthetically on scan 32, is false for the class of successive elementary hyperbolic fibrations used in Corollary 1(b). There is a smooth affine hyperbolic polycurve of dimension two every morphism from which to a semiabelian variety contracts its curve fibres; this occurs even when both levels have genus zero. This gives sweep rows ZAB1 and ZAB6 (149, section 1, manuscript pp. 12, 15 and 17, scans 26, 32 and 36), including the parenthetical claim that genus-zero fibres supply a global torus immersion. B5 proves the stated corollaries without this assertion.

Proof. Work over $k = \overline{\mathbb{Q}}$. Let

$$B: \quad v^2 = u(u-1)(u-2)(u-3)(u-4)$$

be its smooth projective genus-two model. Let P_0, P_1 be the Weierstrass points over $0, 1$. The function $u/(u-1)$ has divisor $2P_0 - 2P_1$ and is not a square in $k(B)$: a square root would have divisor $P_0 - P_1$, giving a degree-one function on a genus-two curve. Let C be the normalization of B in $k(B)(\sqrt{u/(u-1)})$. All valuations of this function are even, so this connected double cover is étale: locally remove the even uniformizer power and take a square root of the resulting unit. Write ι_C for its fixed-point-free deck involution.

Let D be another copy of B, let ι_D be its hyperelliptic involution, and choose a fixed Weierstrass point W. Remove any point b_0 from B, its two preimages from C, and W from D; write the resulting affine curves as $B^\circ, C^\circ, D^\circ$. The diagonal involution on $C^\circ \times D^\circ$ is free, and

$$Y = (C^\circ \times D^\circ)/\langle \iota_C \times \iota_D \rangle \longrightarrow B^\circ$$

is a smooth affine fibration with fibre D°. The quotient is affine because it is a finite-group quotient of an affine scheme; it is smooth because the free quotient map is étale. The family becomes the product after the étale cover $C^\circ \to B^\circ$. Its smooth proper compactification with boundary descends from $C^\circ \times D$ and the invariant section W. Both base and fibre have hyperbolic type $(2, 1)$.

Given $f : Y \to G$ with $1 \to T \to G \to A \to 1$ semiabelian, pull it back to $h : C^\circ \times D^\circ \to G$. The composite into A extends to $C \times D$ by the opened [Mil08, I, Theorem 3.2]: rational maps from nonsingular varieties to abelian varieties extend everywhere. By [Mil08, I, Corollary 1.5], this extension has the form $a(c) + b(d) + a_0$. Diagonal invariance implies that $b(\iota_D d) - b(d)$ is constant in d. After choosing W as origin for the Abel map, $b - b(W)$ is induced by a homomorphism $\beta : J_D \to A$. The hyperelliptic involution acts as -1 on J_D, because $d + \iota_D d$ is linearly equivalent to $2W$. Thus -2β is a constant homomorphism, so $\beta = 0$ (its connected image would otherwise be killed by two). The map into A is independent of d.

For every fixed c, the remaining map $D^\circ \to G$ lies in a torus fibre and is constant. Indeed, every unit on $D \setminus \{W\}$ is constant: its divisor is supported at W and has degree zero. The same is true coordinate by coordinate after identifying the fibre with $\mathbb{G}_m^{\dim T}$. Hence f contracts every curve fibre of $Y \to B^\circ$, and cannot be an immersion. The script checks the five branch values are distinct; the divisor and ramification verifications above are the complete algebraic checks.

Here is the genus-zero example needed to check the additional assertion on scan 36. Let ζ be a primitive cube root of unity and put

$$B_0 = \mathbb{P}^1 \setminus \{0, 1, \infty\}, \quad C_0 = \mathbb{P}^1 \setminus \{0, \infty, 1, \zeta, \zeta^2\}, \quad D_0 = \mathbb{P}^1 \setminus \{1, \zeta, \zeta^2\}.$$

The map $C_0 \to B_0$, $z \mapsto z^3$, is finite étale Galois of degree three. On $C_0 \times D_0$ let the generator act by $(z, w) \mapsto (\zeta z, \zeta w)$, and set

$$Y_0 = (C_0 \times D_0)/\langle\zeta\rangle.$$

As above this is a smooth affine polycurve over B_0, now with both base and geometric fibre of type $(0, 3)$. The boundary of its relative proper compactification is finite étale of degree three; it need not consist of three globally labelled sections.

Every invertible regular function on $C_0 \times D_0$ has the form $f(z)g(w)$. To prove this, consider its divisor on $C_0 \times \mathbb{P}^1$. It is supported on the three irreducible divisors $C_0 \times \{\zeta^i\}$. A restriction at a suitable z_0 has the same three orders. Dividing by that restriction removes the divisor, and the resulting invertible function on $C_0 \times \mathbb{P}^1$ comes from C_0, by properness of the second factor. An invariant function $f(z)g(w)$ satisfies $g(\zeta w)/g(w) = f(z)/f(\zeta z)$, a constant. Thus the divisor of g is invariant under the cyclic permutation of the three cusps. Its three coefficients are equal, and their sum is zero; all vanish. Therefore g is constant. All units on Y_0 are constant on the fibres of $Y_0 \to B_0$.

For a map $Y_0 \to G \to A$, its pullback extends to $\mathbb{P}^1 \times \mathbb{P}^1 \to A$ by the same opened extension theorem. It is constant on each projective-line factor, since a projective line has zero Jacobian, and hence is constant. Translating the image fibre of $G \to A$ identifies the remaining map with a map to a torus. Its coordinates are the units just computed, so it too contracts every curve fibre. This proves the asserted failure of a global torus or semiabelian immersion even for genus-zero towers. □

A7. The field-map image, conservativity, and the limit groupoid *[Proved]*

For a finitely generated field extension L/F, restriction $G_L \to G_F$ has open image G_{F_s}, where F_s is the relative separable algebraic closure of F in L. The map is injective exactly when L/F is algebraic, and is an isomorphism exactly when L/F is finite purely inseparable. Thus the field functor of scans 5–6 is conservative in characteristic zero, and becomes conservative after inverting finite purely inseparable extensions in positive characteristic. For an arbitrary field $F = \varinjlim F_i$, where F_i runs through its finitely generated prime-field subfields, the compatible absolute groups satisfy $G_F \simeq \varprojlim G_{F_i}$; the analogous statements hold for the profinite groupoids and their finite-cover topoi. Keeping the whole pro-system retains the faithfulness of B10, whereas taking only its limit does not. This closes sweep row ZAB2 (149, section 1, manuscript pp. 2–3, scans 5–8).

Proof. Choose one algebraic closure of L and the compatible separable closures. The fixed field in F^{sep} of the restriction image is $L \cap F^{\mathrm{sep}} = F_s$; compactness makes the image closed, and Galois correspondence identifies it with G_{F_s}. The field F_s/F is finite because L/F is finitely generated. The kernel is the absolute group of LF^{sep}, up to its purely inseparable closure, which leaves finite étale extensions unchanged. If L/F is algebraic, this kernel is one. If its transcendence degree is positive, choose a divisorial discrete valuation of the finitely generated field $LF^{\mathrm{sep}}/F^{\mathrm{sep}}$. A uniformizer has a nontrivial separable ℓth-root extension for $\ell \neq \operatorname{char} F$, since its value is one. The kernel is nontrivial. Injectivity plus surjectivity is thus equivalent to algebraicity and $F_s = F$, or to finite pure inseparability.

Inside F^{sep} choose the separable algebraic closure of each F_i. Their directed union is F^{sep}: a separable polynomial and a witness to its nonzero discriminant use finitely many coefficients, as does every finite collection of such polynomials. Compatible automorphisms of these subfields therefore extend uniquely to F^{sep} and fix F, giving the group isomorphism. Conversely restriction supplies every component. It is a homeomorphism by compactness. A finite étale algebra and all its morphisms descend to a sufficiently large F_i by their finite presentations; this proves the groupoid and topos comparisons as well.

Finally, $\mathrm{Ind}(\mathrm{Fields}_{\mathrm{fg}}) \to \mathrm{Fields}$ is an equivalence: every field is such a union, and a map from a finitely generated field into a directed union lands in one stage, with equality detectable at a common later stage. For a faithful contravariant functor on the finitely generated objects, the formula for pro-morphisms, $\mathrm{Hom}_{\mathrm{Pro}}(X_i, Y_j) = \varprojlim_j \varinjlim_i \mathrm{Hom}(X_i, Y_j)$, therefore preserves injectivity on these field maps; equality at a stage in the target can be pulled back by faithfulness. We use the relative identity marking of B10. The limit alone can lose this: separably closed fields have trivial absolute group, while their field embeddings can be distinct. For instance $\overline{\mathbb{Q}}$ has distinct automorphisms but its absolute group is one. □

A8. Cofinal Artin models and the dimension of their groups *[Proved]*

In characteristic zero every smooth integral variety has a dense open which is a tower of affine smooth hyperbolic curves with smooth proper compactifications and finite étale boundary. Thus such opens are cofinal among models of a finitely generated field, and their full geometric groups are successive extensions of nonabelian free profinite groups. A tower of length n has ℓ-cohomological dimension exactly n for every prime. This closes the additional assertions on scans 10–13, sweep row ZAB3 (149, section 1, manuscript pp. 4–6).

Proof. Let V/k be smooth integral of dimension n. Choose a separating transcendence basis $t_1, \dots, t_n$ of $k(V)/k$, and project on $t_1, \dots, t_{n-1}$. Over $F = k(t_1, \dots, t_{n-1})$ the generic fibre has a smooth projective curve model: normalize $\mathbb{P}^1_F$ in the finite extension $k(V)/F(t_n)$. Characteristic zero makes this regular curve smooth. This model and its birational comparison to the generic fibre spread over a nonempty open $B \subset \mathbb{A}^{n-1}_k$. Shrinking B makes the model smooth proper with geometrically connected fibres; regularity of the function-field extension gives geometric connectedness. If the chosen projection initially has a larger constant field in its generic fibre, replace B by its normalization in that finite relative algebraic closure; after shrinking it is smooth and the same construction applies over its function field.

The complement of the comparison open in the generic proper curve is finite. Add finitely many distinct separable closed points so that the complement is nonempty and hyperbolic. Spread these points to disjoint finite étale boundary divisors after shrinking B. Remove any remaining vertical bad locus. Their complement U is a smooth curve fibration contained in V. It is affine over B, since a positive-degree relative divisor on a proper smooth curve is relatively ample. Apply the same construction inductively to a dense affine open of the smooth base B and pull U back to it. Each comparison is the actual restriction to a dense open, so the final tower lies inside the originally prescribed V. Applying this to every dense open proves cofinality. C4 gives the exact sequences; complex comparison and A3 identify the affine hyperbolic fibre groups as finite-rank free profinite groups.

Here is a proof of the exact dimension, not just its upper bound. A free profinite group of positive finite rank has cohomological dimension one: generators lift across every finite surjection, so every finite embedding problem is solvable; equivalently all H^2 with finite modules vanish, and dimension shifting gives the upper bound. Its nonzero $H^1(-, \mathbb{F}_\ell)$ gives equality. Hochschild–Serre gives the additive upper bound for a tower. For the reverse inequality in $1 \to N \to P \to Q \to 1$, where N is free of rank $r > 0$ and $\mathrm{cd}_\ell Q = n - 1$, first take an open subgroup of Q acting trivially on $H^1(N, \mathbb{F}_\ell)$. Its dimension is still $n - 1$ by Serre's finite-index theorem: successive extensions of the torsion-free groups in A3 are torsion-free.

Some further open subgroup Q' has $H^{n-1}(Q', \mathbb{F}_\ell) \neq 0$. Indeed choose a finite $\mathbb{F}_\ell[Q]$-module M with nonzero top cohomology and a surjection onto M from finitely many copies of a permutation module $\mathbb{F}_\ell[Q/Q']$. Vanishing in degree n for its kernel makes the top-cohomology map surjective, and Shapiro's lemma gives the assertion. For the preimage P' of Q', the Hochschild–Serre term

$$E_2^{n-1,1} = H^{n-1}(Q', \mathbb{F}_\ell)^r$$

is nonzero. No differential enters it (the fibre has no degree-two cohomology) or leaves it (the base has dimension $n-1$). Thus $H^n(P', \mathbb{F}_\ell) \neq 0$, proving the lower bound. For $n = 1$ use the initial free group case. This proves the manuscript's finite-model equality as well as the limiting field calculation A1. □

A9. Reconstruction of a centerless extension and uniqueness of lifted maps *[Proved]*

For a profinite extension $1 \to P \to E \to \Gamma \to 1$ with $Z(P) = 1$, the outer action determines the extension, with its identified kernel and quotient, by

$$E \xrightarrow{\sim} \mathrm{Aut}(P) \times_{\mathrm{Out}(P)} \Gamma, \qquad e \longmapsto (\mathrm{int}(e)|_P, \bar{e}).$$

Use the compact-open topology on continuous automorphisms and the quotient topology on outer automorphisms; the displayed pullback is profinite because it is isomorphic to E. If two extension maps over Γ restrict to the same map $P \to P'$, and its image has trivial centralizer in P', the extension maps are equal. Conversely a continuous homomorphism $\varphi : P \to P'$ whose class modulo target inner automorphisms commutes with the two outer Γ-actions, and whose image has trivial centralizer in P', extends uniquely to a continuous homomorphism $E \to E'$ over Γ. These are the formal claims of scans 16–20 and 29, sweep row ZAB4 (149, section 1, manuscript pp. 7–9, 14). They do not require $\mathrm{Aut}(P)$ itself to be profinite for an arbitrary infinitely generated P.

Proof. The displayed homomorphism is injective because its kernel is $Z(P)$. For a pair (a, γ) in the pullback, choose e above γ. The equality of outer classes means $a\,\mathrm{int}(e)^{-1} = \mathrm{int}(p)$ for a unique $p \in P$. Thus pe maps to the pair. Conjugation by elements of E varies continuously in the compact-open topology, and $\mathrm{Inn}(P)$ is the compact, hence closed, image of P. The continuous bijection from compact E to this Hausdorff pullback is a homeomorphism. This verifies both the algebra and topology of the reconstruction. For extension maps F, G, the element $G(e)F(e)^{-1} \in P'$ centralizes their common image of P, by evaluating the two maps on epe^{-1}. The centralizer hypothesis makes it one for every e, proving uniqueness. An outer equality is handled first by the specified inner conjugation, exactly as in B5. For existence, choose $e \in E$ with image $\gamma \in \Gamma$, and any lift $e'_0 \in E'$ of γ. Outer equivariance says exactly that some $q \in P'$ makes $e' = qe'_0$ satisfy

$$\mathrm{int}(e') \circ \varphi = \varphi \circ \mathrm{int}(e)|_P.$$

The centralizer hypothesis makes e' unique. Define $F(e) = e'$. Uniqueness proves $F(e_1e_2) = F(e_1)F(e_2)$ and $F|_P = \varphi$. The graph of F is the closed subset of $E \times E'$ defined by equality of quotient images and the displayed equations for all $p \in P$. Its projection to E is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism; F is continuous.

Finally choose an object in each connected profinite groupoid. The full one-object subcategories have automorphism groups P, P'; they are equivalent to the original groupoids because every other object is isomorphic to the chosen one. A functor is then described by a continuous group homomorphism $\varphi : P \to P'$. A natural transformation to the functor of ψ is precisely an element $q \in P'$ with $\psi = \mathrm{int}(q)\varphi$, and the natural automorphisms of the first functor are $C_{P'}(\varphi(P))$. This proves the rigidity, up to a unique natural isomorphism, asserted in the Scholie on scans15–16. The corresponding finite local systems are the finite continuous P-sets, so passage to their classifying topoi gives the same comparison without changing these natural transformations. □

B12. The semiabelian Albanese homology claim for arbitrary reduced sources *[Refuted]*

For a general reduced connected finite-type scheme, a universal semiabelian recipient of pointed morphisms need not have the same ℓ-adic H_1. Thus the assertion numbered 3 on scan 25 does not justify Corollary 1(a) in all of its stated source generality. B3 supplies a proof in that generality. This is sweep row ZAB5 (149, section 1, manuscript p. 12, scan 25).

Proof. Over $\mathbb{C}$ take the nodal rational cubic

$$X: \quad Y^2Z = X^2(X+Z).$$

Its normalization is $\mathbb{P}^1$, with map

$$[T:S] \longmapsto [S(T^2-S^2) : T(T^2-S^2) : S^3].$$

The two points $[1:1], [-1:1]$ map to the node; elsewhere this is an isomorphism. The displayed equation and the node are checked in `code/AB_checks.py`. A morphism $X \to G$ to any semiabelian variety becomes constant on its normalization: the abelian quotient admits no nonconstant map from $\mathbb{P}^1$, and the remaining torus-valued map on the proper curve is constant. Surjectivity of normalization implies the original map is constant. Thus its universal pointed semiabelian recipient is the zero group.

Topologically, identifying two distinct points of the two-sphere produces a space homotopy equivalent to $S^2 \vee S^1$. Its finite cyclic covers can equivalently be constructed by taking a cyclic chain of n copies of $\mathbb{P}^1$ and identifying successive preimages of the node. Each node has precisely the same two-branch local model, so these are finite étale covers of X. Complex finite-cover comparison therefore gives $\pi_1(X) = \widehat{\mathbb{Z}}$ and $H_1(X, \mathbb{Z}_\ell) = \mathbb{Z}_\ell$. The unique map to the zero semiabelian recipient induces the zero map on this nonzero module. This is the exact failed comparison. □

B13. The toric-kernel disjointness assertion on scan 38 *[Refuted]*

Let $Y = C \setminus D$ be a smooth hyperbolic curve over an algebraically closed field of characteristic zero, and let $J_D \to J_C$ be its compact-Jacobian quotient. For $g(C) \geq 1$, every nonzero $u \in \ker(J_D \to J_C)$ satisfies $(\alpha(Y) + u) \cap \alpha(Y) = \varnothing$. This assertion is false for genus zero. There, for any nonidentity translation, the intersection has at most one geometric point, and its existence is determined by the equations below. This corrects the bracketed claim and the margin "$g \geq 2$?" on scan 38, sweep row ZAB8 (149, section 1, manuscript p. 18).

Proof. For positive genus, projecting $\alpha(s) = \alpha(t) + u$ to J_C^1 gives equality of the compact Abel images of s, t. This forces $s = t$: otherwise a principal divisor $s - t$ would give a degree-one function on C, impossible for positive genus. The original equality then gives $u = 0$. This proves the first assertion, with the explicitly necessary nonzero hypothesis.

For genus zero put three of the missing points at $0, 1, \infty$, and write the other finite missing points as d_i. Up to constant coordinates determined by the choice of Abel origin, the generalized Abel immersion is

$$\alpha(t) = (t, t-1, (t-d_i)_i) \in \mathbb{G}_m^{r-1}.$$

Let $u = (a, b, (c_i)_i)$. A point of the intersection is specified exactly by $t, s \in Y$ satisfying

$$s = at, \qquad s - 1 = b(t-1), \qquad s - d_i = c_i(t - d_i).$$

If $a = b$, the first two equations give $a = b = 1$; then $s = t$ and the remaining equations give every $c_i = 1$, because t is not a cusp. This is the identity translation. Otherwise

$$t = \frac{1-b}{a-b}, \qquad s = \frac{a(1-b)}{a-b}.$$

Thus the intersection has exactly one point precisely when these values avoid all missing points and satisfy the remaining displayed equations; it is empty otherwise. For three punctures take $(a, b) = (2, 3)$: then $t = 2, s = 4$, and $(2, 1)(2, 3) = (4, 3)$. This is the promised counterexample in the entire toric kernel $J_D \to J_C = 0$. The integer calculation is included in `code/AB_checks.py`. A single intersection point supplies no translated open subset, so the criterion is consistent with the genus-zero part of B4. □

9 Additional steps in cote 149, scans 41–104

ZCF1. The open birational Hom statement, and the withdrawn unrestricted version *[Known]*

[§2, manuscript pp. 20–22; scans 41–46.] Let k be a number field, and let L, M be finitely generated regular extensions of k. With compatible separable closures, restriction gives a bijection

$$\mathrm{Hom}_{k\text{-fields}}(M, L) \longrightarrow \mathrm{Hom}^{\mathrm{open}}_{G_k}(G_L, G_M)/\mathrm{Inn}(G_{M\bar{k}}). \tag{ZCF.1}$$

All homomorphisms on the right are continuous. Thus a) and b) of scan 44 are equivalent after the common constant field has been fixed and the geometric kernels are the kernels for that same field. No supplementary compatibility with a filtration by models is required for this open-Hom statement. The version with arbitrary homomorphisms in scans 41–42 is false, as the manuscript itself observes in scan 43.

Proof. The precise input is Mochizuki, Theorem 17.1 [Moc99]: the ground field is sub-p-adic, the two function fields have arbitrary finite transcendence degree, and the ground field is algebraically closed in each, as specified immediately before the theorem. A number field is sub-p-adic via any place over p, so these hypotheses hold. The theorem is exactly (ZCF.1), with the displayed geometric inner-conjugacy ambiguity.

For clarity, the easy direction has its own field-theoretic proof. For $M \hookrightarrow L$, the algebraic closure M_0 of M in L is finite over M, and L/M_0 is regular. Linear disjointness from M_0^{sep} makes $G_L \to G_{M_0}$ surjective. Its composite into G_M therefore has finite-index image G_{M_0}. If $\phi : E' \to E$ is over the identity of G_k, with geometric kernels P', P, then

$$\phi(E') \cap P = \phi(P'), \qquad [E : \phi(E')] = [P : \phi(P')].$$

The equality of intersections follows by applying the quotient map; every coset of $\phi(E')$ has a representative in P, since $\phi(E') \to G_k$ is surjective. This proves the equality of indices and the manuscript's arithmetic/geometric openness comparison, with the directions of the arrows corrected.

Take instead $L = k$ and $M = k(t)$. The coefficient action on compatible Puiseux roots at $t = 0$ gives a continuous section $G_k \to G_{k(t)}$. It is a morphism over G_k, whereas no k-field embedding $k(t) \to k$ exists. The corresponding map of geometric groups is $1 \to G_{\bar{k}(t)}$, so this also refutes the proposed unrestricted geometric-Hom version. The coefficient construction is written out in F9.3. Its image has infinite index, since its intersection with the infinite geometric kernel is trivial.

Finally $G_M = \varprojlim_W \pi_1(W)$, where W runs through nonempty opens of a fixed normal finite-type k-model of M. A finite separable extension of M has finite normalization on that model and is finite étale after deleting its branch locus; the same statement for a compositum and for morphisms of finite extensions proves the equality of Galois categories and then the displayed limit. A8 supplies cofinal elementary models. This proves the limit assertion in scan 46 without using a conjectural preservation of centralizers by inverse limits. □

ZCF2. Recovering an extension map from an equivariant geometric map *[Proved]*

[§2, manuscript p. 21; scans 43–44.] Let $1 \to P_i \to E_i \to G \to 1$ be profinite extensions and let $\phi : P_1 \to P_2$ be continuous. Suppose its conjugacy class is G-equivariant and $C_{P_2}(\phi(P_1)) = 1$. Then ϕ extends uniquely to a continuous homomorphism $F : E_1 \to E_2$ over G. Here equivariance means that, for every $e_1 \in E_1$ and any $e_2 \in E_2$ with the same image in G, the maps $\phi \circ \mathrm{int}(e_1)$ and $\mathrm{int}(e_2) \circ \phi$ differ by an inner automorphism from P_2. The stronger hypothesis involving every open subgroup in scan 43 implies the centralizer hypothesis just stated.

Proof. For e_1 choose e_2 as above. Equivariance supplies $b \in P_2$ such that be_2 satisfies

$$\mathrm{int}(be_2) \circ \phi = \phi \circ \mathrm{int}(e_1).$$

There is only one such element above the given element of G: the ratio of two would centralize $\phi(P_1)$. Define $F(e_1) = be_2$. Uniqueness proves both the multiplication law and $F|_{P_1} = \phi$. Its graph is the closed subset of $E_1 \times E_2$ cut out by equality in G and all the equations $e_2\phi(p)e_2^{-1} = \phi(e_1 p e_1^{-1})$, $p \in P_1$. Projection of this compact graph to E_1 is a continuous bijection to a Hausdorff space, hence a homeomorphism. Thus F is continuous. Changing ϕ by a P_2-inner conjugation changes F by that same conjugation, specifying the outer ambiguity.

The tentative arithmetic centralizer condition c) of scan 44 does not replace openness. For the coefficient section at $t = 0$ above, the attached return's B7 computes the centralizer of the cyclotomic kernel as the local inertia T. For an open subgroup of the full G_k, its centralizer is contained in T, and invariance there is $\chi(g)a = a$. Each ℓ-adic cyclotomic character has infinite image over a number field, so $a_\ell = 0$ for every ℓ. Projection to G_k adds no centralizer, by F8a. Thus all these arithmetic centralizers are trivial although the image is not open. This also refutes the withdrawn assertion $P^{G_{k'}} \neq 1$. The corrected cyclotomic assertion $P^{G^\circ_{k'}} \neq 1$ for an *arbitrary* birational section is retained as Open in ZCF6. □

ZCF3. Geometric homology alone does not determine a morphism *[Refuted]*

[§2, manuscript p. 23; scan 47.] Two morphisms into an elementary hyperbolic variety can induce the same map on every geometric $H_1(-, \mathbb{Z}_\ell)$ and different arithmetic fundamental-group homomorphisms. Consequently the unqualified sentence in scan 47 is false, for either direction of the doubtful source/target notation.

Proof. Set $Y = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$ and take the two morphisms $\mathrm{Spec}\,\mathbb{Q} \to Y$ with values 2 and 3. Both geometric homology maps are zero. The unit x on Y induces $\pi_1(Y) \to \pi_1(\mathbb{G}_m)$ over $G_{\mathbb{Q}}$. On the two sections it gives the Kummer classes of 2 and 3. Their difference is the class of $3/2$; its 2-adic valuation is -1, so it is nonzero in $\mathbb{Q}^*/\mathbb{Q}^{*\ell^n}$ for every $\ell, n > 0$. Thus their arithmetic sections differ even in each geometric pro-ℓ abelian quotient. To have a positive-dimensional source, compose these constants with $Y \to \mathrm{Spec}\,\mathbb{Q}$; restriction at a rational point gives the same distinction. The proved replacements are the arithmetic Kummer injectivity B2 and the dominant-map rigidity B5. The displayed quotient maps explain exactly which information the geometric-homology sentence loses. □

ZCF4. An explicit nondominant map inducing an isomorphism of full fundamental groups *[Refuted]*

[§2, manuscript pp. 24–25; scans 49–50.] Let $C/\mathbb{Q}$ be the smooth projective completion of $y^2 = x^5 - x$, and let $Y = C^3$. On $(\mathbb{P}^1)^3$ take the trihomogeneous polynomial whose affine expression is

$$F = 1 + 7x_1 + 3x_2 + 2x_3 + 13x_1x_2 + 11x_1x_3 + 5x_2x_3 + 19x_1x_2x_3.$$

Let $H \subset Y$ be its inverse-image zero divisor under the product of the three hyperelliptic maps. Then H is a smooth geometrically connected surface and the closed immersion $H \hookrightarrow Y$ induces an isomorphism $\pi_1(H) \simeq \pi_1(Y)$ over $G_{\mathbb{Q}}$. It is not dominant. Thus the open-Hom assertion of scan 50 is false for the stated higher-dimensional elementary targets, already for constant inverse systems.

Proof. The six branch values of $C \to \mathbb{P}^1$ are $B = \{0, 1, -1, i, -i, \infty\}$. We verify smoothness over $\overline{\mathbb{Q}}$. The eight affine charts of $F = 0$ in $(\mathbb{P}^1)^3$ have singularity ideals generated by F and its three first derivatives. Exact rational Gröbner calculation gives the unit ideal in every chart. Fixing any one coordinate to any element of B leaves a nonsingular bilinear form in the other two coordinates: all eighteen 2×2 determinants are nonzero. Fixing all three to branch values gives 216 nonzero values. These exhaustive checks and their exact output are in `code/CF_hyperplane.py` and `code/CF_hyperplane_output.json`.

Here is why these finite checks prove the geometric assertion. At a point of Y, a local hyperelliptic coordinate is unramified unless its x-coordinate is in B; at a branch point it has the form $x - b = u^2$ up to a unit (with $1/x$ at infinity). With no ramified coordinate, smoothness follows from the first check. With one, the nonsingular bilinear restriction makes at least one of the two unramified derivatives nonzero. With two, the remaining linear form is nonzero, since a nonsingular bilinear form cannot vanish identically after fixing one of its variables to a projective point; its zero has nonzero derivative. With three, the third check says that the point is not on H. This proves smoothness everywhere.

The pullback of $\mathcal{O}(1, 1, 1)$ is ample, because the product hyperelliptic map is finite. Its section F is nonzero and defines an effective Cartier divisor in the integral threefold Y. SGA 2 XII, Corollary 3.5 [Gro05] applies: the closed local rings of Y have depth three, and all local rings off the divisor are regular, hence pure. It gives geometric connectedness and $\pi_1(H_{\overline{\mathbb{Q}}}) \simeq \pi_1(Y_{\overline{\mathbb{Q}}})$. The two arithmetic exact sequences have this isomorphism on kernels and the identity on $G_{\mathbb{Q}}$, hence give the asserted arithmetic isomorphism. Dimensions give $\dim H = 2 < 3 = \dim Y$.

The failure mechanism is also explicitly stated in the Remark following Mochizuki's Theorem 17.1. A precise proved replacement for the one-dimensional-target version is his Theorem 16.5 and its following Remark: a smooth pro-variety and a hyperbolic pro-curve over a sub-p-adic field, with birational transition maps, have dominant morphisms in bijection with continuous open maps of their full arithmetic fundamental groups modulo geometric inner automorphisms. Its full profinite comparison is proved there by pulling back every finite cover; it does not claim the false higher-dimensional-target version. □

ZCF5. The centralizer precisions and the necessary half of the fundamental conjecture *[Proved]*

[§2–3, manuscript pp. 23–31; scans 48–49, 53–63.] For a hyperbolic curve U/k, with k finitely generated over $\mathbb{Q}$, and a continuous section $s : G_k \to E_U$, put $G_k^\circ = \ker \chi$. The checked attached return A6–A7 and AE1 [AI-26] gives

$$C_\Pi(s(G_k^\circ)) = \begin{cases} I_c, & s \text{ is cuspidal at its unique lifted cusp } c, \\ 1, & s \text{ is noncuspidal.} \end{cases} \tag{ZCF.2}$$

For every open $\Omega \subset G_k^\circ$ the same formula holds with Ω in place of G_k°. In particular genuine finite point sections have zero cyclotomic centralizer. The same zero centralizer conclusion holds for point sections of hyperbolic polycurves. It proves a)⇒b) in scan 48 for actual morphisms into these varieties.

Proof. We use the proved return, not the unsupported sketch R2. Its A6 uses Stix Theorems 14 and 20 [Sti08a]: cyclotomic normalization detects peripheral subgroups, and the cuspidal packets

are disjoint from each other and from finite-point sections. Its arithmetic hypothesis over $k(\mu_\infty)$ is supplied by the proved absence of divisible abelian-variety points in AE1. A7 passes to any open Ω by writing its fixed field as $k'(\mu_\infty)$ with k'/k finite, and descends the unique cusp using normality after a finite Galois enlargement. We inspected these proofs, including the finite-cover cusp uniqueness step. Thus (ZCF.2) has precisely the needed hypotheses; it says noncuspidal, without asserting that such a section comes from a point.

For a polycurve, project a centralizing element through its curve tower. The base point section has zero centralizer by induction. The element therefore lies in the fibre kernel; C4 identifies that kernel with the full geometric group of the fibre through the point. The last curve case of (ZCF.2) makes it one. The same argument works after each finite field extension and for each open Ω.

Now let $f : V \to U$ be an actual morphism as in scan 48, with V nonempty and of finite type over a number field, and let $H \subset E_V^\circ$ be open. Choose a closed point v of V. Its residue field $k(v)$ is finite over the ground number field, and its point section has image in E_V. The inverse image of H in $G_{k(v)}^\circ$ is open there. The composite point section in E_U is the section of $f(v)$ after this residue-field base change, so its geometric centralizer is one by the preceding paragraph. Any element centralizing $f_*(H)$ centralizes this smaller image. Its arithmetic projection is also one by F8a. Therefore $C_{E_U}(f_*(H)) = 1$, as required.

The lifted-cusp assertion in scan 58 has an explicit group map. Above a fixed geometric cusp the lifted cusps are Π/I_c; the map $aI_c \mapsto aI_ca^{-1}$ is well defined and surjective onto its conjugates. It is injective because $N_\Pi(I_c) = I_c$ (A4/F10a). Distinct base cusps give disjoint conjugacy classes by the same peripheral intersection theorem. This is the claimed recovery of actual lifted cusps, not just their unlabelled orbits.

Finally, for a closed subgroup $\Delta \subset G_k$ containing such an Ω, the cuspidal case gives the exact formula

$$C_\Pi(s(\Delta)) = I_c^{\chi(\Delta)} = \prod_{\ell:\, \chi_\ell|_\Delta = 1} \mathbb{Z}_\ell(1).$$

It follows coordinatewise from $(\chi_\ell(\delta) - 1)a_\ell = 0$ in the integral domain $\mathbb{Z}_\ell$. This is the corrected H1, including the possibility of nonzero invariants with a nontrivial full cyclotomic character. □

ZCF6. The existence assertions retained after the centralizer computation *[Open]*

[§2–3, manuscript pp. 22–34 and 51; scans 45, 48–49, 51, 56–57, 59–63, 66–69, 104.] The following exact assertions are not proved or refuted here. They are recorded separately from the proved uniqueness and centralizer assertions.

1. For $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$, every continuous section $s : G_\mathbb{Q} \to \pi_1(U)$ with $C_\Pi(s(G_\mathbb{Q}^\circ)) = 1$ is the section of a rational point. This is the smallest retained instance of the sufficiency in scan 48 and of Conjecture A in scan 56.
2. Every continuous section $G_\mathbb{Q} \to G_{\mathbb{Q}(t)}$ has image in the decomposition group of a rational place. By the checked return B7, this is exactly the assertion that its geometric cyclotomic centralizer is nonzero. This is the smallest retained birational instance of scan 45.
3. Every section of a hyperbolic curve admits a compatible lift through the maps $\pi_1(U \setminus D) \to \pi_1(U)$ as the finite closed subsets D grow, sufficient to obtain a birational section. The precise lifting problem is about continuous sections and their geometric-conjugacy classes, not just individual lifts of elements of G_k.
4. The assertion of Conjecture B' of scan 63 for arbitrary abstract semilinear $G_\mathbb{Q}$-actions on the universal pro-cover of $\mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$: every such action has a unique $G_\mathbb{Q}^\circ$-fixed compatible geometric point in its completed pro-cover. For actions with an open stabilizer for every rational function, ZCF12 constructs an invariant finite quotient over $\mathbb{Q}$ and reduces the assertion to Conjecture B with its G_k° clause of scan 61. Its fixed-point existence still

contains (1), but no additional finite-level obstruction is left in that continuous case. The extra scope allowing abstract actions without this continuity hypothesis remains unresolved.

Verified deductions, attempts, and exact remaining steps. For (1), (ZCF.2) proves that zero centralizer is equivalent to noncuspidality. It proves neither existence of a point nor existence of a fixed pro-point. The attached return D0b proves at most one fixed pro-point, by passing to a characteristic finite cover whose compactification has positive genus and applying Jacobian Kummer injectivity to two putative fixed points. The maps to that cover and then to its Jacobian are actual maps, so this establishes uniqueness only. The existence of an interior fixed pro-point is precisely the missing step.

We tried three further routes. First, Mochizuki's Hom theorem ZCF1 requires open image; a point section has intersection 1 with the infinite geometric kernel, and so its image is not open. Second, the full topos theorem F9.2 supplies the exact map

$$\pi_0 \operatorname{Hom}^{\mathrm{pin}}_{BG_k}(BG_k, U_{\mathrm{et}}) \longrightarrow \operatorname{Sect}(E_U/G_k)/\Pi.$$

Its source is $U(k)$ and its image is the point-section subset. No construction in the proof lifts an arbitrary noncuspidal section to the source. Establishing that lift for all such sections would finish (1); the full-faithfulness theorem alone does not produce it. Third, the finite-cover Kummer constraints in the attached return D4 separate its specified two points via an explicit degree-81 Fermat cover. They give compatibility requirements on an arbitrary section at every finite level. They do not construct a rational point from those requirements; that construction remains unproved even for the displayed $U/\mathbb{Q}$.

For (2), B7 proves the exact equivalence, section by section, between nonzero centralizer and a unique rational place. Its proof uses a nonzero element z and an open U_0 on which z remains nonzero. It then gets compatible unique cusps on every smaller open. For an arbitrary section no such z has been produced; there is no first finite level to which that proof applies. The t-adic sections show that the positive conclusion is realized, but do not exhaust the sections. The plain full-G_k centralizer condition was refuted in ZCF2 and cannot supply the missing element.

For (3), the canonical surjection $G_{k(U)} \twoheadrightarrow \pi_1(U)$ sends a decomposition-group section at $x \in U(k)$ to the point section, and one at $x \in \bar{U} \setminus U$ to a cuspidal section. This follows by restricting covers to the henselian DVR and applying C2–C3. The kernel of $\pi_1(U \setminus D) \to \pi_1(U)$ is the closed normal subgroup generated by the inertia groups at D, by normalization and purity for smooth curves. These facts specify the quotients but do not give sections of their pullbacks by an arbitrary s. Solving the finite embedding problems separately would also leave a compatibility problem. This is the missing cuspidalization step in the proposed "forage de trous" of scan 67.

For (4), ZCF12 supplies the previously missing comparison for the explicit topology of pointwise convergence on the discrete rational-function field. A finite orbit of a set of field generators gives a G_k-stable finite quotient, and effective descent gives a curve over k itself. Thus, with this continuity clause, Conjecture B with its G_k° strengthening of scan 61 and Conjecture B', quantified over all hyperbolic curves at the same base field, are equivalent. Their existence conclusion has not thereby been proved.

The remaining abstract-action problem has literal equations. Fix the usual tangential continuous action ρ_0 on this universal pro-cover and put $A = \operatorname{Aut}_{\overline{\mathbb{Q}}}(\mathcal{U})$. Every abstract semilinear action is uniquely $\rho(g) = a_g\rho_0(g)$ with

$$a_{gh} = a_g\rho_0(g)a_h\rho_0(g)^{-1}.$$

We have neither constructed nor excluded such a cocycle whose action normalizes none of the profinite deck groups of finite hyperbolic quotients. The finite-orbit construction in ZCF12 excludes this possibility for every action with open function stabilizers. Without that hypothesis the orbit of a field generator can be infinite, and its generated field need not be a finite

covering level; this is the exact failed step of that construction. No counterexample or substitute construction is supplied here for those abstract actions. The further fixed-point assertion in (4) remains unproved. ZCF7 refutes the additional proposed extension to a cyclotomic ground field even for the continuous actions coming from sections.

Thus the general reconstruction programme of scans 51 and 66–69 has these explicit remaining existence steps. F9.2 reconstructs schemes from their full pinned *étale topoi*; the displayed forgetful map proves how this theorem relates to the smaller fundamental-group input and locates exactly what is missing. The text's predicate f), "pas trop loufoque", has no specified mathematical definition and therefore supplies no further test. □

ZCF7. The proposed extension to cyclotomic ground fields *[Refuted]*

[§3, manuscript pp. 30–31; scans 62–63.] Let $F = \mathbb{Q}(\mu_\infty)$. The smooth projective normalization C/F of $z^6 + 2y^3 + 4 = 0$ has genus four, no F-rational point, and a continuous section of $\pi_1(C) \twoheadrightarrow G_F$. This refutes the claim that all the preceding existence statements extend to fields finitely generated over a cyclotomic field.

Proof. This is the checked attached return D5 [AI-26]; we give the exact mechanism. The map $(z, y) \mapsto (z^2 : y : 1)$ extends to a morphism to the smooth cubic $D : X^3 + 2Y^3 + 4Z^3 = 0$. In a finite cyclotomic field $\mathbb{Q}(\mu_N)$ the ramification degree e at 2 is a power of two. With integral-normalized valuation, the three summands in the cubic have residues $0, e, 2e$ modulo three. The least nonzero valuation is therefore attained once, which is incompatible with their sum being zero; the same argument applies when a coordinate vanishes. Hence $D(F)$ and then $C(F)$ are empty. The degree-three map $C \to \mathbb{P}^1_z$ is ramified to index three at the six simple zeros of $z^6 + 4$, and unramified at infinity. Riemann–Hurwitz gives $2g - 2 = -6 + 12 = 6$.

D5 proves projectivity of G_F by showing that every Brauer class over every finite extension of F is killed by a cyclotomic extension, followed by the finite embedding-problem argument. We checked that proof, including its reduction from finite modules to Sylow subgroups and the Frattini-kernel induction. To apply it here, take a cofinal descending sequence of characteristic open subgroups N_j of the finitely generated geometric group Π_C. Starting at $E_C/\Pi_C = G_F$, projectivity lifts the section through each finite-kernel quotient $E_C/N_{j+1} \to E_C/N_j$. The compatible inverse limit is a continuous section of $E_C \to G_F$. As C is proper it has no cuspidal packet, and as $C(F)$ is empty the section is not a point section. This is the actual extension and the actual section-existence argument, not a deduction from a section conjecture. □

ZCF8. The commensurability assertion for universal algebraic pro-covers *[Proved]*

[§3, manuscript p. 30; scan 62.] For smooth connected curves U, V over an algebraically closed field of characteristic zero, their universal connected finite-étale pro-covers are isomorphic as schemes over that field exactly when some connected finite étale cover of U is isomorphic to a connected finite étale cover of V. No compatibility of an initial isomorphism with either deck group is assumed.

Proof. Write the universal pro-covers as inverse limits of their connected finite covers; transition maps are finite, hence affine, and every level is quasi-compact and quasi-separated. An isomorphism $\widetilde{U} \simeq \widetilde{V}$ followed by projection to U descends to a map $f : V_i \to U$, by the finite-presentation limit theorem, Stacks Proposition 32.6.1 [The26a]. Using the inverse isomorphism, the induced map $\widetilde{U} \to V_i$ descends to $g : U_j \to V_i$. After increasing j, the equality on the inverse limit descends to $fg = p_j : U_j \to U$, the finite étale transition map. The map f is nonconstant and surjective, since fg is; g is nonconstant as well. For integral curves these maps are quasi-finite. Moreover g is proper: its graph is closed because $V_i \to U$ is separated, and U_j is proper over

U. Properness and quasi-finiteness also make g finite. The image of g is closed and dense in V_i, hence all of V_i. Properness descends along this proper surjection: after any base change, the image of a closed subset of V_i is the image of its closed inverse image in U_j. Thus f is universally closed; being separated and of finite type, it is proper. Quasi-finiteness then makes f finite.

At a geometric closed point the ramification indices for the finite maps multiply. Their product for fg is one, so the index for f is one at every point, since g is surjective. Residue extensions are separable in characteristic zero. A finite map of smooth curves is flat and unramified when all these indices are one, hence finite étale. Thus V_i is the desired common cover. Conversely, the covers of any connected finite étale cover W are cofinal among covers of its base: intersect a subgroup of $\pi_1(W)$ with its finitely many conjugates in the base group. The resulting connected cover is finite étale over both. The universal pro-cover of W therefore identifies with the universal pro-covers of both U and V. This proves the converse and the claimed equivalence of actual pro-schemes. □

ZCF9. A basis of the more special elementary neighborhoods *[Proved]*

[§3, manuscript pp. 33–34; scans 68–69.] Let X be smooth over a characteristic-zero field k. Every point has a fundamental system of Zariski-open neighborhoods which are towers of hyperbolic curve fibrations, each curve fibration being a finite étale cover of a relative punctured projective line with finite étale boundary of degree at least three. The bottom field may be a finite extension of k, accounting for geometric connected components. This proves the geometric neighborhood assertion of scan 68 in its characteristic-zero context; it does not construct the additional sections needed in scan 67.

Proof. We work inside any prescribed affine open of X. The exact Artin-neighborhood input is Panin–Stavrova–Vavilov, Proposition 1 [PSV09]: over an infinite field, a smooth geometrically irreducible variety and a finite set of closed points admit a Zariski-open neighborhood with an elementary fibration to an open of projective space of dimension one less. Definition 1 there gives a smooth proper geometrically connected curve compactification with nonempty finite étale boundary. Passing to the finite constant field of a connected component gives the indicated bottom field; in characteristic zero this component is geometrically irreducible. For a nonclosed point choose a closed specialization in the prescribed affine open; every open containing the specialization also contains the original point.

It remains to prove the more special form of a single fibration, retaining the chosen closed point x. Write $C \to B$ for its smooth proper compactification and D for its nonempty finite étale boundary; let b be the image of x. On the smooth proper curve $C_b/k(b)$ choose $L = \mathcal{O}(nD_b)$ with n large. There is a finite map $h_b : C_b \to \mathbb{P}^1$ defined by a pair of sections of L, with these properties: its branch points are simple and have pairwise distinct images; it is unramified at x and at D_b; the images of D_b are pairwise distinct and avoid all branch values and $h_b(x)$; and the entire fibre over $h_b(x) = 0$ is unramified.

Here are details of this choice, to justify the retained-point condition. Impose that the numerator vanish simply at the finite reduced subscheme x and that the denominator not vanish there. For n sufficiently large, Riemann–Roch makes sections with these conditions surject onto arbitrary jets of order at most two at any other point and first jets at any pair of distinct other points. The pair has no common zero on a nonempty open of its parameter space: a common zero imposes two independent conditions at a variable point of a curve. Ramification of order at least three imposes two conditions on its two-jet, also at a one-dimensional variable point. Two distinct simple critical points with the same value impose three independent conditions on their two first jets, where the pair of points has dimension two. A critical point above zero, or above the image of a point of D_b, similarly imposes two conditions with only one variable point. At the finitely many prescribed points the simple-zero, unramified, and distinct-value conditions are

nonempty open conditions on their jets and values. These dimension counts use the specified surjections on jets; their bad incidences have images of strictly smaller dimension. They leave a nonempty open in the affine parameter space over the infinite field $k(b)$, so a $k(b)$-defined pair exists.

For the same large n, proper cohomology and base change identifies sections of $\mathcal{O}_C(nD)$ on the fibre with the fibre of a locally free pushforward. Lift the chosen pair to a neighborhood of b. Properness allows us to shrink that neighborhood so that the pair has no common zero and defines a finite map $h : C \to \mathbb{P}^1_B$. Its ramification divisor is finite étale over B: its points were simple on the fibre at b, and the failure locus is closed after taking its proper image in B. The distinct branch images and the images of D consequently form disjoint finite étale divisors after further shrinking. Add enough disjoint sections of $\mathbb{P}^1_B$ to make their total degree at least three, avoiding $h_b(x)$ and all those values at b, and shrink again. Their existence follows by choosing finitely many distinct values in the infinite field $k(b)$ and lifting their coordinates locally on B.

Let V be the complement of this divisor in $\mathbb{P}^1_B$ and let $U' = h^{-1}(V)$. The map $U' \to V$ is finite étale; U' contains x and lies in $C \setminus D$. The reduced complement of U' in C is finite étale over B as well. At a simple ramification point the local equation is a unit times z^2 after étale localization; its reduced inverse image is the ramification section, disjoint from the unramified inverse images. At the other removed values the inverse image is already étale. This proves the last assertion without confusing a reduced inverse image with the generally nonreduced branch fibre. The fibre Euler characteristic is multiplied by the degree of the finite étale cover, so it is negative.

Apply induction to a neighborhood of b in the smooth base B and pull U' back to it. All changes are restrictions to actual Zariski opens retaining the prescribed point; hence the final tower is inside the initially prescribed open of X. This proves the claimed fundamental-system assertion. □

ZCF10. Generic cusps, closed-point determination, and higher-codimension decomposition groups *[Proved]*

[§3, manuscript pp. 32, 36, 45–49; scans 65, 73, 91–99.] The following additional assertions used without proof in the section argument hold under its hypotheses of a smooth integral finite-type base in characteristic zero and a smooth relative hyperbolic curve.

1. A rational point of the generic finite étale boundary extends uniquely to a boundary section over the normal base.
2. Two rational sections which agree at all closed points of a common nonempty open domain are equal as rational sections.
3. For $A = \mathcal{O}_{S,s}$, compatible henselizations inside K^{sep}, and $K^h = \mathrm{Frac}(A^h)$, $K^{\mathrm{sh}} = \mathrm{Frac}(A^{\mathrm{sh}})$, the decomposition and inertia groups are

$$N_s = G_{K^h}, \qquad I_s = G_{K^{\mathrm{sh}}}, \qquad 1 \to I_s \to N_s \to G_{k(s)} \to 1. \tag{ZCF.3}$$

The groups depend on the chosen prolongation by conjugation. For an actual rational section f_K, its map $G_K \to E_U$ kills I_s exactly when f_K extends on a neighborhood of s. The induced quotient section is then the section of $f(s)$.

Proof. For (1), the connected component of the finite étale boundary containing the generic point in question is finite birational over the normal base, hence isomorphic to it: its finite coordinate algebra lies in the fraction field and is integral over the integrally closed base ring. This supplies the section. For (2), the equalizer of two sections is closed because the target is separated. Closed points are dense in every closed subset of a finite-type scheme over a field. Thus its underlying

set is the entire common domain. The ideal defining the equalizer vanishes in every prime and therefore is zero on the reduced domain. The sections agree as morphisms, hence rationally.

For (3), normalize S in finite Galois subextensions of K^{sep}. The stabilizer of a chosen point above s acts on its residue field, with kernel inertia. Passing to the limit, the unramified algebraic local extensions are precisely the étale neighborhoods defining the henselization, and their strictly local extensions are precisely those defining the strict henselization. Hensel's lemma lifts each finite separable residue extension, so the residue map is surjective. This proves every term and map in (ZCF.3). The doubled tilde on the strict field in scan 93 is mathematically necessary: using K^h for both groups would erase the nontrivial quotient $G_{k(s)}$.

If f extends near s, functoriality and the strict local base make its map on I_s trivial. Conversely, base change to $\operatorname{Spec} A^{\mathrm{sh}}$. The rational map to the pulled-back curve has trivial map on the absolute group of its fraction field: its image lies in the geometric fibre group, whose map to E_U is injective by C4. At each height-one point its inertia map is therefore trivial. C2 extends the point over every such DVR; finite presentation spreads these extensions and C1 extends across the codimension-at-least-two complement. The extension over A^{sh} is unique, agrees over the fraction field with the given K-point, and descends by faithfully flat descent to A. Finite presentation spreads it to a neighborhood of s. C4 identifies the two arithmetic extensions for the henselian base and its closed fibre: they have the same geometric kernel and quotient $G_{k(s)}$. The descended section restricts to the actual point $f(s)$ under this identification.

In scan 73 the map $G_K \to \pi_1(\mathcal{O}_{S,s}) \to \pi_1(S)$ is surjective at both stages: a connected finite étale cover of a normal integral scheme is normal integral, and its generic and local restrictions remain integral and connected. This is the finite-cover criterion for surjectivity. In scans 98–99, composition with a decomposition-group section gives the finite point section when f extends, by the preceding argument; when it meets the boundary it gives the tangential section calculated in C3 and C9. The Tate factor in scan 91 is $T = \widehat{\mathbb{Z}}(1)$ with cyclotomic action. The word "trivial" cannot be interpreted as a trivial $G_{k(s)}$-module over a number field: the actual conjugation formula is $g\tau g^{-1} = \tau^{\chi(g)}$. □

ZCF11. The exact meaning of the inner-action reformulation in scan 102 *[Proved]*

[§3, manuscript pp. 50–51; scans 101–104.] Let $1 \to \Pi \to E \to G_K \to 1$ be the arithmetic extension of a hyperbolic curve and let $s : G_K \to E$ be a continuous section. For an inertia group $T \subset G_K$, write $\alpha_s : T \to \mathrm{Aut}(\Pi)$ for conjugation by s. Triviality of its outer action is equivalent to the existence of a continuous homomorphism $u_s : T \to \Pi$ with $\alpha_s(\tau) = \mathrm{int}(u_s(\tau))$. This homomorphism is unique. The outer condition is independent of s, so one such section suffices and every such section then has this property.

Proof. The map $\Pi \to \mathrm{Inn}(\Pi)$ is an isomorphism of topological groups because Π is centre-free (A3): it is a continuous bijection from a compact group to a Hausdorff group. Vanishing in $\mathrm{Out}(\Pi)$ places $\alpha_s(T)$ in this image and permits its unique continuous inverse. The inverse preserves multiplication, which proves that u_s is a homomorphism. The converse follows by projecting inner automorphisms to the outer group. Two sections differ by a Π-valued crossed homomorphism, so their conjugation actions differ by inner automorphisms. This proves the last assertion. The actual map $s|_T : T \to E$ does not itself land in Π: its composite with $E \to G_K$ is the inclusion of T. The map u_s just constructed is the precise comparison required by the manuscript's phrase "induit un hom $T \to \pi$".

The suggestion that a criterion using only the Jacobian's p-divisible group should detect good reduction of a proper curve has the same obstruction as the abelian criterion: the compact-type degeneration F17.2 has a Jacobian extending to an abelian scheme, whose p-power kernels form an extending Barsotti–Tate group for every p, while the curve is singular in its stable reduction. The same counterexample exists with the prime equal to the residue characteristic:

take $y^2 = (x^3 - 5^6)(x^3 - 1)$ over $\mathbb{Q}_5$. The equations of F17.2 with $t = 5$ give a stable model over $\mathbb{Z}_5$, with node $VW = 5$ and the two displayed elliptic components over $\mathbb{F}_5$. Their discriminants are units at 5, so both components are smooth. The compact-type Jacobian extends to an abelian scheme by the relative-Picard construction and descent checked in F17.2 [DR17, §1]; its kernels of multiplication by 5^n form an extending Barsotti–Tate group, whereas the stable curve has a node. The nonabelian criterion is F17.1. The further construction of a nongeometric genuine action whose outer action already is geometric, in scan 104, is the unresolved section-existence problem ZCF6(1); F17.4 instead constructs an outer action which is itself nongeometric. □

ZCF12. Continuous semilinear actions descend to a finite quotient over the base field *[Proved]*

[§3, manuscript pp. 29–33; scans 58–59, 62–63, 66–67.] Let k have characteristic zero. A continuous section of $E_U \to G_k$ for a hyperbolic curve U/k defines a semilinear action on its universal geometric pro-cover, equivariant for the ordinary scalar action on $U_{\bar{k}}$. These are exactly the continuous semilinear actions equivariant for that specified projection.

More generally, let $\mathcal{U}$ be the universal pro-cover of a hyperbolic curve over $\bar{k}$, and let Ω be the union of the function fields of its finite covering levels. Give the semilinear automorphism group the topology of pointwise convergence on the discrete field Ω. Every semilinear G_k-action for which each $f \in \Omega$ has open stabilizer admits a finite hyperbolic quotient W/k: there is an equivariant projection $\mathcal{U} \to W_{\bar{k}}$, its deck group is normalized by all of G_k, and the action comes from a continuous section of $E_W \to G_k$. The geometric quotient can be chosen as a finite étale cover of any prescribed initial level.

Consequently the family of Conjecture B with the G_k° strengthening of scan 61, and that of Conjecture B' at the same field k, each quantified over all hyperbolic curves and with this continuity clause in B', are equivalent. This statement uses G_k°-fixed compatible geometric pro-points on both sides. Every semilinear automorphism also extends canonically to the completed pro-cover. The natural G_k-action on germs of geometric-conjugacy classes of sections is well defined; the point-section map to those germs is equivariant and injective for k finitely generated over $\mathbb{Q}$.

Proof. Fix a geometric fibre functor. The function field Ω of the geometric universal cover of U is also the union of the finite arithmetic covering fields of U/k: a geometric cover and its map descend to a finite coefficient extension, and every arithmetic cover becomes a geometric cover after extending the constants. Thus $E_U = \mathrm{Aut}_{k(U)}(\Omega)$ with its Krull topology. A section acts semilinearly on the directed system of finite covers and hence on its inverse limit; a cofinal system of finite arithmetic Galois covering fields is invariant. The Krull topology and continuity of the section give open stabilizers for every element of Ω. Conversely an action equivariant for the scalar action on $U_{\bar{k}}$ fixes $k(U)$, giving a homomorphism $G_k \to E_U$ whose projection is the identity. Open function stabilizers are exactly the continuity condition for this homomorphism. Equivalently the construction is the action on finite continuous Π-sets and their fibre functor. This identifies actual sections after the fibre functor has been fixed; changing it gives precisely geometric inner conjugation.

We next prove the descent for an arbitrary action ρ with the stated topology. Start with any level $U_0/\bar{k}$ and choose field generators $L_0 = \bar{k}(U_0) = \bar{k}(f_1, \dots, f_m) \subset \Omega$. An open subgroup of the profinite group G_k has finite index. Hence each orbit $\{\rho(g)f_j : g \in G_k\}$ is finite. Put

$$L = \bar{k}\big(\rho(g)f_j : g \in G_k,\ 1 \le j \le m\big), \qquad P = \mathrm{Aut}_L(\Omega). \tag{ZCF.4}$$

The extension L/L_0 is finite: its finitely many generators lie in the algebraic extension Ω/L_0. It corresponds to a connected finite étale cover $W_{\bar{k}} \to U_0$, because Ω is the union of precisely those covering fields. More explicitly, put all the generators in one finite Galois étale covering

field; its intermediate field L is the function field of the finite étale quotient by the corresponding subgroup. Thus P is an open subgroup of the initial deck group, and $\mathcal{U} \to W_{\bar{k}}$ is its universal geometric pro-cover. The field L is G_k-stable, so $\rho(g)P\rho(g)^{-1} = P$ for every $g \in G_k$.

This normalization descends the action on the actual open curve, not merely its function field. Write $p : \mathcal{U} \to W_{\bar{k}}$. Regarding a semilinear map as a linear map to the appropriate scalar-conjugate curve, $p\rho(g)$ factors through a finite covering level by Stacks, Proposition 32.6.1 [The26a]. Refine that level to a finite Galois cover $W' \to W_{\bar{k}}$ corresponding to an open normal subgroup $P' \subset P$. The descended map $h : W' \to g^*W_{\bar{k}}$ is invariant under P/P': for $a \in P$ the equality $p\rho(g)a = p(\rho(g)a\rho(g)^{-1})\rho(g) = p\rho(g)$ holds, and the projection from $\mathcal{U}$ to W' is faithfully flat, being an inverse limit of finite étale surjections. Finite faithfully flat descent along $W' \to W_{\bar{k}}$ therefore gives a morphism $W_{\bar{k}} \to g^*W_{\bar{k}}$. In affine descent charts the descended coordinate functions are exactly the invariants in the finite Galois algebra; uniqueness glues these morphisms. The inverse action gives its inverse, and the action law follows after pullback along p. We have therefore obtained the full semilinear action on the open curve. It extends uniquely to its smooth proper compactification and preserves its boundary.

For completeness we make the arithmetic descent effective. The finite-generator argument in the checked prior return D11 [AI-26] works for any prescribed level as follows. Choose a finite coefficient extension k_0/k over which the smooth compactification, its boundary, field generators, and their finite defining relations are defined. Intersect G_{k_0} with the stabilizers of those generators and take the normal core in G_k. The result is an open normal subgroup $H = G_{k_i}$. It fixes the generators and acts on the coefficients by their usual scalar action, so the level has a k_i-form on which this is exactly the standard H-action. Equality on the function field gives equality of the projection maps: one can factor them through a common finite level and use integrality of that level and separatedness of the target. In particular the projection is equivariant, including its boundary. This proves directly the prior return's claim for every initial level after an open restriction.

Apply this construction to the G_k-stable level just obtained. Its field of H-invariants is $k_i(W_i)$. Normality of H implies that every $\rho(g)$ preserves that field, with its scalar action on k_i. Thus the finite group $G_k/H = \mathrm{Gal}(k_i/k)$ acts semilinearly on the smooth proper marked pair $(\bar{W}_i, D_i)$ over k_i. The action is an actual Galois descent datum. It is effective: $\omega_{\bar{W}_i}(D_i)^{\otimes 3}$ is very ample for a hyperbolic marked curve, is canonically linearized by every isomorphism of the pair, and embeds both the curve and its marked divisor into the projective space of its sections. Finite Galois vector-space descent identifies that section space with $k_i \otimes_k$ its invariant space; the trace-dual basis proof and the descent of all homogeneous ideal pieces are exactly those written in F17.7. Descending these ideals gives a projective pair $(\bar{W}, D)/k$ whose base change is the displayed pair. Smoothness and the finite étale property of D descend by faithful flatness. Its complement W is hyperbolic because its geometric curve is a finite étale cover of the initial hyperbolic curve. This proves the claimed quotient over k itself.

The field Ω is the union of the finite arithmetic covering fields of this W, by the same coefficient descent used in the first paragraph. The action fixes $k(W)$ and has its prescribed action on $\bar{k}$. It therefore gives the actual continuous section in

$$1 \longrightarrow P \longrightarrow \mathrm{Aut}_{k(W)}(\Omega) = E_W \longrightarrow G_k \longrightarrow 1. \tag{ZCF.5}$$

Continuity again follows from the open stabilizers of finite sets of field elements. This proves that every action in the stated continuous form of B' is an ordinary section action for a quotient curve over the same field. Conversely every section action satisfies that continuity clause.

The comparison of completed geometric pro-points is explicit. The pointed finite covers of $W_{\bar{k}}$ form a cofinal subsystem of the pointed finite covers of U_0: composition gives a cover of U_0, and the component through the chosen lift in $V \times_{U_0} W_{\bar{k}}$ refines any cover $V \to U_0$. Extending the

maps to proper compactifications therefore gives a bijection

$$\varprojlim_{V \to U_0} \bar{V}(\bar{k}) \xleftarrow{\sim} \varprojlim_{V \to W_{\bar{k}}} \bar{V}(\bar{k}).$$

In group notation, if Δ_0 is the original deck group, the opens $M \subset P$ are cofinal among opens $N \subset \Delta_0$, by taking $M \subset N \cap P$. The inverse bijection sends the coordinate at such an M to the level for N; common refinement proves independence of this choice. It is G_k-equivariant: both actions extend the same action on $\mathcal{U}$, and extension of every finite-level map to its proper compactification is unique. Hence the two assertions about a unique G_k°-fixed compatible geometric pro-point concern exactly the same actions and points. This proves their fixed-base equivalence without proving either existence assertion. For a general characteristic-zero field no replacement of G_k° by G_k is part of this argument. Over a field finitely generated over $\mathbb{Q}$, the checked uniqueness statement D0b [AI-26] supplies the additional comparison: a unique G_k°-fixed point is G_k-fixed by normality, while a G_k-fixed point is the unique G_k°-fixed point. In that arithmetic range the plain existence formulation of B has the same fixed-point problem.

Every finite map between smooth curves extends uniquely between their smooth proper compactifications, by the DVR valuative criterion for the proper target. Hence transition maps and the action extend compatibly to the inverse limit of compactifications. For an isomorphism of universal pro-covers without a specified deck group, the descent argument of ZCF8 supplies finite-level comparison maps in both directions; they extend to the proper compactifications, and their inverse-limit composites are the identity. This proves the claimed canonical extension of such an isomorphism. For a semilinear isomorphism, apply the same argument to the corresponding scalar-conjugate curve.

For the assertion on germs, a section over an open $H \subset G_k$ is identified with its restrictions to smaller open subgroups and with its conjugates by Π. For $g \in G_k$, choose $e \in E_U$ above g and send s to $h \mapsto e\, s(g^{-1}hg)\, e^{-1}$ on gHg^{-1}. Another choice of e differs by Π and gives the same class; restriction commutes with this operation. The group law is valid modulo exactly the same conjugation, so it is an action on germs. A geometric point $x \in U(\bar{k})$ is defined over a finite extension; its section defines a germ, and functoriality gives equivariance. Equality of two such germs is equality of the two point-section classes over a common finite extension. B5 then implies equality of the points. The assertion that the noncuspidal germs are all in this image is precisely the unresolved surjectivity in ZCF6, not a consequence of this equivariant injection. □

10 Additional steps in cote 149, scans 105–136

The following statements record the complete sweep of §4. The formulas already answered in D1–D10 and E1–E6 are not duplicated. We retain a genuinely unresolved assertion as an explicitly identified obstruction; no conjectural assertion is used in any preceding proof.

ZD1. A simply connected scheme as a fiber functor *[Proved]*

Let Y be connected and locally noetherian, and let $Z \to Y$ be nonempty and connected with trivial *étale* fundamental group. Then

$$F_Z : \mathrm{FEt}(Y) \longrightarrow \mathrm{FinSets}, \qquad E \longmapsto \pi_0(E \times_Y Z)$$

is a fiber functor. A map of such pointed data gives a genuine continuous homomorphism of its automorphism groups. For a morphism $f : X \to Y$, the relative kernel depends on the choice of connecting path through conjugation by $\pi_1(X)$; that action need not be inner as an action on the kernel itself.

Passages: cote 149, §4, manuscript pp. 52–56, scans 105–114, especially “on devrait pouvoir ècrire” at scan 107.

Proof. Every finite *étale* cover of Z is a disjoint union of copies of Z. Choose a geometric point z of Z. Each of these copies has exactly one point above z, giving a natural bijection $F_Z(E) \simeq E_z$. Hence F_Z is isomorphic to the usual geometric fiber functor, and is exact and faithful. This identification also constructs every claimed comparison map. A different connecting path between two geometric fiber functors is obtained by the action of an element of $\pi_1(X)$, so the comparison of kernels is its restricted conjugation. It is generally an outer automorphism of the kernel: for $X = \mathbb{G}_{m,\mathbb{Q}}$ and $Y = \operatorname{Spec} \mathbb{Q}$, conjugation acts cyclotomically on the abelian kernel T, whereas every inner automorphism of T is trivial. Thus the manuscript's qualification on scan 113 is necessary, and "automorphismes intérieurs" on scan 114 refers to the ambient group. □

ZD2. Relative kernels, universal covers, and the missing left exactness *[Proved]*

For connected locally noetherian pointed schemes $f : X \to Y$, put $H = \operatorname{im}(\pi_1(X) \to \pi_1(Y))$. The pointed connected pro-finite-*étale* cover $Y_H \to Y$ corresponding to H admits a unique pointed lift of f, the map $\pi_1(X) \to \pi_1(Y_H) = H$ is surjective, and its kernel is unchanged. Pullback of the universal pro-cover of Y_H has fundamental group that kernel. The identification of this kernel with a geometric fiber group additionally requires injectivity of the fiber homomorphism; an exact homotopy sequence beginning only at that fiber group does not supply it.

Passages: cote 149, §4, manuscript pp. 54–56, scans 109–113, formulas (12)–(20).

Proof. The finite quotients of the coset space $\pi_1(Y)/H$, with its distinguished coset, define Y_H. Pullback to X has that coset fixed by $\pi_1(X)$, hence a compatible section at every finite level; this constructs the lift as a morphism to the pro-cover. The homomorphism to H is surjective by its definition, and inclusion $H \hookrightarrow \pi_1(Y)$ is injective, proving equality of the two kernels. Pulling the universal cover back along this surjection gives the connected pro-cover of X corresponding to the kernel. Its fundamental group is that closed subgroup: a finite *étale* cover of the inverse limit descends to a finite level because its defining algebra and *étaleness* involve finitely many equations, and the usual finite-cover Galois correspondence then gives precisely the finite continuous sets for the subgroup. Pullback over Y instead gives components indexed by $\pi_1(Y)/H$; the chosen pointing selects the component just described. These are the comparison maps in (14)–(18). For a geometric fiber $X_{\bar{y}}$ the homomorphism $\pi_1(X_{\bar{y}}) \to \pi_1(X)$ has image the relative kernel whenever the usual homotopy exactness holds. It identifies the two groups precisely when it is also injective. The punctured tautological line bundle $\mathcal{O}_{\mathbb{P}^1}(-1)^\times = \mathbb{A}^2 \setminus \{0\}$ over an algebraically closed characteristic-zero field proves that this extra point matters: its base and total space have trivial fundamental group, by the projective-line computation and purity, whereas the fiber has group T. The map is $T \to 1$. This is also the explicit example in [BES15, Remark 9]. Thus the phrase "suite exacte d'homotopie universelle" must include the missing left injectivity for (12), (16), and (19) in the manuscript to have their asserted isomorphism meaning. Over a field the ordinary geometric exact sequence does include this injection, so the application at scan 110 is valid. □

ZD3. Recovering the model quotient and the function field *[Proved]*

For a regular integral finite-type model U of L/K, the kernel of $G_L \to \pi_1(U)$ is the closed normal subgroup generated by the inertia groups of its codimension-one local rings. It is equally generated by the inertia groups of all valuations with a center on U. For the nonempty open subschemes of one fixed normal model,

$$G_L \simeq \varprojlim_U \pi_1(U), \qquad P_{L/K} \simeq \varprojlim_U \pi_1(U_{\overline{K}})$$

when the models are geometrically connected; in general the second formula uses the kernels of the maps to G_K. For $U = X \setminus Z$ with X proper regular, the kernel of $\pi_1(U) \to \pi_1(X)$ is generated by the boundary-divisor inertia images.

Passages: cote 149, §4, manuscript pp. 57–60, scans 116–121, (21)–(25) and the first paragraph of scan 121.

Proof. A finite Galois quotient of G_L killing the listed codimension-one inertias gives a finite normalization of U unramified in codimension one; D1's purity argument makes it finite *étale*. Conversely every such cover kills the inertia of every valuation with a center, because its pullback to a strictly henselian valuation ring is trivial. Taking all finite quotients proves equality of the closed normal subgroups. The same argument applied to the boundary divisors of X proves the last assertion. Every finite separable extension of L is finite *étale* over a sufficiently small nonempty affine open: choose integral equations for field generators, invert their denominators and a nonzero discriminant, and take the resulting finite algebra. Thus the finite quotients in the inverse system exhaust those of G_L. Restriction of a connected finite *étale* cover of a normal integral scheme to a nonempty open remains connected, because that cover is normal integral. This proves the surjectivity of the transition maps and of each generic-point map. Equality of the finite quotients gives the first isomorphism as an actual isomorphism with the fixed generic fiber functor. Taking kernels of the compatible maps to G_K commutes with the inverse limit and proves the second formula. No section-conjecture input is involved. □

ZD4. Model opens, finite unions, and when the Riemann–Zariski space is a scheme *[Refuted]*

The points of manuscript codimension one are exactly the divisorial valuation rings, as asserted on scans 121–122. The Riemann–Zariski space is the proper normal curve for transcendence degree one, a point for degree zero, and has no nonempty affine open in its locally ringed-space structure for transcendence degree at least two. The claim on scan 123 that the opens supplied by models are not stable under finite unions is false: these opens are precisely the quasicompact opens, and they are stable under finite unions and finite intersections.

Passages: cote 149, §4, manuscript pp. 60–61, scans 121–123.

Proof. Write $n = \operatorname{trdeg}_K L$. Suppose a nontrivial valuation has residue transcendence degree $n-1$. Choose units $u_1, \dots, u_{n-1}$ with algebraically independent residues and an element t of positive value. The least-value argument of D4 makes these n elements algebraically independent, so L is finite over $K(u_1, \dots, u_{n-1}, t)$. Its restricted valuation is the t-adic valuation with coefficient field $K(u_1, \dots, u_{n-1})$: independence of the residues ensures that every nonzero rational coefficient has value zero. The fundamental inequality for this finite field extension makes both the value-group index and the residue-field degree finite. The value group is consequently discrete, and the residue field finitely generated of transcendence degree $n-1$. Normalize $K[u_1, \dots, u_{n-1}, t]$ in L. The center lies above the prime (t) and has height one; its local normal ring is a DVR, dominated by the given DVR of the same fraction field, hence equal to it. This realizes the valuation as a divisor on a model. Conversely a codimension-one point of a normal finite-type model has exactly this residue transcendence degree by the dimension formula. The valuation of manuscript codimension zero is trivial by D4's inequality. For $n = 1$ every nontrivial valuation is the local DVR of its center on the proper normal curve, proving the assertion about the whole locally ringed space by D2. For the topology, an open of the form $p_X^{-1}(U)$, for a model open $U \subset X$, consists exactly of the valuations with center on U. Such opens form a basis and are quasicompact. Indeed, the valuation-ring description embeds the space as a closed subspace of the compact product $\{0,1\}^L$ in the constructible topology, by the valuation-ring axioms. For an affine model open $\operatorname{Spec} A$, having a center there says that the finitely many algebra generators of

A/K lie in the valuation ring. This condition is clopen in the compact constructible topology, proving the required quasicompactness. A general model open is a finite union of affine ones. A quasicompact open is a finite union of such basic opens. Dominate their finitely many proper models by the closure of the common generic point in their product. On that common model their union, or intersection, is the inverse image of the union, or intersection, of the corresponding model opens. This proves the asserted description and stability, and directly corrects the sentence on scan 123. Finally suppose a nonempty affine open W existed in the Riemann–Zariski space for $n \geq 2$. By what we just proved, $W = p_X^{-1}(U)$ for a nonempty open of a normal proper model. Its ring of sections is

$$A = \Gamma(W, \mathcal{O}) = \bigcap_{V \in W} V = \Gamma(U, \mathcal{O}_U).$$

The first equality with the intersection follows from D2's local-ring description: membership of a rational function in a valuation ring is witnessed on a neighborhood of its center on some model. The last equality follows by testing the function at all codimension-one DVRs of the normal scheme U, which is the local criterion for regularity on a normal scheme. Pick a smooth closed point $a \in U$. There are distinct valuation rings centered at a: blow up a, take the valuation of its exceptional divisor, and refine that valuation by a nontrivial valuation of the exceptional divisor's positive-transcendence-degree function field. Both lie in W. Both contract their maximal ideals to the same prime $\mathfrak{m}_a \cap A$, because $A \subset \mathcal{O}_{U,a}$ and both dominate this local ring. This is impossible in $W = \operatorname{Spec} A$, where a point is exactly its contracted prime. Thus no such affine open exists. This proves the nonscheme claim without resolution of singularities. □

ZD5. Nonuniqueness of the valuation normalized by a section *[Refuted]*

The uniqueness and the decomposition-group inference in conjecture (a) of scans 126–127 are false for arbitrary valuations, even with nontrivial inertia and residue field exactly K. The normalization condition records the coarsening w of D5, and need not recover its residual refinement v/w.

Passage: cote 149, §4, manuscript pp. 62–63, scans 126–127, "et ce V est uniquement" and "en fait la section provient".

Proof. Let $K = \mathbb{Q}$, $L = \mathbb{Q}(z, x, y)$, and let w be the z-adic valuation. On its residue field $\mathbb{Q}(x, y)$ use the two valuations r_0, r_1 of D3 with value group $\mathbb{Q}$ and residue field $\mathbb{Q}$, obtained by substituting $x - a = t$, $y = 2 + \sum t^{j+1/j!}$ for $a = 0, 1$. They are distinct because their centers on $(\mathbb{P}^1)^2$ are $(0, 2)$ and $(1, 2)$. Compose them with w to obtain v_0, v_1. Their value groups are $\mathbb{Z} \oplus_{\text{lex}} \mathbb{Q}$, their residue fields are $\mathbb{Q}$, and D5 gives

$$I_{v_0} = I_{v_1} = I_w = T \neq 1, \qquad N_{G_L}(I_{v_a}) = D_w.$$

There is a section $G_{\mathbb{Q}} \to D_w$: first construct a section $G_{\mathbb{Q}} \to G_{\mathbb{Q}(x,y)}$ by successive decomposition splittings along the rational flag $y = 2$, $x = 0$, and then use the splitting for w in D4. Its image normalizes both displayed identical inertia groups, proving nonuniqueness with nonzero inertia. This section is not contained in D_{v_1}. Restriction to $\mathbb{Q}(x)$, followed by the quotient for $C = \mathbb{P}^1 \setminus \{2, 3, \infty\}$, is the point section at 0. Containment in D_{v_1} would make it the point section at 1, since $r_1(x - 1) > 0$. Point-section injectivity for this hyperbolic curve, as proved in B and in the prior return's Kummer argument, excludes equality or conjugacy of those sections. Thus actual containment in a decomposition group does not follow from normalization of inertia. Without the nonzero-inertia qualification, the two valuations r_0, r_1 themselves already have $I = 1$ and are normalized by every section. These examples explain both failures by the explicit homomorphisms, rather than by a change of terminology. □

ZD6. The overly broad centralizer margin *[Refuted]*

The assertion in the margin of scan 127 that every subgroup of N_v with open image in N_v/I_v has centralizer I_v is false. Cyclotomic triviality is necessary even for inclusion of I_v in that centralizer. For decomposition sections of rational places in a one-variable function field, the corrected cyclotomic-kernel statement is the one proved in D9 and the prior return B7–B8.

Passage: cote 149, §4, manuscript p. 63, scan 127, "ou simplt ∀ s-groupe de N_V dont l'image ... soit ouvert".

Proof. Take $L = \mathbb{Q}(t)$ and the t-adic valuation. A decomposition section $s : G_{\mathbb{Q}} \to D_v$ has image projecting onto all of $G_{\mathbb{Q}} = D_v/I_v$. The conjugation formula of D4 is

$$s(g)\tau s(g)^{-1} = \chi(g)\tau \quad (\tau \in T = I_v).$$

Complex conjugation has $\chi(g) = -1$ and does not centralize any nonzero ℓ-adic inertia element for an odd prime ℓ. Therefore I_v is not even contained in the claimed centralizer. For $s(G^{\circ}_{\mathbb{Q}})$ the action on I_v is trivial, and the reverse inclusion is the independently proved curve result cited in D9. The two assertions cannot be equated without restricting the subgroup in the margin. □

ZD7. Boundary inertia images do not determine the valuation or form a model-independent list *[Refuted]*

Distinct valuations can have the same inertia image in $\pi_1(U)$, and even the collection of local boundary-inertia images, up to conjugacy and commensurability, changes under a blow-up. Thus the intrinsicness asserted at scan 132 does not follow and is false in that form.

Passages: cote 149, §4, manuscript pp. 63–66, scans 128, 131–132.

Proof. All valuations centered in U have zero inertia image by D7. D3 gives distinct valuations centered outside U with the same zero image. It also gives distinct refinements with exactly equal nonzero inertia subgroups by ZD5. These already disprove injectivity of the assignment $v \mapsto \Delta_v$. For dependence of the local-image collection, take the product compactification of $U = (\mathbb{P}^1 \setminus \{0, 1, \infty\})^2$ and blow up the crossing $(0, 0)$. At a general point of the exceptional divisor use $x = u$, $y = uv$ with v a nonzero finite unit. Its meridian maps into the original cusp product $T_x \times T_y$ by $a \mapsto (a, a)$. On the original compactification the nonzero local images are a cusp group in one curve factor or a full product of two cusp groups. A diagonal group has nontrivial projections into both factors, so cannot be conjugate or commensurable with a one-factor group. It has infinite index in a two-factor product: already on each ℓ-adic factor the quotient has rank one. Thus the blow-up creates an image not commensurable with any original local image. The homomorphism is the Kummer map $v(x) = v(y) = 1$ of E3, which verifies every comparison used here. □

ZD8. What the boundary incidence complex does describe *[Proved]*

For a smooth complex variety, the incidence complex of a simple normal crossings boundary has the same connected components as the boundary. It must have a cell for every connected component of each intersection, not merely one simplex for each nonempty set of labels. It does not in general give the homotopy type of a punctured neighborhood, even when all its strata are points. The pro-system of complements of compact subsets of $U(\mathbb{C})$ is independent of its smooth proper compactification; punctured boundary neighborhoods are a cofinal system in it.

Passage: cote 149, §4, manuscript p. 66, scan 133, "quelle est sa signification topologique?" and the withdrawn contractibility proposal.

Proof. A connected component of the boundary is precisely a maximal set of its connected smooth components joined by a chain of nonempty intersections. This is exactly connectivity of the incidence graph and hence of the incidence complex. To see why intersection components must be retained, in $\mathbb{P}^1 \times \mathbb{P}^1$ take the graphs $y = x$ and $y = 1/x$. They meet transversely at $(1, 1)$ and $(-1, -1)$. Their actual incidence complex has two vertices joined by two edges, whereas a single simplex per label set gives only one edge and loses that circle. For $U = \mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\}$ all boundary strata are points. The incidence complex is three discrete points. A small punctured neighborhood of the boundary is three punctured discs, each with fundamental group $\mathbb{Z}$; it cannot have that homotopy type. The torus factors and their Chern gluing in E4 are precisely the additional data in this example and its higher-dimensional counterparts. Finally, for any compact subset $C \subset U(\mathbb{C})$, the open set $X(\mathbb{C}) \setminus C$ contains the boundary. Conversely the complement of a boundary neighborhood is a closed subset of the compact space $X(\mathbb{C})$ disjoint from the boundary, hence a compact subset of $U(\mathbb{C})$. These inclusions identify the two cofinal systems directly and prove the stated intrinsic end pro-system. The connected components of sufficiently small punctured tubular neighborhoods agree with those of the boundary, since the normal punctured-disc fibers are connected and tubes over intersecting branches meet. We assert this exact comparison, not an identification with the bare incidence complex. □

ZD9. Conjugacy along strata and the arithmetic stabilizer *[Proved]*

On a geometrically connected open stratum with labelled geometric boundary branches, its homomorphisms $T^J \to \pi_1(U_{\overline{K}})$ are conjugate by geometric paths. Arithmetic transport acts on the labels by permutation and on each T by the cyclotomic character. An individual geometric branch carries canonical equivariance under its Galois stabilizer; the full Galois group preserves the orbit of branches, and invariance of an individual image can fail. Thus scan 136's full-group assertion needs geometric connectedness or this stabilizer correction.

Passages: cote 149, §4, manuscript p. 67, scans 135–136, (38)–(40).

Proof. The local Kummer constructions of E1 glue along the normal-bundle torsor of E4. A path in the connected geometric stratum identifies its two torus fibers. A lift of this path to the total space conjugates their fundamental-group homomorphisms; a second lift changes the conjugator by an element of the torus fiber, which is abelian and acts trivially on its own group. Composition with the morphism to U gives the claimed geometric conjugacy, whether or not the meridian map is injective. On geometric fibers the branch labels are fixed. An arithmetic path permutes the geometric branches and sends a root of unity ζ_n to $\zeta_n^{\chi(g)}$, so its action is exactly the stated permutation and cyclotomic action. This proves the action on actual finite Kummer covers, and passing to their inverse limit proves the statement for T^J. For a concrete failure without the correction, take $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{\sqrt{2}, -\sqrt{2}, \infty\}$. The degree-two closed boundary point is irreducible and connected over $\mathbb{Q}$, but has two geometric branches. An element of $G_{\mathbb{Q}}$ interchanging them interchanges their cusp subgroups in the geometric free profinite group of rank two. These subgroups are not conjugate, by the distinct-cusp theorem of A. Thus neither individual branch homomorphism is fixed up to geometric conjugacy by all of $G_{\mathbb{Q}}$. Its stabilizer is $G_{\mathbb{Q}(\sqrt{2})}$, and the two-element orbit has the full equivariance claimed above. □

ZD10. The remaining birational lifting assertion and a smallest obstruction *[Open]*

The following exact assertion from conjecture (b) is not proved in this return: for $K = \mathbb{Q}$ and $U = \mathbb{P}^1 \setminus \{0, 1, \infty\}$, every continuous section of $\pi_1(U) \to G_K$ lifts through the continuous quotient $G_{K(t)} \to \pi_1(U)$ to a continuous section of $G_{K(t)} \to G_K$. Likewise, after the false uniqueness clause in conjecture (a) is removed, the assertion that every section of $G_{K(t)} \to G_K$ is contained

in a rational-place decomposition group remains unproved here. These unproved assertions are not premises of D or E.

Passages: cote 149, §4, manuscript pp. 62–63, scans 126–127, conjectures (a) and (b).

Attack and exact point of failure. All point sections and all cuspidal sections in this example do lift: at an interior rational point or a rational cusp, the decomposition sequence has the continuous splitting of D4, and its image under $G_{K(t)} \to \pi_1(U)$ is precisely the point or tangential section. For an arbitrary section s, form the actual profinite pullback

$$E_s = G_{K(t)} \times_{\pi_1(U),s} G_K, \qquad 1 \longrightarrow \ker(G_{K(t)} \to \pi_1(U)) \longrightarrow E_s \longrightarrow G_K \longrightarrow 1.$$

The required lift is exactly a continuous splitting of this extension. ZD3 describes its kernel by the interior-point inertia normal closure, but that description neither produces a complement nor kills the finite embedding obstructions of E_s. The normalizer attack identifies decomposition containment for a valuation already supplied, by D5 in rank one; it gives no existence of a normalized inertia subgroup for an arbitrary section. Passing to the abelianized kernel retains only the abelian lifting obstructions and supplies no implication back to a splitting of E_s. Thus these are exact remaining obstructions, not a proof by linearization. The source search checked the opened Borne–Emsalem–Stix theorem: its Theorem 27 lifts sections over $\mathbb{Q}$ for positive-genus proper curves and divisors that are unions of torsion packets to a specified cuspidally central quotient. That quotient is explicitly smaller than $G_{K(t)}$, its base curve has positive genus, and it supplies no splitting of the displayed E_s for the thrice-punctured line. The attached prior return also records these cases as unclosed. No claimed universal lift or arbitrary-section classification follows from those results.

ZD11. The universal lifting claim has the strength of the section conjecture *[Known]*

For a fixed smooth geometrically connected hyperbolic curve C/k, with k finitely generated over $\mathbb{Q}$, the following two universal assertions are equivalent: every section of C_F/F is geometric or cuspidal for every finitely generated extension F/k; and every such section lifts through $G_{F(C)} \to \pi_1(C_F)$. Thus even the curve case of the unrestricted conjecture (b) in scan 127 contains the full section-conjecture assertion.

Passage: cote 149, §4, manuscript p. 63, scan 127, conjecture (b).

Proof. This is Bresciani, Theorem B, with its hypotheses exactly as stated above [Bre24]. The comparison underlying its use here is explicit. For an indeterminate t, geometric invariance of the curve fundamental group identifies

$$\pi_1(C_{F(t)}) \simeq \pi_1(C_F) \times_{G_F} G_{F(t)}.$$

The base change of a section s is the continuous map $g \mapsto (s(\mathrm{res}(g)), g)$ into this fiber product. Bresciani's Theorem A states, for a smooth geometrically connected curve over a field finitely generated over $\mathbb{Q}$, that s is geometric or cuspidal exactly when this base-changed section is birationally liftable. Universal lifting applies in particular to the finitely generated extension $F(t)/k$, so that theorem gives the first universal assertion. Conversely a geometric section lifts using the rational-place decomposition sequence of D4; a cuspidal section already factors through that sequence, and its identical local splitting defines the birational lift. Quotienting by the geometric conjugacy ambiguity and lifting a conjugator through the surjection $G_{F(C)} \to \pi_1(C_F)$ makes this an assertion about actual continuous lifts as well as section classes. These maps prove the claimed equivalence and locate the unresolved step of ZD10 exactly. □

11 Additional steps in the opening twenty-two scans of cote 142

Z142a. Orientation signs and compactness of the Hom spaces *[Proved]*

For the full, possibly orientation-reversing, finite-cover isotopy category, the orbits under $M^+(X)$ are indexed by $R_X(Y) \times \{+1, -1\}$, where $R_X(Y)$ is the finite set of unoriented isomorphism classes of covers of Y homeomorphic to X. In the oriented completed category, each Hom space is compact and metrizable. The unqualified assertion that it has no isolated points is false: for $X = Y = \mathbf{P}^1(\mathbf{C}) \setminus \{0, 1, \infty\}$ the Hom space consists of the six elements of S_3. With individually labelled punctures it consists of one element. More generally, a nonempty completed right orbit has no isolated points exactly when $\widehat{M^+(X)}$ is infinite.

Manuscript passage: Cote 142, manuscript pp. 3–6, scans 8–9 and 13.

Proof. Every connected cover of an oriented surface has the pulled-back orientation. A homeomorphism from X to its source has one of two constant signs. Precomposition by an orientation-preserving map preserves the sign; two markings of the same sign differ by such a map. An orientation-reversing homeomorphism of a finite-type oriented surface exists: realize the compact surface as a symmetric connected sum of tori, place the punctures on its reflection locus, and transfer this reflection by a homeomorphism. Both signs therefore occur. G4 proves that $R_X(Y)$ is finite. G6 and G8 express each oriented Hom space as a finite union of compact quotients $S_u\backslash\widehat{M^+(X)}$ with S_u finite.

The mapping-class group is finitely generated: for genus zero, [FM, Theorem 4.9] gives finitely many twists generating the pure subgroup; adding lifts of generators of the finite puncture permutation group gives the full group. For positive genus, [FM, Theorem 4.11] gives finite generation of the pure mapping-class group of $S_{g,n}$ for every $g \geq 1, n \geq 0$; adding the finite permutation quotient again gives the full group. A group generated by r elements has only finitely many homomorphisms to each finite group, since the images of the r generators determine a homomorphism. There are countably many finite groups. Its profinite completion therefore has a countable neighbourhood basis and is metrizable, by the metric determined by any nested cofinal list of open normal subgroups. Finite unions and quotients by finite groups preserve compact metrizability.

A point of $S_u\backslash\widehat{M}$ is isolated precisely when its finite inverse image is open in $\widehat{M}$. A profinite group with a finite open subset has an open identity subgroup and is compact discrete, hence finite. Conversely a finite quotient space is discrete. For the three-punctured sphere, cutting along three arcs shows that a homeomorphism fixing all three punctures is isotopic to the identity: the complementary discs have fixed boundary after straightening the arcs, and the disc isotopy is $h_s(z) = sh(z/s)$ on $|z| \leq s$, with identity outside, in a disc coordinate. Thus its full oriented mapping class group is S_3. Euler characteristic forces every self-cover to have degree one. Its completed Hom space is consequently S_3. The finite permutation calculation is recorded in `code/G_checks.py`. □

11.1 An algebraic description in characteristic zero

We spell out the path construction proposed in scans 13–16, including the correspondence that transports a finite cover. For a hyperbolic type $a = (g, n)$, let $\mathcal{M}_a$ be the Deligne–Mumford stack of smooth proper genus-g curves with a reduced unordered divisor of n points. It is the quotient of the ordered marked-curve stack by S_n. A geometric point is the same datum as a smooth open curve of type a. We use the *étale fundamental groupoid*: arrows between geometric points are natural isomorphisms of their fibre functors on representable finite étale covers. A path has a profinite topology, being a torsor under either endpoint's profinite fundamental group.

For fixed degree and source and target types, let $\mathcal{H}$ be the stack of connected finite étale maps of the open curves. Equivalently, its objects are finite maps of the proper curves whose ramification is contained in the target marked divisor, with the source marked divisor its full

inverse image. It has maps

$$s : \mathcal{H} \longrightarrow \mathcal{M}_a, \qquad t : \mathcal{H} \longrightarrow \mathcal{M}_b. \tag{Z.1}$$

The arrows in this stack retain isomorphisms of covers; a cover with a deck group consequently retains that stabilizer. This convention specifies the groupoid in (Z.1), including its finite automorphisms.

Z142b. The characteristic-zero algebraic category and its comparison *[Known]*

Over an algebraically closed field k of characteristic zero there is a category $\mathcal{T}_k$ described by the following arrows:

$$X \longrightarrow Y : \quad (u : X' \longrightarrow Y,\ \alpha : X \rightsquigarrow X' \text{ in } \Pi_1^{\acute{e}t}(\mathcal{M}_a)), \tag{Z.2}$$

modulo isomorphisms of finite covers that carry the marking α to the other marking. Composition transports the first cover along the second marking using (Z.1), and then composes the finite maps. Over $\mathbf{C}$ this category is naturally equivalent, with its profinite Hom topologies, to the category of G8 restricted to complex algebraic curves. Base extension along any embedding $k \hookrightarrow k'$ of algebraically closed characteristic-zero fields induces an equivalence $\mathcal{T}_k \to \mathcal{T}_{k'}$. Its object isomorphism classes are precisely the hyperbolic types (g, n), and its automorphism group at a complex object is the full profinite completion of the oriented mapping-class group.

Manuscript passage: Cote 142, manuscript pp. 6–7, scans 13–16.

Proof. We specify and verify the literature inputs. In the opened January 2007 version of Bertin–Romagny [BR07], Theorem 6.9 constructs the marked tame Hurwitz stacks as smooth proper Deligne–Mumford stacks over $\mathbf{Z}[1/|G|]$; their smooth-curve loci are the loci used here. Proposition 6.14 and its proof identify the target discriminant map as étale over smooth pointed curves. Theorem 6.20 constructs the nongalois-cover stack by quotienting the Galois-cover stack of fixed monodromy data (G, H), with H core-free, by the finite group of the monodromy data. These hypotheses apply: we are in characteristic zero, each Galois closure has finite group G, the subgroup fixing the connected cover is core-free, and all branch points lie in the marked target divisor. Additional unramified marked points are allowed by Definition 6.8. Hyperbolicity makes both marked curves stable. Thus (Z.1) exists algebraically and is étale locally over the target a finite groupoid of covers with finite deck stabilizers. Finite degree gives finitely many such covers over a complex target, by G4. The quotient description retains deck groups; we do not replace this groupoid by its set of isomorphism classes.

Over $\mathbf{C}$, take a marked oriented surface S of type b. Its Teichmüller space is contractible: Fenchel–Nielsen coordinates identify it with $\mathbf{R}^{6g-6+2n}$ for every hyperbolic type [FM, Theorem 10.6 and section 10.6.3]. A marking is an orientation-preserving homeomorphism modulo isotopy. Changing that marking is exactly the $M(S)$-action, so the associated analytic moduli stack is $[\mathrm{Teich}(S)/M(S)]$. Here the quotient-stack assertion can be checked on an arbitrary family: local topological trivializations give markings, two such trivializations differ by a mapping class, and their change maps satisfy the descent cocycle. Conversely the universal marked family and that descent cocycle recover the family. The contractible space therefore gives the topological fundamental group $M(S)$.

Fix a topological cover $u : X \to Y$. Pulling a complex structure on Y through u gives the unique complex structure on X making u holomorphic, including the local completion $z \mapsto z^e$ at every end. Changing the two markings preserves this cover precisely for

$$G_u = \{(a, b) \in M(X) \times M(Y) : bu = ua\}.$$

The corresponding analytic Hurwitz component is consequently $[\mathrm{Teich}(Y)/G_u]$, with the action through the target projection. Its fundamental group is G_u, and its maps on fundamental groups

are exactly the two projections in G8. Noohi's opened [Noo05, Theorem 20.4] states that, for an algebraic stack locally of finite type over $\mathbf{C}$, representable finite étale covers correspond to finite covering stacks of the associated topological stack. Hence its étale fundamental group is the profinite completion of this topological fundamental group. Applied to the two stacks just described, this identifies their groups and maps with $\widehat{M(S)}$ and $\widehat{G}_u$ and its endpoint projections. G8 proves the injectivity of the latter stabilizer in the product; thus the finite deck stabilizer is retained under completion.

SGA 1 XII, Theorem 5.1 algebraizes each finite topological cover of Y. By the preceding path identification, (Z.2) has Hom space $\coprod_u S_u \backslash \widehat{M(X)}$, exactly (G.1). For explicit composition, write the second arrow as $(v : Y' \to Z, \beta : Y \rightsquigarrow Y')$. Lift β in the finite-cover groupoid (Z.1), starting at u. It ends at $u' : X'' \to Y'$ and produces a source path $\alpha' : X' \rightsquigarrow X''$. The product is $(vu', \alpha'\alpha)$. Existence of the lift follows either from the locally finite étale groupoid description or, over $\mathbf{C}$, from the open target image $\widehat{H}_u$ of $\widehat{G}_u$ and its finite cosets. Two lifts differ by the unique deck comparison respecting their markings, by G8; they give the same arrow. This is precisely (G.3). Its associativity, continuity, and identity laws were proved there. The identification proves the claimed comparison.

For base extension we use the opened Bresciani [Bre21, Lemma A.22 and Proposition A.23]. They give invariance of the étale fundamental gerbe of a geometrically connected concentrated fibred category under characteristic-zero field extension; the lemma identifies maps to every finite étale stack over an algebraically closed extension. Our moduli and fixed-degree Hurwitz stacks are of finite type with quasi-compact separated atlases and therefore satisfy the concentration hypothesis. We apply the geometrically connected statement to each component of a fixed-degree Hurwitz stack separately. The lemma identifies their fundamental groupoids and all path torsors used in (Z.2). Applied to the curve Y, it also identifies maps to every finite classifying stack BG. Every connected finite cover is a quotient of its Galois closure; hence the category of finite covers of Y is unchanged. In particular the covers u in (Z.2) descend, together with their isomorphisms. These identifications preserve composition because the maps commute with base change. The resulting functor is fully faithful. Every point over k' belongs to a type component containing a point over k; that component is connected, as its complex Teichmüller description shows after descent to a finitely generated characteristic-zero field. The path torsor between those points is nonempty: its finite quotient torsors have the finite intersection property, and their inverse limit is compact. It supplies an isomorphism in (Z.2), proving essential surjectivity. This argument identifies types, rather than asserting that the individual algebraic curves descend up to scheme isomorphism. Conservativity from G6 ensures that no arrow of degree greater than one becomes an isomorphism. Degree-one markings give all isomorphisms within a type. This proves every remaining assertion. □

Z142c. The positive-characteristic base-invariance premise *[Refuted]*

The assertion on scan 16 that the full geometric fundamental group of a curve is invariant under every extension of algebraically closed fields is false in characteristic $p > 0$. The counterexample already occurs on the hyperbolic curve $U = \mathbf{P}^1 \setminus \{0, 1, \infty\}$. The valid replacements are: invariance for curves in characteristic zero; invariance for proper curves in arbitrary characteristic; and invariance of the maximal prime-to-p geometric fundamental group of a smooth curve in characteristic p.

Manuscript passage: Cote 142, manuscript p. 7, scans 15–16, parenthetical justification.

Proof. Let $k = \overline{\mathbf{F}}_p$, let t be transcendental over k, and let k' be an algebraic closure of $k(t)$. The equation

$$y^p - y = tx \tag{Z.3}$$

defines a degree-p finite étale $\mathbf{Z}/p$-torsor over $U_{k'}$: the polynomial is monic in y and its derivative is -1. Its class does not lie in the image of $H^1_{\acute{e}t}(U, \mathbf{Z}/p) \to H^1_{\acute{e}t}(U_{k'}, \mathbf{Z}/p)$. Indeed Artin–Schreier theory expresses a torsor by a rational function, modulo $h^p - h$. A rational function can be reduced at every pole by successively removing terms whose pole orders are divisible by p, using the pth roots of their leading coefficients. Over k all those roots remain in k. The reduced nonconstant principal parts have orders prime to p, and they are unique: a nonconstant h has a pole of order $m > 0$, so $h^p - h$ has a pole of order pm, whereas the largest pole order in a nonzero reduced principal part is prime to p. Constants disappear over an algebraically closed field. The unique reduced principal part of (Z.3) at infinity is tx, whose coefficient lies outside k. Thus it cannot come from a class over k. It is nonzero by the same pole-order argument, so the torsor is connected. Even changing its chosen $\mathbf{Z}/p$-action multiplies t by an element of $\mathbf{F}_p^*$ and does not make it lie in k.

The natural continuous map $\pi_1(U_{k'}) \to \pi_1(U_k)$ would induce a bijection on continuous maps to $\mathbf{Z}/p$ if it were an isomorphism. The displayed cohomology failure proves that it is not an isomorphism. For the replacements, SGA 1 X, Corollary 1.8 gives invariance for proper schemes over algebraically closed fields, and XIII, Corollaries 2.9 and 2.12 give the prime-to-characteristic presentation and comparison for smooth curves [G+71]. In characteristic zero the same presentation is the full one; equivalently the finite-cover comparison and descent used in Z142b apply. These statements have exactly the indicated properness or prime-to-p hypotheses. The counterexample refutes the stated premise for the proposed wild category. It does not identify an unspecified replacement for that category: its required path-transport construction is recorded separately in Z142g below. □

Z142d. Field automorphisms and the actual outer ambiguity *[Proved]*

Every automorphism σ of an algebraically closed characteristic-zero field k induces a functor on $\mathcal{T}_k$ by base change, and these functors compose with their canonical base-change isomorphisms. After choosing one object of each type and paths to its conjugates, the induced automorphisms of its profinite automorphism group are defined up to inner automorphisms. Changing a path by a changes the induced automorphism by conjugation by a.

Manuscript passage: Cote 142, manuscript pp. 7–8, scans 16–17.

Proof. Send X to X^σ, the cover u to u^σ, and a natural isomorphism of fibre functors to its pullback along σ. Base change preserves finite étale covers and the stack maps (Z.1); it therefore preserves the composition already constructed. The canonical isomorphism $(X^\sigma)^\tau \simeq X^{\tau\sigma}$, with the order fixed by this formula, satisfies the cocycle identity by the associativity of tensor products. These maps give the promised action without identifying a curve with its conjugate. A chosen path $c : X \rightsquigarrow X^\sigma$ identifies the endpoint groups by $b \mapsto c^{-1}bc$. Replacing c by ca conjugates this identification by a^{-1}. This explicit formula proves the claimed ambiguity and gives a well-defined outer action. □

Z142e. Path isotopy after allowing extra omitted branch points *[Refuted]*

For maps obtained by composing finite covers and inclusions that fill punctures, a continuous path through étale maps need not lie in a single two-sided orbit of the identity components of the source and target homeomorphism groups. The following example is an explicit composite of the two allowed sorts of arrows, has degree six, and has fixed source type $(0, 20)$ and target type $(0, 3)$. For a continuous family of holomorphic maps of compact curves with fixed ramification data and fixed incidence partition of the images of all marked points, local path isotopies do lie in such a two-sided orbit. The incidence hypothesis is essential in the example.

Manuscript passage: Cote 142, manuscript pp. 8–9, scans 18–19; “isotopie raisonnable”.

Proof. For $0 \le t \le 1/10$ set

$$h_t(z) = z^3 - 3z + t, \qquad f_t(z) = h_t(z)^2, \qquad Y = \mathbf{P}^1 \setminus \{9, 16, \infty\}.$$

Define the intermediate curves and the source by

$$\begin{aligned} Y_3 &= \mathbf{P}^1 \setminus \{0, 9, 16, \infty\}, \\ Y_2 &= \mathbf{P}^1 \setminus \{0, 3, -3, 4, -4, \infty\}, \\ Y_{1,t} &= Y_2 \setminus \{t-2, t+2\}, \qquad X_t = h_t^{-1}(Y_{1,t}). \end{aligned}$$

The restriction of f_t is the actual composite

$$X_t \xrightarrow[\text{finite étale, degree 3}]{h_t} Y_{1,t} \hookrightarrow Y_2 \xrightarrow[\text{finite étale, degree 2}]{w \mapsto w^2} Y_3 \hookrightarrow Y. \tag{Z.4}$$

All five curves are hyperbolic. The cubic has finite critical points $1, -1$ and critical values $t-2, t+2$, both removed from $Y_{1,t}$; infinity is removed as well. This proves that the first finite map is étale. The removed sets for the square map are exactly the inverse images of $0, 9, 16, \infty$, proving the same assertion for the second finite map. Each inclusion fills the stated punctures.

The source has twenty punctures: the fifteen simple preimages under h_t of $0, 3, -3, 4, -4$, the four points $1, -2, -1, 2$, and infinity. Indeed,

$$h_t(z) - (t-2) = (z-1)^2(z+2), \qquad h_t(z) - (t+2) = (z+1)^2(z-2).$$

For each $c \in \{0, 3, -3, 4, -4\}$ the program checks

$$\operatorname{disc}_z(h_t - c) = 27\big(4 - (t-c)^2\big).$$

These five discriminants never vanish on the chosen interval: $|t| < 2$, whereas $|t - c| > 2$ for the four nonzero values of c. Also $t-2, t+2$ are distinct and avoid that five-element set. Thus all twenty punctures remain distinct. Tracking them and applying disjoint-disc isotopies gives homeomorphisms $\eta_t : X_0 \to X_t$ with $\eta_0 = 1$. The maps $f_t \eta_t : X_0 \to Y$ form a continuous path of local homeomorphisms. They belong to the category generated by the arrows in (Z.4), including the source homeomorphism η_t.

The derivative is $f_t' = 6h_t(z)(z^2 - 1)$. Consequently its finite critical values are exactly

$$0, \quad (t-2)^2, \quad (t+2)^2.$$

The last two differ by $8t$. At $t = 0$ there are two distinct finite critical values, $0, 4$; at every $0 < t \le 1/10$ there are three. This calculation and the displayed factorizations are checked exactly in `code/G_checks.py`. A homeomorphism of a finite-type surface extends uniquely to the finite-end compactification. A two-sided conjugacy of two maps on these punctured surfaces therefore extends to their compact spheres and carries their sets of critical values bijectively. The two maps cannot be in the same two-sided orbit, because those sets have different cardinalities (including infinity adds one on both sides). This is an explicit failure for the ordinary path interpretation of "isotopie". The manuscript does not define the adjective "raisonnable", so this example does not select a final reading of that adjective.

For the corrected local assertion, track the finitely many distinct marked images and branch values in disjoint discs; the fixed incidence partition ensures that the same labels remain together. An ambient isotopy of the target carries all these points back to their initial positions. Away from those values the family is a family of finite unramified covers of a fixed punctured compact curve. The implicit function theorem gives the covering property also in the parameter direction; properness over compact subsets makes it a finite cover of the product with an interval. Path lifting in the interval trivializes that cover. Extend this trivialization over each omitted point using its unchanged local degree $z \mapsto z^e$. The fixed incidence data tracks the remaining marked preimages as well. It gives a source isotopy and the desired two-sided equivalence. Thus fixed Hurwitz data supplies the exact transfer map that an arbitrary path in the example lacks. □

Z142f. Filling punctures and the necessary source containment *[Proved]*

For finite maps $\overline{f} : \overline{X} \to \overline{Y}$ of smooth proper curves in characteristic zero, and finite reduced sets S, T, the restriction $X = \overline{X} \setminus S \to Y = \overline{Y} \setminus T$ is defined and étale exactly when $\overline{f}^{-1}(T) \subset S$ and every ramification point of $\overline{f}$ lies in S. Composing finite étale maps and puncture fillings produces such maps, including maps whose omitted ramification points have images inside Y. For ordered punctures, forgetting a specified set I induces a functor on étale fundamental groupoids of the corresponding moduli stacks whenever the remaining type is hyperbolic. Successive forgettings compose with their canonical natural isomorphisms.

Manuscript passage: Cote 142, manuscript pp. 8–10, scans 18, 20–21.

Proof. The first containment is exactly the condition for the restriction to land in Y. In local parameters, a finite map of smooth curves has $y = x^e$ times a unit; in characteristic zero it is étale at that point exactly when $e = 1$. This proves the second condition and both necessity and sufficiency. For a concrete composite, take

$$\mathbf{P}^1 \setminus \{0, \infty, 1, -1, i, -i\} \xrightarrow{z \mapsto z^2} \mathbf{P}^1 \setminus \{0, 1, -1, \infty\} \hookrightarrow \mathbf{P}^1 \setminus \{1, -1, \infty\}.$$

The first arrow is finite étale of degree two; the second fills the puncture 0. Its composite omits the ramification point 0 although its branch value 0 is inside the final target. All three open curves are hyperbolic. The inverse images and derivative are checked in `code/G_checks.py`.

On families of smooth proper marked curves, the forgetting map keeps the underlying family and the remaining disjoint sections. With a hyperbolic remaining type these are again objects of the indicated moduli stack, and the operation commutes with every base change and isomorphism of families. This defines an algebraic morphism of stacks. Pullback of representable finite étale covers defines its functor on fibre functors and their natural isomorphisms, hence on the étale fundamental groupoids. The composite forgetting two disjoint sets is the same operation on families as forgetting their union; the resulting natural isomorphisms satisfy associativity. Unordered remaining points are handled by their finite permutation quotient. The hyperbolicity condition is necessary to stay in the category under discussion: forgetting one puncture of type $(0, 3)$ produces type $(0, 2)$. This gives the exact type restriction omitted in scan 20. □

Z142g. The uncompleted parts of the later programme *[Open]*

The following objectives in scans 15–21 are not established here: (i) a wild positive-characteristic version of (Z.2) with the stated finite-cover factorization, path transport, and base-extension equivalence; (ii) the requested intrinsic description of the entire category together with the image of the arithmetic action, using only the specified profinite and combinatorial data; (iii) a completion theorem for the enlarged category containing arbitrary puncture fillings, with the unspecified "reasonable" isotopy of scan 19.

Manuscript passage: Cote 142, manuscript pp. 7–10, scans 15–21.

Recorded attacks and exact obstructions. For (i), we constructed the characteristic-zero category and its comparison in Z142b, then tested the asserted base-invariance premise on degree-p covers of $U_{0,3}$. Equation (Z.3) proves that this premise fails. The next required datum is a replacement path-lifting rule for the wild Hurwitz stack, compatible with its families of covers of a fixed target; the finite discrete transport argument in Z142b does not give that rule. We have not identified a category satisfying all three requested properties in this setting.

For (ii), Z142d constructs the arithmetic functors and explicitly computes their inner ambiguity, and G6–G8 give the entire completed finite-cover category from the discrete surface category. These constructions supply no criterion deciding which compatible profinite automorphisms are

induced by field automorphisms. The remaining task is an image computation for the displayed base-change action. Completing mapping-class groups does not compute that image: the functor $\sigma \mapsto (-)^{\sigma}$ is additional data, and its formula in Z142d still uses the field. No surjectivity claim about this action is made in this paper.

For (iii), we first tested ordinary path isotopy; Z142e refutes its identification with two-sided isotopy. We then proved the local comparison with fixed Hurwitz incidence data, and Z142f constructs the actual puncture-forgetting functors. These do not specify the relations among arbitrary composites when branch-image incidence changes. A free category on these functors can be constructed as the set of finite composable words, modulo chosen relations, but no relations were given in the manuscript that imply the proposed comparison. The next mathematical step is an explicit choice of the isotopy relation and a theorem identifying its groupoid on each ramification stratum and across its boundary. We retain this missing comparison as an exact unfinished objective; we do not turn the programmatic wording into an asserted theorem. □

Bibliography and source verification

Every entry includes an explicit verification note. A manuscript or an attached AI return is identified as such; verification of its contents is not a claim of independent literature authority.

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